

Explicit upper bounds for the Erdős–Surányi function $g(n)$

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Abstract

For $n \geq 1$ let $g(n)$ be the least integer g such that for every set $\{a_1 < \dots < a_n\}$ of integers greater than 1 and every $x \geq 0$, some g of the integers $x+1, \dots, x+a_n$ have product divisible by $a_1 \dots a_n$. Erdős and Surányi (1959) proved $g(3) = 4$ and $g(n) \geq (2 - o(1))n$, and Erdős asked (offering \$100) whether $g(n) \leq (2 + o(1))n$, or even $g(n) \leq 2n$ (Erdős Problem #708). To our knowledge no upper bound depending on n alone has been published. We prove

$$g(n) \leq n(\lceil \log_2 2n \rceil + \lceil \log_2 \lceil \log_2 2n \rceil \rceil + 2) \quad (n \geq 2), \quad g(n) \leq \left\lceil 48n \sqrt{\frac{81 + \ln n}{\ln(81 + \ln n)}} \right\rceil,$$

and $g(n) \leq (2 + o(1))n\sqrt{\ln n / \ln \ln n}$, and, by a different method (linear-programming duality and an elementary sieve lemma for intervals), $g(n) \leq (16 + 20 \ln \ln(8n^3))n = O(n \ln \ln n)$. The duality reduces a linear bound to a “hinge inequality” for weighted prime-factor counts on intervals; we prove it for 0/1 weights with the absolute threshold 4 for every set of primes, by explicit counting certificates; for fractional weights we prove it with threshold 65 whenever the small-modulus part of the weight has mean at least $\frac{17}{16}$, which reduces a linear bound

***Disclosure of the production pipeline.** Every mathematical statement in this note was produced by automated systems inside an autonomous research pipeline: the proofs of Theorems 1.1 and 1.2 were found in two independent runs of OpenAI GPT-5.6 at “Pro” effort through the web interface (26 and 90 minutes of reasoning respectively, on 2026-09-03) and the proofs of Theorems 1.3 and 1.4 by Qwen3.8-Max (Alibaba) through its web interface in two runs on 2026-09-03/04, all working from briefs written by an orchestrating agent (Claude Fable 5.1 through 2026-09-07 and Claude Opus 5 from 2026-09-08, both Anthropic) that selected the problem, pinned the statement to Erdős’s 1992 text, verified the lower-bound construction by exact computation, re-derived every step of both proofs, commissioned independent referee runs (Claude Opus 5 agents in fresh contexts, which also brute-forced the lemmas on hundreds of thousands of instances), and drafted this document. The proof of Theorem 14.1 (Section 14) was found by OpenAI GPT-6 Astra through the Codex command-line interface in a single 33-minute run on 2026-09-07, from a brief by the orchestrating agent; it was self-checked by exact arithmetic, re-derived line by line by the orchestrator, and refereed by a Claude Opus 5 agent in a fresh context (hinge, counting and value lemmas brute-forced on hundreds of instances). The Lean formalisation described at the end of Section 14 was likewise written by GPT-6 Astra from the paper’s statements and checked by the orchestrator. The threshold-17 proof of Section 15 was found by GPT-6 through the ChatGPT web interface at “Pro” effort (34 minutes), refereed by a Claude Opus 5 agent, and formalised by GPT-6 Astra. The rounding-cost reduction of Section 16 was found by GPT-6 Astra (15 minutes) and formalised by it (14 minutes); the threshold- $\frac{97}{10}$ proof of Section 17 was found by GPT-6 (web interface, “Pro”, 45 minutes) and refereed by a Claude Opus 5 agent with an independent exact recomputation of every constant; it was formalised by GPT-6 Astra (108 modules). The threshold- $\frac{56501}{6400}$ proof and the barrier of Section 18 were found by GPT-6 (web interface, “Pro”, 50 minutes) and refereed by a Claude Opus 5 agent; they are not formalised. The four constructions were implemented and exercised on more than seven million random instances, and every numerical inequality in the explicit parameter choices was checked by machine (scripts in the accompanying repository). No human supplied a definition, a lemma, a proof idea, or a correction. The problem itself remains open: the conjectured bound $g(n) \leq 2n$ is not proved here.

$g(n) \leq 81n$ to a “sparse core” statement; we prove that statement for all m by a signed divisor certificate (dyadic rounding of the weights, prefix carriers of mass between 2 and 4, and a capped-mass count of the carriers through a point), so that $g(n) \leq 81n$ for all n , and, with tightened constants, the threshold 17; a sharper count of the primes in the rounding step then gives $g(n) \leq 19n$ for all n (and $18n + o(n)$), and a four-mantissa rounding with mass-dependent moment orders brings the threshold to $\frac{97}{10}$, so that $g(n) \leq 12n$ for all n (all of this verified in Lean 4); a further refinement, refereed but not formalised, gives the threshold $\frac{56501}{6400}$ and $g(n) \leq 11n$, while an explicit family shows that no threshold below 2 holds, so this route cannot give less than $4n$; the two-sided interval counts alone cannot prove it, and the certificate’s value bound uses one further arithmetic input, a factorial-moment bound. The proofs of both linear bounds (Sections 5, 7, 13, 14 and 15) have been formalised in Lean 4 and checked by the kernel: the final theorems depend only on the three standard axioms. We also show that the conjectured bound holds for long intervals: if $a_n \geq 8n^3$ then $2n$ integers always suffice, so Erdős’s question is reduced to the range $a_n < 8n^3$; more generally, for every integer $k \geq 2$, kn integers suffice whenever $a_n \leq kn$ or $a_n^{k-1} \geq (kn)^{k+1}$; hence for every fixed $\delta > 0$ a bound linear in n (with a constant depending on δ) holds whenever $a_n \geq n^{1+\delta}$, e.g. $12n$ integers suffice whenever $a_n \geq n^2$. All proofs are elementary and constructive: prime-power atoms of the a_i that are large relative to a_n are placed one per interval element by a per-prime greedy rule, the small atoms are packed into bins, and a counting argument bounds the number of bins. We also record $g(4) \geq 5$ and $g(5) \geq 6$ (and, in Section 19, $g(10) \geq 14$ and $g(12) \geq 17$) by exact computation.

1 Introduction

Gallai observed that among any a_2 consecutive integers one can always find two whose product is a multiple of $a_1 a_2$ (for $1 < a_1 < a_2$), but that there are $a_1 < a_2 < a_3$ and a run of a_3 consecutive integers among which no three have product divisible by $a_1 a_2 a_3$. Erdős and Surányi [1] studied the general situation. Following Erdős’s formulation [2, §1], let $g(n)$ be the smallest integer such that, whenever $1 < a_1 < \dots < a_n$ are integers and $x \geq 0$, there are integers $x < u_1 < \dots < u_{g(n)} \leq x + a_n$ with $u_1 \dots u_{g(n)} \equiv 0 \pmod{a_1 \dots a_n}$. (Since enlarging B inside the interval preserves divisibility, “ $g(n)$ integers” may be read as “at most $g(n)$ integers”; we use the latter, which also covers the case $a_n < g(n)$.) They proved $g(3) = 4$ and

$$g(n) \geq (2 - \varepsilon)n \quad (n > n_0(\varepsilon)), \quad (1)$$

the lower bound coming from the set of all products $p_i p_j$ of ℓ primes satisfying $2p_1^2 > p_\ell^2$, with the interval positioned so that a single integer u , divisible exactly once by each p_i , is the only common multiple of every $p_i p_j$ in it. Erdős [2] asked whether $g(n) < (2 + \varepsilon)n$ for large n , “or perhaps even $g(n) \leq 2n$ ”, and offered \$100 for a proof or disproof; the question is Problem #708 on the Erdős problems site [3]. Neither [2] nor [3] records an upper bound for $g(n)$, and we are not aware of one in the literature (the original paper [1], in Hungarian, was not accessible to us). Note that a bound in terms of n alone is not obvious: the trivial estimate $g \leq \sum_i \omega(a_i)$ (one interval element per prime factor of each a_i , see Lemma 2.3) depends on a_n , which is unbounded.

Throughout, $v_p(k)$ is the exponent of the prime p in k , $\omega(k)$ the number of distinct prime factors of k , and $\Lambda(t) = \lceil \log_2 t \rceil$ is the least $r \geq 0$ with $t \leq 2^r$.

Theorem 1.1. *For every $n \geq 2$,*

$$g(n) \leq F(n) := n(\Lambda(2n) + \Lambda(\Lambda(2n)) + 2) = n \log_2 n + n \log_2 \log_2 n + O(n).$$

In particular $F(2) = 10$, $F(3) = 21$, $F(4) = 28$, $F(10) = 100$, $F(100) = 1300$, $F(1000) = 17000$.

Theorem 1.2. *For every $n \geq 1$, with $L = 81 + \ln n$,*

$$g(n) \leq \left\lceil 48n\sqrt{L/\ln L} \right\rceil, \quad \text{and} \quad g(n) \leq (2 + o(1))n\sqrt{\frac{\ln n}{\ln \ln n}} \quad (n \rightarrow \infty).$$

Theorem 1.1 gives the smaller value for every n below about 10^{104} ; Theorem 1.2 is asymptotically stronger. Neither is close to the conjectured $2n$. For long intervals, however, the conjecture holds:

Theorem 1.3. *If $a_n \geq 8n^3$, then for every $x \geq 0$ some $2n$ of the integers $x+1, \dots, x+a_n$ have product divisible by $a_1 \cdots a_n$ (at most $2n$ are needed, and since $a_n \geq 2n$ the set may be padded); in fact Conjecture 7.4 below holds for such A . Consequently $g(n) \leq 8n^3$ as well, and the question whether $g(n) \leq 2n$ is equivalent to the same question restricted to $a_n < 8n^3$.*

Theorem 1.3 is the case $k = 2$ of a one-parameter family:

Theorem 1.4. *Let $k \geq 2$ be an integer and suppose that $a_n \leq kn$ or $a_n^{k-1} \geq (kn)^{k+1}$. Then for every $x \geq 0$ some (at most) kn of the integers $x+1, \dots, x+a_n$ have product divisible by $a_1 \cdots a_n$. In particular $a_n \geq 9n^2$ gives $3n$ and $a_n \geq 1024^{1/3}n^{5/3}$ gives $4n$; for every $\delta > 0$ and every integer k with $\delta(k-1) > 2$, $a_n \geq n^{1+\delta}$ gives kn as soon as $n^{\delta(k-1)-2} \geq k^{k+1}$. Taking for each n the smallest admissible k (and Theorem 1.1 where none is admissible) gives, for every $n \geq 1$: $7n$ integers suffice whenever $a_n \geq n^3$, $12n$ whenever $a_n \geq n^2$, and $15n$ whenever $a_n \geq n^{7/4}$.*

Thus for every fixed $\delta > 0$ a bound linear in n holds when $a_n \geq n^{1+\delta}$, with a constant depending on δ ; for a fixed constant C the range not covered by Theorem 1.4 is $Cn < a_n < (Cn)^{(C+1)/(C-1)}$, a full power of n . The Erdős–Surányi family, with $a_n \asymp n(\ln n)^2$, lies just above the trivial range. The gap between $n\sqrt{\ln n/\ln \ln n}$ and $2n$ is discussed in Section 7.

Finally, Section 8 reduces a linear bound to a single inequality about additive functions on intervals (Conjecture 8.1): if it holds, then $g(n) \leq 18n$ (Theorem 8.2). We prove the inequality in two special cases and show, with explicit certificates, that three natural strengthenings of it are false because intervals can contain more integers free of small primes than $[1, m]$ does. Section 9 shows that the inequality follows, by linear-programming duality, from the two-sided count $\lfloor m/d \rfloor \leq N_I(d) \leq \lfloor m/d \rfloor + 1$ of multiples alone in every instance we could compute, observes that for the full set of primes the 0/1 case already follows from the trivial bound $\sum_I (w-1)^+ \geq \sum_{k \leq m} w(k) - m$ (Corollary 9.2), gives an explicit anchor certificate for prime sets without small primes, where that bound fails, and proves the 0/1 case for all m and all prime sets with the threshold 2 replaced by $20 \ln \ln m$ (Theorem 9.4); Section 12 replaces $20 \ln \ln m$ by the absolute threshold 4, for every prime set, with two explicit certificates (Theorem 12.1). Section 10 extends this to arbitrary weights with threshold $1 + 149e \ln(1 + \ln m)$ (Theorem 10.1), which through the same duality gives $g(n) = O(n \ln \ln n)$ (Corollary 10.2). Section 11 sharpens the threshold to $20 \ln \ln m$ by a geometric refinement of the same sieve (Theorem 11.1), giving $g(n) \leq (16 + 20 \ln \ln(8n^3))n$ for all n , which is below the bound of Theorem 1.2 for every n . Section 13 returns to the fractional case: the inequality with threshold 65 holds whenever the small-modulus atoms of the weight have mean at least $\frac{17}{16}$ (Theorem 13.2), so a linear bound $g(n) \leq 81n$ follows from a “sparse core” statement (Corollary 13.3); we show that the two-sided interval counts alone cannot prove it (Proposition 13.5), prove it for $m \leq 10^{2887}$ by a nonnegative certificate (Theorem 13.7, Corollary 13.8), and finally, in Section 14, prove it for all m by a certificate with coefficients of both signs (Theorem 14.1), which gives $g(n) \leq 81n$ for all n (Theorem 14.11); Section 15 tightens the constants to the threshold 17, giving $g(n) \leq 33n$

(Theorem 15.1); Section 16 replaces the rounding cost $16n$ by $2n$ (the primes dividing $\prod A$ number at most $n + \pi(\sqrt{a_n})$, and the k -split with $k = 3$ removes $a_n \geq 9n^2$), giving $g(n) \leq 19n$ and $18n + o(n)$ (Corollary 16.2, Theorem 16.6); Section 17 proves the threshold $\frac{97}{10}$, giving $g(n) \leq 12n$ (Corollary 17.2); Section 18 (refereed, not formalised) reaches $\frac{56501}{6400}$ and $g(n) \leq 11n$ (Corollary 18.2), and shows that no threshold below 2 holds (Proposition 18.9).

2 The common construction

Fix $A = \{a_1 < \dots < a_n\} \subset \{2, 3, \dots\}$, put $m = a_n$, $I = \{x + 1, \dots, x + m\}$ and $P = a_1 \dots a_n$. We must find $B \subseteq I$ with $P \mid \prod_{b \in B} b$ and bound $|B|$. Note that B is a set: an interval element contributes its valuations once, however many a_i it may be a multiple of.

Lemma 2.1 (counting). *For every integer $d \leq m$: $\#\{a \in A : d \mid a\} \leq \lfloor m/d \rfloor \leq \lfloor (x+m)/d \rfloor - \lfloor x/d \rfloor = \#\{b \in I : d \mid b\}$.*

Proof. The multiples of d in $\{1, \dots, m\}$ are $d, 2d, \dots, \lfloor m/d \rfloor d$, and the a_i are distinct elements of $\{2, \dots, m\}$. Any m consecutive integers contain $\lfloor (x+m)/d \rfloor - \lfloor x/d \rfloor \geq \lfloor m/d \rfloor$ multiples of d . $\square$

Lemma 2.2 (whole interval). *P divides $\prod_{b \in I} b$. Hence if $m \leq H$ then $B = I$ has $|B| = m \leq H$.*

Proof. $P \mid m!$ because the a_i are distinct elements of $\{2, \dots, m\}$, and $\prod_{b \in I} b = m! \binom{x+m}{m}$. $\square$

Atoms, large and small. Fix an integer budget $H \geq 1$ and assume $m > H$; put $T = m/H > 1$. For $a \in A$ and $p \mid a$ call $q_p(a) = p^{v_p(a)}$ an *atom* of a ; the atoms of a are pairwise coprime and multiply to a , and a has $\omega(a)$ atoms. An atom is *large* if $q > T$ and *small* if $q \leq T$.

Lemma 2.3 (large atoms, one element each). *Let $\mathcal{L}$ be the multiset of all large atoms of all $a \in A$ (atoms of different a are distinct members, so $r = |\mathcal{L}|$ counts with multiplicity). There is a set $U \subseteq I$ with $|U| \leq r$ such that for every prime p*

$$v_p\left(\prod_{u \in U} u\right) \geq \sum_{q \in \mathcal{L}, q \text{ a power of } p} v_p(q).$$

More precisely, one can assign to each $q \in \mathcal{L}$ an element $\beta(q) \in I$ with $q \mid \beta(q)$ such that atoms of the same prime receive distinct elements; then $U = \beta(\mathcal{L})$.

Proof. Fix a prime p and list the large atoms of p as $p^{e_1}, \dots, p^{e_s}$ with $e_1 \geq \dots \geq e_s$. Each $a \in A$ has at most one atom of p , so the first j atoms come from j distinct elements of A , all divisible by p^{e_j} ; by Lemma 2.1, I contains at least j multiples of p^{e_j} . Assign the atoms in this order, giving the j -th one a multiple of p^{e_j} in I different from the $j - 1$ elements already used for p . Elements used for different primes may coincide; this only decreases $|U|$. For a fixed p the assigned elements are distinct, so their p -adic valuations add up to at least $\sum_j e_j$. $\square$

Lemma 2.4 (assembly). *Suppose the small atoms of all $a \in A$ (a multiset; atoms of different a are distinct members) have been partitioned into bins, each bin being a set of small atoms whose product f satisfies $f \leq T$ (a bin may contain atoms of different a). Let s be the total number of bins and $D = r + s$. If $D \leq H$, then there is $B \subseteq I$ with $|B| \leq D$ and $P \mid \prod_{b \in B} b$.*

Proof. Take U from Lemma 2.3, $|U| \leq r$. Order the bins $f_1, \dots, f_s$ arbitrarily. Since $f_j \leq T = m/H$ and H is an integer, Lemma 2.1 gives at least $\lfloor m/f_j \rfloor \geq H$ multiples of f_j in I . When bin j is processed, the elements to avoid are those of U and the $j-1$ elements chosen for earlier bins, at most $r+s-1 = D-1 \leq H-1$ in number; so an unused multiple v_j of f_j exists. Put $B = U \cup \{v_1, \dots, v_s\}$. By construction $U \cap \{v_1, \dots, v_s\} = \emptyset$ and the v_j are pairwise distinct, so $|B| \leq r+s = D$ and (this is the only place where the valuations of different elements are added)

$$\prod_{b \in B} b = \prod_{u \in U} u \cdot \prod_j v_j,$$

whose p -adic valuation is at least (sum over large atoms of p) + $v_p(\prod_j f_j)$, which equals $v_p(P)$ because every atom of every a is either large or lies in exactly one bin. $\square$

Thus everything reduces to producing bins and a budget H with $D = r+s \leq H$. Theorems 1.1 and 1.2 do this in two different ways.

3 Proof of Theorem 1.1

Next-fit bins. For each $a \in A$ list its small atoms in increasing order of the prime and pack them by multiplicative next-fit: keep a current bin, initially empty (product 1); if multiplying the next atom keeps the product $\leq T$, add it; otherwise close the bin and start a new one containing that atom. Every small atom is $\leq T$, so every bin has product $\leq T$.

Lemma 3.1. *If $d_1, \dots, d_u$ are the bin products of one a in the order created, then $d_j d_{j+1} > T$ for $1 \leq j < u$.*

Proof. Bin j was closed because the next atom q did not fit: $d_j q > T$. That atom is the first element of bin $j+1$, so $d_{j+1} \geq q$. $\square$

Lemma 3.2 (budget). *Let $H, \kappa \geq 1$ be integers with*

$$(B1) \ H \geq n(2\kappa + 1), \quad (B2) \ H^{(\kappa+1)n} \leq 2^{\kappa H}. \quad (2)$$

Then the next-fit construction with threshold $T = m/H$ produces $D = r+s \leq H$ demands.

Proof. Case $m^\kappa \leq H^{\kappa+1}$. Binning never increases the count, so $D \leq W := \sum_{a \in A} \omega(a)$. Since every atom is at least 2, $2^{\omega(a)} \leq a$, hence $2^W \leq \prod_a a \leq m^n$ and $2^{\kappa W} \leq m^{\kappa n} \leq H^{(\kappa+1)n} \leq 2^{\kappa H}$ by (B2). So $W \leq H$.

Case $m^\kappa > H^{\kappa+1}$. Then $T^{\kappa+1} = m^{\kappa+1}/H^{\kappa+1} > m$. Fix $a \in A$ with ℓ large atoms and u bins $d_1, \dots, d_u$. The large atoms and the products $d_1 d_2, d_3 d_4, \dots$ of consecutive bin pairs are pairwise coprime divisors of a , each exceeding T (Lemma 3.1). If $\ell + \lfloor u/2 \rfloor \geq \kappa + 1$, the product of $\kappa + 1$ of them would exceed $T^{\kappa+1} > m \geq a$ while dividing a ; so $\ell + \lfloor u/2 \rfloor \leq \kappa$, and since $u \leq 2\lfloor u/2 \rfloor + 1$, $\ell + u \leq 2(\ell + \lfloor u/2 \rfloor) + 1 \leq 2\kappa + 1$. Summing over a , $D \leq n(2\kappa + 1) \leq H$ by (B1). $\square$

Lemma 3.3. *For $n \geq 2$ let $r = \Lambda(2n)$, $s = \Lambda(r)$, $C = r + s + 2$, $H = nC = F(n)$ and $\kappa = \lfloor (C-1)/2 \rfloor$. Then (B1) and (B2) hold.*

Proof. (B1): $2\kappa + 1 \leq C$. (B2), $n \geq 3$: then $2n \geq 6$, so $r \geq 3$ and $s \geq 2$; by definition $2n \leq 2^r$, i.e. $n \leq 2^{r-1}$, and $r \leq 2^s$; also $s+2 \leq 2^s$ for $s \geq 2$. Hence $C = r + s + 2 \leq 2^{s+1}$ and $H = nC \leq 2^{r-1} 2^{s+1} = 2^{C-2}$. Moreover $C \leq 2(\kappa + 1)$, i.e. $(C-2)(\kappa + 1) \leq \kappa C$, so $H^{(\kappa+1)n} \leq 2^{(C-2)(\kappa+1)n} \leq 2^{\kappa C n} = 2^{\kappa H}$. For $n = 2$: $r = 2$, $s = 1$, $C = 5$, $H = 10$, $\kappa = 2$, and $10^6 \leq 2^{20}$. $\square$

Proof of Theorem 1.1. Let $H = F(n)$. If $m \leq H$ use Lemma 2.2. Otherwise Lemmas 3.2 and 3.3 give $D \leq H$, and Lemma 2.4 gives B with $|B| \leq D \leq F(n)$. $\square$

4 Proof of Theorem 1.2

Here the bins are formed by *maximal merging*: start with each small atom (of any a) in its own bin and, as long as two bins have product $\leq T$, merge them. Each merge preserves the property “product $\leq T$ ” and decreases the number of bins, so the process terminates, and at termination any two bins have product $> T$.

Lemma 4.1. *With $W = \sum_a \omega(a)$ and $D = r + s$ as before,*

$$(i) \ D \leq W, \quad (ii) \ D \leq \frac{2n \ln m}{\ln(m/H)} + 1.$$

Proof. (i) is clear. (ii): any two bins have product $> T$, so at most one bin has product $\leq \sqrt{T}$, and that bin still has product $\geq 2 > 1$. Hence, if $s \geq 1$, $P = \prod_{\text{large}} q \cdot \prod_j f_j > T^r T^{(s-1)/2}$ strictly, and $P \leq m^n$; thus $r + (s-1)/2 < n \ln m / \ln T$ and $D = r + s < 2n \ln m / \ln T - r + 1 \leq 2n \ln m / \ln(m/H) + 1$. If $s = 0$, $P > T^r$ gives $D = r < n \ln m / \ln T$. $\square$

Lemma 4.2. *For real $X > 1$, $\sum_{p \leq X} 1/p \leq e \ln(1 + \ln X)$ (sum over primes).*

Proof. Put $\sigma = 1 + 1/\ln X$. For $p \leq X$, $p^{\sigma-1} \leq X^{1/\ln X} = e$, so $1/p \leq e p^{-\sigma}$. Using $-\ln(1-t) \geq t$ and the finite Euler product, $\sum_{p \leq X} p^{-\sigma} \leq \ln \prod_{p \leq X} (1 - p^{-\sigma})^{-1} \leq \ln \zeta(\sigma)$, and $\zeta(\sigma) \leq 1 + \int_1^\infty t^{-\sigma} dt = 1 + 1/(\sigma - 1) = 1 + \ln X$. $\square$

Lemma 4.3 (average number of prime factors). *For every real $t > 1$,*

$$\frac{1}{n} \sum_{i=1}^n \omega(a_i) \leq \frac{\ln(m/n) + (t-1) e \ln(1 + \ln m)}{\ln t}.$$

Proof. For every positive integer k , $t^{\omega(k)} = \prod_{p|k} (1 + (t-1)) = \sum_{d|k, d \text{ squarefree}} (t-1)^{\omega(d)}$. All terms are nonnegative because $t > 1$. Summing over $k \leq m$,

$$\sum_{k \leq m} t^{\omega(k)} = \sum_{\substack{d \leq m \\ d \text{ sqfree}}} (t-1)^{\omega(d)} \left\lfloor \frac{m}{d} \right\rfloor \leq m \prod_{p \leq m} \left(1 + \frac{t-1}{p}\right) \leq m \exp\left((t-1) \sum_{p \leq m} \frac{1}{p}\right) \leq m e^{(t-1)e \ln(1 + \ln m)}$$

by Lemma 4.2. The a_i are distinct integers in $[2, m]$, so $\sum_i t^{\omega(a_i)} \leq \sum_{k \leq m} t^{\omega(k)}$, and by the AM–GM inequality $t^{\frac{1}{n} \sum_i \omega(a_i)} \leq \frac{1}{n} \sum_i t^{\omega(a_i)}$. Taking logarithms gives the claim (note $m \geq n + 1$, so $\ln(m/n) > 0$). $\square$

Lemma 4.4 (explicit parameters). *Let $L = 81 + \ln n$, $\ell = \ln L$, $R = \sqrt{L/\ell}$ and $H = \lceil 48nR \rceil$. If $m > H$ then $D < H$.*

Proof. Elementary facts: $\ell > 4$ (as $L \geq 81 > e^4$); $\ell \leq L/4$ (the function $y/4 - \ln y$ is increasing and positive for $y \geq 81$); $\ln \ell \leq \ell/2$ (as $y/2 - \ln y > 0$ is increasing for $y \geq 4$); hence $R > 2$. Put $y = \ln m$, $z = \ln H$, $d = y - z > 0$, $c = z - \ln n$. Since $48nR > 1$, $H \leq 96nR$, so

$c \leq \ln 96 + \ln R < 7 + \ell/2$ (using $\ln R = (\ell - \ln \ell)/2 \leq \ell/2$) and $z < \ln n + 7 + \ell/2 < 2L$. Put $\Delta = 4\sqrt{L\ell} = 4R\ell \leq 2L$ (as $4\ell \leq L$).

Case $d \leq \Delta$. Then $y \leq 4L$. Apply Lemma 4.3 with $t = R$: $\ln R = (\ell - \ln \ell)/2 \geq \ell/4$ and $e \ln(1 + y) \leq e \ln(5L) < 3(\ell + 3) < 6\ell$. Since $\ln(m/n) = d + c$,

$$\frac{W}{n} \leq \frac{d + c + (R - 1) \cdot 6\ell}{\ell/4} < \frac{4R\ell + 7 + \ell/2 + 6R\ell}{\ell/4} = 40R + \frac{28}{\ell} + 2 < 45R,$$

so by Lemma 4.1(i), $D \leq W < 45nR < H$.

Case $d > \Delta$. By Lemma 4.1(ii), $D \leq 2n \frac{z+d}{d} + 1 < 2n + \frac{2n \cdot 2L}{4\sqrt{L\ell}} + 1 = 2n + nR + 1 < 3nR < H$, using $R > 2$. $\square$

Proof of Theorem 1.2, explicit part. Let $H = \lceil 48nR \rceil$. If $m \leq H$ use Lemma 2.2; otherwise Lemma 4.4 gives $D < H$ and Lemma 2.4 gives $|B| \leq D < H$. $\square$

Proof of Theorem 1.2, asymptotic part. Fix $\varepsilon > 0$, put $Q_n = \sqrt{\ln n / \ln \ln n}$ and $H = \lceil (2 + \varepsilon)nQ_n \rceil$; we show $D < H$ for all $m > H$ once n is large, which suffices as before. With $y = \ln m$, $z = \ln H$, $d = y - z$, $c = z - \ln n$ we have $z = \ln n + O(\ln \ln n)$, hence $\sqrt{z / \ln z} = (1 + o(1))Q_n$ and $c = O(\ln z)$. Put $\Delta = \sqrt{z \ln z}$. If $d > \Delta$, Lemma 4.1(ii) gives $D/n \leq 2 + 2z/d + 1/n \leq (2 + o(1))\sqrt{z / \ln z}$. If $d \leq \Delta$, then $y = (1 + o(1))z$; apply Lemma 4.3 with $t = \sqrt{z} / (\ln z)^2$, so $\ln t = (\frac{1}{2} + o(1)) \ln z$: the three terms of the numerator divided by $\ln t$ are at most $(2 + o(1))\sqrt{z / \ln z}$, $O(1)$ and $O(\sqrt{z} / (\ln z)^2)$ respectively, so $D/n \leq W/n \leq (2 + o(1))\sqrt{z / \ln z}$. In both cases $D \leq (2 + \varepsilon/2)nQ_n < H$ for all $n \geq n_0(\varepsilon)$, so $g(n) \leq \lceil (2 + \varepsilon)nQ_n \rceil$ for $n \geq n_0(\varepsilon)$; as $\varepsilon > 0$ was arbitrary, $\limsup_n g(n)/(nQ_n) \leq 2$. $\square$

Remark 4.5. No attempt was made to optimise the constant 48; the same argument with the case threshold and the prime-sum estimate tightened (e.g. an explicit Mertens bound in place of Lemma 4.2) gives a single-digit constant. Theorems 1.1 and 1.2 are proved by the same construction and differ only in how the number of bins is counted; Theorem 1.3 uses the same two ingredients (per-prime placement of large atoms, private multiples for small parts) with the budget $2n$ made possible by the length of the interval.

5 Long intervals: proof of Theorem 1.3

Assume $m = a_n \geq 8n^3$ and put $C = m/(2n)$. Then $C \geq m^{2/3}$ (equivalent to $m^{1/3} \geq 2n$), $C \geq 4n^2 \geq 2n$, and, since $m \geq 8n^3 \geq 4n^2$, also $C^2 = m^2/(4n^2) \geq m$. In the language of Section 2 we take the integer budget $H = 2n$ and threshold $T = C = m/H$ (note $m > H$).

Lemma 5.1 (two-bin packing). *Let $C \geq 1$ be real and let $q_1, \dots, q_r > 1$ be integers with $q_i \leq C$ and $P = q_1 \cdots q_r \leq C^{3/2}$. Then some sub-product s of the q_i (over a subset, possibly empty or everything) satisfies $P/C \leq s \leq C$; consequently the q_i can be partitioned into two classes with products $\leq C$.*

Proof. If $P \leq C$ take $s = P$. Otherwise $1 < P/C \leq \sqrt{C}$. If some $q_i > \sqrt{C}$, take $s = q_i$: then $s \leq C$ and $s > \sqrt{C} \geq P/C$. If all $q_i \leq \sqrt{C}$, multiply them in any order and stop at the first partial product $s \geq P/C$ (this happens at the latest at $s = P$); the previous partial product, possibly the empty product $1 < P/C$, was $< P/C$, so $s < (P/C)\sqrt{C} = P/\sqrt{C} \leq C$. In every case the complementary product is $P/s \leq C$. $\square$

Lemma 5.2 (split). *Every $a \in A$ has at most one atom exceeding C , and admits a factorisation $a = d_1 d_2$ into coprime parts such that either $d_1, d_2 \leq C$, or d_1 is an atom $> C$ and $d_2 < 2n \leq C$.*

Proof. Two atoms $> C$ are coprime, so they would give $a > C^2 \geq m$, impossible. If $q = p^e > C$ is an atom, put $d_1 = q$, $d_2 = a/q < m/C = 2n$. Otherwise all atoms of a are $\leq C$ and their product $a \leq m \leq C^{3/2}$, so Lemma 5.1 splits them into two coprime classes with products $\leq C$. $\square$

Proof of Theorem 1.3. Split every $a \in A$ by Lemma 5.2; parts equal to 1 impose no condition and may be assigned to any element, so we discard them. This gives at most $2n$ demands, each either a *large* atom $> C$ or a *small* integer $\leq C$. (Lemma 2.4 with $H = 2n$ already gives $|B| \leq 2n$; we repeat the argument because Conjecture 7.4 asks for the explicit assignment.) Assign the large demands as in Lemma 2.3: for each prime p , the large atoms of p (one per a) receive distinct multiples of the required powers of p ; different primes may share elements. Let U be the set of elements used, $|U| \leq$ number of large demands. Now assign the small demands one at a time: a small demand $d \leq C = m/(2n)$ has at least $\lfloor m/d \rfloor \geq 2n$ multiples in I by Lemma 2.1 ($2n$ is an integer), while the forbidden elements (U together with the elements already given to earlier small demands) number at most $2n - 1$, because there are at most $2n$ demands in all and the current one is not yet placed; so d receives a multiple of its own, outside U and different from all other small representatives. Let B be the set of all elements used, $|B| \leq 2n$. At an element of U the demands present are prime powers of distinct primes, each dividing the element; at every other element of B exactly one demand d is present and d divides the element. Hence the product of the demands at any $b \in B$ divides b , and since the demands of each a multiply to a , $P \mid \prod_{b \in B} b$. This is exactly an assignment as in Conjecture 7.4. Finally, if $a_n < 8n^3$ then Lemma 2.2 gives a solution with at most $a_n < 8n^3$ elements, so $g(n) \leq 8n^3$. $\square$

6 The k -split lemma: proof of Theorem 1.4

Lemma 6.1 (k -split). *Let $k \geq 1$ and let $H \leq m$ be positive integers. Let Q be a finite multiset of integers > 1 with product A such that $A^2 H^{k+1} \leq m^{k+1}$. Then Q can be partitioned into at most k pieces (none if $Q = \emptyset$), each of which is either a single element of Q or a bin: a nonempty $F \subseteq Q$ whose product f satisfies $fH \leq m$.*

Proof. Induction on k ; if $Q = \emptyset$ there is nothing to do, so let $Q \neq \emptyset$. For $k = 1$ the hypothesis reads $A^2 H^2 \leq m^2$, so $AH \leq m$ and Q itself is one bin. Let $k \geq 2$.

Case 1: some $q \in Q$ has $q^2 H > m$. Make q a singleton piece. The product $A' = A/q$ of the rest satisfies $A'^2 H^k = A^2 H^{k+1} / (q^2 H) < m^{k+1} / m = m^k$, so the induction hypothesis (with $k - 1$) splits the rest into at most $k - 1$ pieces.

Case 2: $q^2 H \leq m$ for all $q \in Q$. Build one bin greedily: start with $f = 1$ and, while Q is not exhausted and $f^2 H < m$, multiply the next element into f . The invariant $fH \leq m$ holds at the start ($H \leq m$) and is preserved, because before a step $f^2 H < m$ and $q^2 H \leq m$ give $(fqH)^2 < m^2$. The first element is always absorbed unless $H = m$, in which case the hypothesis forces $A = 1$, i.e. $Q = \emptyset$; so the bin is nonempty. If the process exhausts Q , all of Q is one bin. Otherwise it stops with $f^2 H \geq m$; then the product $A' = A/f$ of the remaining elements satisfies $A'^2 H^k = A^2 H^{k+1} / (f^2 H) \leq m^k$, and the induction hypothesis splits them into at most $k - 1$ further pieces. $\square$

Proof of Theorem 1.4. Put $H = kn$. If $m \leq H$, Lemma 2.2 gives $B = I$ with $|B| = m \leq kn$. Otherwise $m > H$ and $m^{k-1} \geq H^{k+1}$; for every $a \in A$, $a^2 H^{k+1} \leq m^2 \cdot m^{k-1} = m^{k+1}$. Apply

Lemma 6.1 to the atoms of each $a \in A$ (pairwise coprime, product a , all $\leq m$): each a splits into at most k pieces, hence at most $kn = H$ pieces in all. Call a singleton piece q *large* if $qH > m$; every other piece (a bin, or a singleton with $qH \leq m$) is a bin f with $fH \leq m$ in the sense of Section 2 with threshold $T = m/H$. The large pieces are atoms of distinct elements of A for each fixed prime (an element of A has one atom per prime), so Lemma 2.3 applies to them, and Lemma 2.4 (with r = number of large pieces, s = number of bins, $r + s \leq H$) gives $B \subseteq I$ with $|B| \leq H = kn$ and $P \mid \prod_{b \in B} b$; note that the p -adic valuation of a bin comes from the single p -atom of the element of A it was cut from. For the corollaries: $k = 2$ is $m \geq 8n^3$; $k = 3$ is $m^2 \geq 81n^4$; $k = 4$ is $m^3 \geq 1024n^5$; for $\delta > 0$ and $\delta(k-1) > 2$, $m \geq n^{1+\delta}$ gives $m^{k-1} \geq n^{(1+\delta)(k-1)} \geq k^{k+1}n^{k+1}$ as soon as $n^{\delta(k-1)-2} \geq k^{k+1}$. The last sentence of the theorem is a finite computation: for each exponent e and each n take the least $k \geq 2$ with $e(k-1) \ln n \geq (k+1) \ln(kn)$, or $F(n)/n$ from Theorem 1.1 if there is none; for $e = 3, 2, 7/4$ some $k \leq 10$ is admissible for all $n \geq 5$, $n \geq 38$, $n \geq 207$ respectively, and the maximum over n is attained at $n = 4, 36, 165$ with the values 7, 12, 15; the script is in the repository. $\square$

Remark 6.2. Lemma 6.1 is the sharp form of the same atom-binning that underlies Theorem 1.1 (which needs no hypothesis on a_n); what is new here is the exact threshold $\sqrt{m/H}$ and the explicit family of thresholds $a_n^{k-1} \geq (kn)^{k+1}$. The threshold is essentially sharp for the lemma: $k+1$ atoms each with $q^2H > m$ cannot share bins and violate the hypothesis. With $k = 2$ the lemma is Lemma 5.1 in a different normalisation, so Section 5 is the case $k = 2$; it was kept because it also exhibits the explicit assignment required by Conjecture 7.4. Choosing k as a function of m/n does not improve the global bounds of Theorems 1.1 and 1.2: for $m = ne^{\sqrt{\ln n \ln \ln n}}$ both routes give about $n\sqrt{\ln n / \ln \ln n}$.

7 Small values, obstructions and remarks

Proposition 7.1. $g(4) \geq 5$ and $g(5) \geq 6$.

Proof. Let $A = \{10, 12, 15, 20\}$ and $x = 50$, so $I = \{51, \dots, 70\}$ and $P = 2^5 \cdot 3^2 \cdot 5^3$. No element of I is divisible by 25, so a solution contains at least three of the multiples 55, 60, 65, 70 of 5; any set S of multiples of 5 in I has $v_2(\prod S) \leq 3$ and $v_3(\prod S) \leq 1$. Hence a further element with $v_2 \geq 2$ and $v_3 \geq 1$, i.e. a multiple of 12, is needed; the only multiple of 12 in I is 60, and using it still leaves $v_2 \leq 3 < 5$. So at least five elements are needed, and $\{55, 63, 64, 65, 70\}$ works. For $A = \{5, 10, 12, 15, 20\}$ and $x = 50$ we have $P = 2^5 \cdot 3^2 \cdot 5^4$, so all four multiples of 5 are forced, supplying $v_2 = 3$ and $v_3 = 1$; two more elements are then needed for 2^2 and 3 (no element of I other than 60 is divisible by 12), and $\{55, 60, 63, 64, 65, 70\}$ works. Both minima were also confirmed by an exhaustive dynamic programme over capped valuation vectors, which further shows that no other x gives a larger minimum for these two sets. $\square$

Remark 7.2 (why the obvious approaches fail). (a) Choosing a multiple of each a_i and multiplying does not work, because the chosen multiples may coincide and a shared element contributes its valuations once; in the Erdős–Surányi family every a_i has the same unique multiple u . (b) One cannot in general match the $a_i \leq m/2$ to *distinct* multiples inside an interval of length m : van Doorn, Li and Tang [4] showed that an interval of length $2 \max A$ guarantees only $\min(n, \lceil 2\sqrt{n} \rceil)$ disjoint pairs $(a, \text{multiple of } a)$, and this is optimal. (c) An induction that removes a_n , takes a solution B' for the rest and repairs it with two new elements fails: for $A' = \{2, 4, 8, 10, 12\}$, $I = \{1, \dots, 16\}$, $B' = \{8, 12, 15, 16\}$ and $a_6 = 16$, the missing factor 2^4 cannot be supplied

by two elements of $I \setminus B'$. Lemma 2.4 avoids all three pitfalls by giving every bin at least H candidates and letting different primes share an element only through the per-prime bookkeeping of Lemma 2.3.

Remark 7.3 (the gap to $2n$). Both proofs pay one interval element per large atom and per bin. In the balancing region of Lemma 4.1 an a_i may have about $\sqrt{\ln n / \ln \ln n}$ atoms of “medium” size, none of which can be merged, and this is exactly the loss in Theorem 1.2. A linear bound would require medium atoms of *different* a_i to share representatives. A natural sufficient statement is the following.

Conjecture 7.4 (split assignment). For every A and I as above one can choose for each $a \in A$ a factorisation $a = d_1(a)d_2(a)$ into coprime parts ($d_2 = 1$ allowed) and assign each of the at most $2n$ parts to an element $\beta(d) \in I$ so that for every $b \in I$ and every prime p , $\sum_{d: \beta(d)=b} v_p(d) \leq v_p(b)$.

The conjecture implies $g(n) \leq 2n$ (take $B = \beta(\text{parts})$), and Theorem 1.3 proves it whenever $a_n \geq 8n^3$. Since the capacity constraint couples two parts only if they share a prime, the assignment always exists when every $a \in A$ has at most two distinct prime factors (use the finest splitting and Lemma 2.3 prime by prime); a counterexample would need elements with three or more prime factors competing for composite multiples. An exact search found the assignment on the Erdős–Surányi instances with $\ell = 3, 4$, on the instances of Proposition 7.1, and on more than 10^6 random instances with $n \leq 6$ and $a_n \leq 210$; we found no counterexample.

8 A conditional linear bound

Theorem 1.3 settles the range $a_n \geq 8n^3$ with the conjectured value $2n$, so a bound $g(n) \leq Cn$ with an absolute constant C is a statement about $a_n < 8n^3$ only. In this section we reduce such a bound to a single inequality about additive functions on intervals of integers, we prove the inequality in two special cases, and we record three natural strengthenings of it that are *false*, with explicit certificates.

Throughout, a *weight* is a family $z = (z_p)_p$ indexed by the primes with $0 \leq z_p \leq 1$ and only finitely many $z_p \neq 0$, and $w_z(k) := \sum_p z_p v_p(k)$ is the associated completely additive function. We write $t^+ = \max(t, 0)$.

Conjecture 8.1 (hinge inequality). For every $m \geq 1$, every $x \geq 0$ and every weight z , with $I = \{x+1, \dots, x+m\}$,

$$\sum_{k=1}^m (w_z(k) - 2)^+ \leq \sum_{b \in I} (w_z(b) - 1)^+. \quad (3)$$

Theorem 8.2. *If Conjecture 8.1 holds, then $g(n) \leq 18n$ for all $n \geq 1$. More precisely, it suffices that (3) holds for $m = a_n$ and every weight supported on the primes dividing $\prod A$.*

The proof is a linear-programming argument. Fix $A = \{a_1 < \dots < a_n\}$ with $m = a_n < 8n^3$, an interval I of length m , the set $\mathcal{P}$ of primes dividing $P = \prod A$, and $R_p := \sum_{a \in A} v_p(a)$ for $p \in \mathcal{P}$. Consider the fractional cover problem

$$\tau^* := \min \left\{ \sum_{b \in I} y_b : \sum_{b \in I} v_p(b) y_b \geq R_p \ (p \in \mathcal{P}), \ 0 \leq y_b \leq 1 \right\}. \quad (4)$$

It is feasible, since $y \equiv 1$ works: $\prod A$ divides $m!$, which divides the product of any m consecutive integers, so $\sum_{b \in I} v_p(b) \geq R_p$.

Lemma 8.3 (duality). $\tau^* = \max_{z \geq 0} \left[\sum_{a \in A} w_z(a) - \sum_{b \in I} (w_z(b) - 1)^+ \right]$, where the maximum is over $z = (z_p)_{p \in \mathcal{P}}$ with $z_p \geq 0$, and it is attained at some z with all $z_p \leq 1$.

Proof. The dual of (4) has variables $z_p \geq 0$ (one per covering constraint) and $u_b \geq 0$ (one per upper bound $y_b \leq 1$) and reads $\max \sum_p z_p R_p - \sum_b u_b$ subject to $\sum_p z_p v_p(b) - u_b \leq 1$ for every $b \in I$. For fixed z the best choice is $u_b = (w_z(b) - 1)^+$, and $\sum_p z_p R_p = \sum_{a \in A} w_z(a)$; strong duality holds because (4) is feasible and bounded. For the last claim fix all $z_q, q \neq p$, and let $z_p \geq 1$ vary: every b with $p \mid b$ then has $w_z(b) \geq 1$, so the right derivative of the objective in z_p equals $R_p - \sum_{b \in I} v_p(b) \leq 0$. Hence the objective does not increase on $z_p \geq 1$. $\square$

Lemma 8.4 (rounding). *There is a set $B \subseteq I$ with $\prod A \mid \prod B$ and $|B| \leq \tau^* + |\mathcal{P}|$.*

Proof. Let y^* be an optimal vertex of the polytope in (4). A vertex of a polytope in $\mathbb{R}^m$ is determined by m linearly independent tight constraints, at most $|\mathcal{P}|$ of which are covering constraints, so at most $|\mathcal{P}|$ coordinates of y^* lie strictly between 0 and 1. Put $B = \{b : y_b^* > 0\}$. Then $\sum_{b \in B} v_p(b) \geq \sum_b v_p(b) y_b^* \geq R_p$ for every p , i.e. $\prod A \mid \prod B$, and $|B| \leq \#\{b : y_b^* = 1\} + |\mathcal{P}| \leq \tau^* + |\mathcal{P}|$. $\square$

Lemma 8.5 (few primes). *If $m < 8n^3$ then $|\mathcal{P}| < 16n$.*

Proof. The i -th prime is at least $i+1$, so if $|\mathcal{P}| \geq 16n$ then $\prod_{p \in \mathcal{P}} p \geq (16n+1)! > (16n/e)^{16n}$, while $\prod_{p \in \mathcal{P}} p \leq \prod A \leq m^n < (8n^3)^n$. Taking logarithms, $16 \ln(16n/e) = 28.3 \dots + 16 \ln n > \ln 8 + 3 \ln n$ for every $n \geq 1$, a contradiction. $\square$

Proof of Theorem 8.2. If $a_n \geq 8n^3$, Theorem 1.3 gives $2n \leq 18n$ elements. Otherwise let z be a maximiser in Lemma 8.3 with $z_p \leq 1$, supported on $\mathcal{P}$. Since $w_z(a) \leq 2 + (w_z(a) - 2)^+$ and $A \subseteq [1, m]$,

$$\sum_{a \in A} w_z(a) \leq 2n + \sum_{k=1}^m (w_z(k) - 2)^+ \leq 2n + \sum_{b \in I} (w_z(b) - 1)^+$$

by (3), so $\tau^* \leq 2n$. Lemmas 8.4 and 8.5 give $|B| \leq 2n + 16n - 1 < 18n$. $\square$

Equivalent forms and a per-prime-power sufficient condition

Subtracting $\sum w_z$ over both ranges, (3) is equivalent to

$$\sum_{b \in I} \min(w_z(b), 1) \leq \sum_{k=1}^m \min(w_z(k), 2) + \Delta, \quad \Delta := \sum_{b \in I} w_z(b) - \sum_{k=1}^m w_z(k) = \sum_q z_q \delta_q \geq 0, \quad (5)$$

where q runs over prime powers, $z_{p^j} := z_p$, and $\delta_q := \#\{b \in I : q \mid b\} - \lfloor m/q \rfloor \in \{0, 1\}$. Writing $\min(w, 1) = w \min(1, 1/w)$ and $\min(w, 2) = w \min(1, 2/w)$ and expanding $w_z(n) = \sum_{q \mid n} z_q$, both sides of (5) become sums over prime powers, and (3) follows if for every prime power q with $z_q > 0$

$$\sum_{b \in I, q \mid b} \min\left(1, \frac{1}{w_z(b)}\right) \leq \delta_q + \sum_{k \leq m, q \mid k} \min\left(1, \frac{2}{w_z(k)}\right). \quad (6)$$

Substituting $b = qi'$, $k = qi$ and $w_z(qi) = a + w_z(i)$ with $a := w_z(q)$, the multiples of q in I correspond to an interval J of $L := \lfloor m/q \rfloor$ or $L + 1$ consecutive integers, and (6) follows from

$$\sum_{i \in J} f_a(w_z(i)) \leq \sum_{i=1}^L g_a(w_z(i)), \quad f_a(u) := \min\left(1, \frac{1}{a+u}\right), \quad g_a(u) := \min\left(1, \frac{2}{a+u}\right), \quad (7)$$

for intervals J of length L (the possible extra element of J is paid by δ_q , as $f_a \leq 1$).

Proposition 8.6. *Inequality (7) holds for every interval J of length L and every $a \geq 0$ in each of the following cases: (i) z is supported on a single prime; (ii) $a \geq 2$ and $aL \geq w_z(L!)$, in particular whenever $a \geq \max\{2, \sum_{p \leq L} z_p/(p-1)\}$.*

Proof. (i) Let z be supported on p and $\phi(j) := f_a(z_p j)$, a nonincreasing function of j , with increments $D_j := \phi(j-1) - \phi(j) \geq 0$. For any set S of L integers, $\sum_{i \in S} \phi(v_p(i)) = L\phi(0) - \sum_{j \geq 1} D_j \#\{i \in S : p^j \mid i\}$. Since an interval of length L contains at least $\lfloor L/p^j \rfloor = \#\{i \leq L : p^j \mid i\}$ multiples of p^j , the sum over J is at most the sum over $[1, L]$, and $f_a \leq g_a$ pointwise. (ii) For $a \geq 2$ the minima are inactive; the left side is at most L/a , and by the Cauchy–Schwarz inequality the right side is $2 \sum_{i \leq L} (a + w_z(i))^{-1} \geq 2L^2/(aL + w_z(L!)) \geq L/a$. Finally $w_z(L!) = \sum_p z_p v_p(L!) \leq L \sum_{p \leq L} z_p/(p-1)$ by Legendre’s formula. $\square$

Three strengthenings that fail

Each of the following statements would imply (3) or (7) by a short argument, and each is false. The common mechanism is that an interval of length L can contain more integers free of all primes $\leq L$ than $\pi(L) + 1$: the pattern of such integers is an admissible set, and by the Chinese remainder theorem every admissible pattern occurs, so their maximal number is the Hensley–Richards function $\rho^*(L)$, which exceeds $\pi(L)$ for $L \geq 3159$ and by an unbounded amount as $L \rightarrow \infty$ [5, 6]. Explicit windows were built by a centred sieve (residue class 0 modulo every $p \leq \sqrt{L}$, then the least populated class modulo each larger prime) and the Chinese remainder theorem; the starting points x and verification scripts are in the repository.

Proposition 8.7. (a) (weight-free transport) *There need not exist an injection $\varphi : [1, m] \rightarrow I$ such that for every k some prime $p \mid k$ satisfies $(k/p) \mid \varphi(k)$. For $m = 20000$ there is an x (with 4298 digits) for which I contains 2270 integers free of all primes $\leq m/2$; every k related to such an integer is 1 or a prime, so Hall’s condition fails because $2270 > \pi(m) + 1 = 2263$. (Such an injection would give (3) at once, since $w_z(\varphi(k)) \geq w_z(k) - z_p \geq w_z(k) - 1$.)*

(b) (single-threshold level sets) *The inequality $\#\{i \leq L : w_z(i) \geq 2\tau\} \leq \#\{i \in J : w_z(i) \geq \tau\}$ fails for $L = 10^5$, $z_p = 1$ for all $p \leq L/2$, at $\tau = 1$ (by 142) and at $\tau = 1/2$ (by 5275), although (7) holds on the same window with room to spare. (Integrating it over τ would give (7).)*

(c) (Laplace form) *The inequality $s \sum_{i \in J} s^{w_z(i)} \leq \sum_{i \leq L} s^{w_z(i)}$ for all $s \in (0, 1]$, which would give (7) for all $a \geq 2$ by integrating against $s^{a-1} ds$, fails for $L = 10^6$, $z_p = 1$ for all $p \leq L$, and $s \in [0.005, 0.015]$: the window (with x of 433636 digits) contains 80436 integers free of all primes $\leq L$, against $\pi(L) + 1 = 78499$, and at $s = 0.005$ the two sides are 405.03 and 398.77.*

Remark 8.8 (evidence and the remaining programme). Conjecture 8.1 was tested on 2.1×10^6 random and adversarial instances with $m \leq 600$ (0/1, fractional and $p^{-\theta}$ weights; $x = 0$, random x , $x \equiv -1$ modulo products of weighted primes, windows centred at highly divisible numbers) and, with nine weight choices, on the dense windows of Proposition 8.7 of lengths $2 \cdot 10^4$ and 10^5 , where the slack is at least $0.18m$; (7) was tested on 3.5×10^5 instances over all prime powers. Equality in (3) occurs only for $x = 0$ and single primes. For 0/1 weights $z = \mathbf{1}_{\mathcal{Q}}$ the inequality is equivalent to $B_0 - A_0 \leq \Delta + B_{\geq 2}$, where B_0 and A_0 count the integers in $[1, m]$ and in I

free of the primes of $\mathcal{Q}$, and $B_{\geq 2}$ counts the $k \leq m$ with at least two prime factors from $\mathcal{Q}$ (with multiplicity); by inclusion–exclusion $B_0 - A_0$ is an alternating sum of the surpluses δ_d over squarefree products d of primes of $\mathcal{Q}$. In the layer-cake form of (7) the integrand is positive only where the count on $[1, L]$ is stuck at a low level, and its positive part is bounded by the number of integers in a window free of the weighted primes, i.e. by an explicit large sieve bound, while the negative part is a counting quantity on $[1, L]$. Making this quantitative for all weights is the remaining step; the failures in Proposition 8.7 show that it cannot be replaced by a weight-free or single-threshold argument.

9 Counting certificates and the 0/1 case

All three failures in Proposition 8.7 have a common feature: they use only the lower bound $N_I(d) \geq \lfloor m/d \rfloor$ on the number $N_I(d) := \#\{b \in I : d \mid b\}$ of multiples of d in the interval. The upper bound $N_I(d) \leq \lfloor m/d \rfloor + 1$, which is exactly what the dense windows violate in spirit, was never used. Using both bounds, linear programming duality turns the hinge inequality into a finite question, and this question has a positive answer in every case we could compute; for 0/1 weights it also has an explicit answer with a slightly larger threshold.

The certificate principle

Proposition 9.1 (counting certificates). *Let z be a weight and let (c_d) be real numbers, finitely many nonzero, indexed by positive integers d , such that*

$$\sum_{d|n} c_d \leq (w_z(n) - 1)^+ \quad \text{for every positive integer } n. \quad (8)$$

Put $V(c) := \sum_{c_d > 0} c_d \lfloor m/d \rfloor - \sum_{c_d < 0} |c_d| (\lfloor m/d \rfloor + 1)$. Then for every $x \geq 0$, $\sum_{b \in I} (w_z(b) - 1)^+ \geq V(c)$. In particular (3) holds for (m, z) and every x as soon as $V(c) \geq W(m, z) := \sum_{k \leq m} (w_z(k) - 2)^+$ for some c satisfying (8).

Proof. Summing (8) over $b \in I$ gives $\sum_{b \in I} (w_z(b) - 1)^+ \geq \sum_d c_d N_I(d)$, and $\lfloor m/d \rfloor \leq N_I(d) \leq \lfloor m/d \rfloor + 1$ for every d because I consists of m consecutive integers. $\square$

Corollary 9.2 (the trivial regime). *For every weight z , every $m \geq 1$ and every $x \geq 0$, $\sum_{b \in I} (w_z(b) - 1)^+ \geq \sum_{k \leq m} w_z(k) - m$. Hence the hinge inequality with threshold $c \geq 1$ holds for (m, z) and every x whenever $\sum_{k \leq m, w_z(k) < c} (c - w_z(k)) \leq (c - 1)m$; for 0/1 weights, (3) holds whenever $\#\{k \leq m : w_z(k) = 0\} \leq \#\{k \leq m : w_z(k) \geq 2\}$, in particular for P the set of all primes up to m and every m .*

Proof. This is Proposition 9.1 for the affine certificate $c_1 = -1$, $c_{p^j} = z_p$ ($j \geq 1$), which satisfies (8) with $\sum_{d|n} c_d = w_z(n) - 1$; directly, $\sum_I (w_z - 1)^+ \geq \sum_I (w_z - 1) = \sum_{p,j} z_p N_I(p^j) - m \geq \sum_{k \leq m} w_z(k) - m$. Since $\sum_{k \leq m} (w_z(k) - c)^+ = \sum_{k \leq m} w_z(k) - cm + \sum_{w_z(k) < c} (c - w_z(k))$, the stated condition is exactly $\sum_K (w_z - c)^+ \leq \sum_K w_z - m$. For 0/1 weights and $c = 2$ the condition reads $2\#\{w_z = 0\} + \#\{w_z = 1\} \leq m$. For the full set of primes $\#\{w_z = 0\} = 1$ and $\#\{w_z \geq 2\} \geq 1$ for $m \geq 4$, while for $m \leq 3$ the left side of (3) vanishes. $\square$

So the hinge inequality is nontrivial only when integers of weight below the threshold dominate $[1, m]$: for 0/1 weights, prime sets without small primes (for P the primes in $[53, 653]$ and $m = 10^8$

one has $\#\{\omega_P = 0\} = 61,911,251$ against $\#\{\omega_P \geq 2\} = 7,944,229$, and the trivial bound is negative); for fractional weights, weights spread over primes of many sizes with $\sum_{k \leq m} w_z(k)$ of order m . This is also the regime of the dense windows of Proposition 8.7.

The best certificate is the optimum of a linear programme: the dual of minimising $\sum_t n_t(w_z(t) - 1)^+$ over nonnegative real “multisets” (n_t) of valuation patterns t subject to $\lfloor m/d \rfloor \leq \sum_{t: d|t} n_t \leq \lfloor m/d \rfloor + 1$ for all d . We solved this programme (for general weights with the exponents capped at $\lfloor \log_p m \rfloor + 1$, which is legitimate because a certificate supported on capped d satisfies (8) for all n ; for 0/1 weights with c supported on pairs and triples of primes, adding the constraints (8) by separation) and found $V \geq W$ in every instance: for all weights $z \in \{1, 0.7, 0.4\}^P$ and mixed weights with P the primes up to 17 and $m \leq 200$ (e.g. $V = 114$ against $W = 64$ for $m = 100$, $z \equiv 1$), and for 0/1 weights with P the set of all primes up to m for $m \in \{300, 500, 1000\}$ ($V = 223.5, 407.4, 907.0$ against the distinct-prime targets $W_\omega = 59, 124, 321$ defined below; these are the values of the programme restricted to pairs and triples, and the affine certificate of Corollary 9.2 alone gives 277, 507, 1125) as well as for various subsets of the primes at $m = 400$. The optimal certificates found (they are not unique) are small: $c_{p^j} = 1$ for prime powers with $j \geq 2$, $c_{pq} = 1$ for pairs containing 2, 3 or 5, and $c_{pqr} = -1$ for a few triples (no coefficient at $d = 1$ or at single primes). The scripts are in the repository.

An explicit family for 0/1 weights

For 0/1 weights $z = \mathbf{1}_P$ write $\omega_P(n)$ for the number of primes of P dividing n , $\Omega_P(n) = \sum_{p \in P} v_p(n)$ and $E_P := \Omega_P - \omega_P = \sum_{p \in P} \sum_{j \geq 2} [p^j \mid n]$. Since $(\Omega_P - 1)^+ = E_P + (\omega_P - 1)^+$, $(\Omega_P - c)^+ \leq E_P + (\omega_P - c)^+$ for every $c \geq 0$, and $\sum_{b \in I} E_P(b) \geq \sum_{k \leq m} E_P(k)$ (count the multiples of each p^j), it suffices to certify the distinct-prime inequality $\sum_{k \leq m} (\omega_P(k) - 2)^+ \leq \sum_{b \in I} (\omega_P(b) - 1)^+$, whose left side we denote $W_\omega(m)$ (for $z = \mathbf{1}_P$ one has $\bar{W}(m, z) \geq W_\omega(m)$, and the certificates below are for the distinct-prime version).

Lemma 9.3. *Let R be the set of the r smallest primes of P . Put $c_d := (-1)^{|A|}$ for $d = \prod A$ with $A \subseteq R$, $|A| \geq 2$, and $c_d := (-1)^{|A|+1}$ for $d = q \prod A$ with $q \in P \setminus R$ and $\emptyset \neq A \subseteq R$; all other $c_d = 0$. Then (8) holds with $(\omega_P(n) - 1)^+$ on the right, and it keeps holding if all coefficients belonging to any fixed $q \in P \setminus R$ are set to 0. For $r = 2$, $R = \{2, 3\}$ and the coefficients of the primes $q > m/2$ removed, the value is*

$$V_2(m) := \left\lfloor \frac{m}{6} \right\rfloor + \sum_{\substack{q \in P \\ 5 \leq q \leq m/2}} \left(\left\lfloor \frac{m}{2q} \right\rfloor + \left\lfloor \frac{m}{3q} \right\rfloor - \left\lfloor \frac{m}{6q} \right\rfloor - 1 \right). \quad (9)$$

Proof. Let n have prime set $T = A_0 \cup Q_0$ with $A_0 \subseteq R$, $Q_0 \subseteq P \setminus R$. Then $\sum_{d|n} c_d = \sum_{A \subseteq A_0, |A| \geq 2} (-1)^{|A|} + \sum_{q \in Q_0} \sum_{\emptyset \neq A \subseteq A_0} (-1)^{|A|+1} = (|A_0| - 1)^+ + |Q_0| [A_0 \neq \emptyset] \leq (|T| - 1)^+$, using $\sum_{A \subseteq A_0} (-1)^{|A|} = [A_0 = \emptyset]$. Removing the terms of one q removes a nonnegative contribution. The value follows from the definition of V . $\square$

With $\mathcal{A}(x) := \sum_{p \leq x} \lfloor x/p \rfloor = \sum_{n \leq x} \omega(n)$ and P the set of all primes up to m one has the exact identities $W_\omega(m) = \mathcal{A}(m) - 2m + 2 + \#\{\text{prime powers} \leq m\}$ and $V_2(m) = \mathcal{A}(m/2) + \mathcal{A}(m/3) - \mathcal{A}(m/6) - \lfloor m/4 \rfloor - \lfloor m/6 \rfloor - \lfloor m/9 \rfloor + \lfloor m/12 \rfloor + \lfloor m/18 \rfloor - \pi(m/2) + 2$, so $V_2(m) \geq W_\omega(m)$ was checked exactly for every $m \leq 10^7$ (at $m = 10^3, \dots, 10^7$ the ratio V_2/W_ω is 2.72, 2.00, 1.71, 1.55, 1.44). Since $\mathcal{A}(x) = x \ln \ln x + Mx + O(x/\ln x)$ [7, Thm. 430], $V_2(m) - W_\omega(m) = -\frac{1}{3}m \ln \ln m + (\frac{29}{18} - \frac{M}{3})m + o(m)$, so this single certificate proves the 0/1 case of (3) for the full set of primes for every

$m \leq 10^7$ (checked exactly for each m) and fails for all sufficiently large m (the leading-order crossing is at $\ln \ln m = \frac{29}{6} - M \approx 4.57$, i.e. $m \approx 10^{42}$). For the full set of primes this is of course weaker than Corollary 9.2, which settles that case for every m ; the point of the family is that it does not need small primes in P , where the affine certificate fails. The number of anchors must depend on P : for P the primes in $[53, 653]$ and $m = 10^8$ the two-anchor certificate gives $1,595,615 < W_\omega = 1,602,217$, while three anchors give $2,266,053$.

The 0/1 case with a larger threshold

Theorem 9.4. *Let P be a finite set of primes, $m \geq 3$, $x \geq 0$, and $c^*(m) := 5 + \max(2, \lceil 2e^2 \ln(1 + \frac{\ln m}{6}) \rceil)$. Then*

$$\sum_{n \leq m} (\Omega_P(n) - c^*(m))^+ \leq \sum_{b \in I} (\Omega_P(b) - 1)^+, \quad (10)$$

and $c^*(m) \leq 20 \ln \ln m$ for $m \geq 30$.

The proof rests on an elementary sieve lemma which, in the language of Proposition 9.1, is a certificate with positive coefficients on single primes and negative coefficients on pairs.

Lemma 9.5. *For $y \geq 2$, $\sum_{p \leq y} 1/p \leq e \ln(1 + \ln y)$.*

Proof. With $\sigma = 1 + 1/\ln y$ one has $p^{\sigma-1} \leq e$ for $p \leq y$, so $1/p \leq e p^{-\sigma}$; and $\sum_p p^{-\sigma} \leq \ln \zeta(\sigma) \leq \ln(1 + 1/(\sigma - 1)) = \ln(1 + \ln y)$. $\square$

Lemma 9.6 (sieve lemma). *Let $L \geq 64$, let Q be a finite set of primes, let J be any set of L consecutive integers and $s(L) := \max(2, \lceil 2e^2 \ln(1 + \frac{\ln L}{6}) \rceil)$. Then*

$$\#\{n \leq L : \omega_Q(n) \geq s(L) + 5\} \leq \#\{n \in J : \omega_Q(n) \geq 1\}.$$

Proof. Put $y = L^{1/6}$, $R = \{q \in Q : q \leq y\}$, $H = \sum_{q \in R} 1/q$ and $s = s(L)$. An integer $n \leq L$ has at most five prime factors exceeding y , so $\omega_Q(n) \geq s + 5$ forces $\omega_R(n) \geq s$, and $N_s := \#\{n \leq L : \omega_R(n) \geq s\} \leq \sum_{S \subseteq R, |S|=s} \lfloor L / \prod S \rfloor \leq L H^s / s!$ by the bound $s! e_s \leq H^s$ on elementary symmetric functions. By Lemma 9.5, $H \leq e \ln(1 + \frac{\ln L}{6})$, so $s \geq 2eH$ and, with $s! \geq (s/e)^s$, $N_s \leq L(eH/s)^s \leq L 2^{-s} \leq L/4$.

For the right-hand side we use that any set J of L consecutive integers satisfies $L/d - 1 < N_J(d) < L/d + 1$. If $H \geq 1/2$, add primes of R one at a time until the reciprocal sum first reaches $1/2$; the resulting R' has $1/2 \leq H' \leq 1$ (the last prime added contributes at most $1/2$) and $t' = |R'| \leq y$. By the two-term Bonferroni inequality, $\#\{n \in J : \omega_Q(n) \geq 1\} \geq \#\{n \in J : \omega_{R'}(n) \geq 1\} \geq \sum_{q \in R'} N_J(q) - \sum_{q < q'} N_J(qq') \geq L(H' - H'^2/2) - t'(t' + 1)/2 \geq \frac{3}{8}L - \frac{3}{4}y^2 \geq \frac{3}{8}L - \frac{1}{8}L = \frac{1}{4}L \geq N_s$, using $y(y + 1)/2 \leq \frac{3}{4}y^2$ for $y \geq 2$ and $\frac{3}{4}L^{1/3} \leq L/8$ for $L \geq 64$. If $0 < H < 1/2$, the same estimate with $R' = R$ and $H - H^2/2 \geq \frac{3}{4}H$ gives $\#\{n \in J : \omega_R(n) \geq 1\} \geq \frac{3}{4}LH - \frac{3}{4}y^2 \geq \frac{1}{4}LH$ (because $H \geq 1/y$ and $\frac{3}{4}y^2 \leq L/(2y)$ for $L \geq 64$), while $N_s \leq LH^2/2 \leq LH/4$. If $H = 0$ then $N_s = 0$. $\square$

Proof of Theorem 9.4. Order P increasingly and let $\omega_{<P}(n)$ count the primes of P below p dividing n . For every integer $c \geq 0$,

$$(\omega_P(n) - c)^+ = \sum_{p \in P, p|n} [\omega_{<P}(n) \geq c], \quad (11)$$

since the right side counts all but the c smallest primes of P dividing n . Fix $p \in P$ and put $L = \lfloor m/p \rfloor$. The multiples $n = pi \leq m$ of p have $\omega_{<p}(pi) = \omega_{<p}(i)$ with $i \leq L$, and the multiples $b = pi$ of p in I have i ranging over a set J_p of L or $L + 1$ consecutive integers. If $L \geq 64$, Lemma 9.6 applied to $Q = \{q \in P : q < p\}$ and to L consecutive elements of J_p gives $\#\{i \leq L : \omega_{<p}(i) \geq c^*(m)\} \leq \#\{i \in J_p : \omega_{<p}(i) \geq 1\}$, because $c^*(m) \geq s(L) + 5$. If $L < 64$ the left side is 0, as $\omega_{<p}(i) \leq 3 < 7 \leq c^*(m)$ for $i \leq 63$. Summing over p and using (11) with $c = c^*(m)$ on the left and $c = 1$ on the right proves (10) with ω_P in place of Ω_P ; the peeling identities stated before Lemma 9.3 then give (10). Finally $G(t) := 20 \ln t - 7 - 2e^2 \ln(1 + t/6)$ is increasing for $t > 0$ (as $2e^2 < 20$) and $G(\ln 30) > 0$, which gives $c^*(m) \leq 20 \ln \ln m$ for $m \geq 30$. $\square$

Remark 9.7 (fractional weights). Theorem 9.4 does not imply (3) for fractional weights, and integrating it over weight thresholds cannot (Proposition 8.7 and the identity $\int_0^1 (\#\{p \mid n : z_p > t\} - 1)^+ dt = \sum_{p \mid n} z_p - \max_{p \mid n} z_p$). Writing $w_z = E + S$ with $S(n) = \sum_p \min(z_p v_p(n), 1)$ and $E = \sum_p (z_p v_p(n) - 1)^+$, one has $(w_z - 1)^+ = E + (S - 1)^+$ (if $S < 1$ then every $z_p v_p < 1$, so $E = 0$), $(w_z - c)^+ \leq E + (S - c)^+$, and $\sum_I E \geq \sum_K E$ because $(z_p v_p(n) - 1)^+ = \sum_{j \geq 1} \Delta_{p,j} [p^j \mid n]$ with nonnegative increments $\Delta_{p,j} = (z_p j - 1)^+ - (z_p(j-1) - 1)^+$, so (3) for all weights follows from its analogue for the bounded function S ; for rational weights $z_p = a_p/N$ this is the integer statement $\sum_{k \leq m} (R(k) - 2N)^+ \leq \sum_{b \in I} (R(b) - N)^+$ with $R(n) = \sum_p \min(a_p v_p(n), N)$, of which the 0/1 case ($a_p \in \{0, N\}$) is the one treated above. The linear programme of Proposition 9.1 certifies the fractional inequality in every instance we solved; an explicit certificate for all m and all weights would complete the proof of $g(n) \leq 18n$.

10 The fractional case and an $O(n \ln \ln n)$ bound

Theorem 9.4 used only 0/1 weights. In this section the same method, combined with a random-thinning device, handles arbitrary weights, at the price of a larger threshold; through Lemma 8.3 this gives the first bound for $g(n)$ that is linear up to a $\ln \ln n$ factor.

Theorem 10.1. *For every weight z , every $m \geq 16$ and every $x \geq 0$,*

$$\sum_{k \leq m} (w_z(k) - c_m)^+ \leq \sum_{b \in I} (w_z(b) - 1)^+, \quad c_m := 1 + 149e \ln(1 + \ln m), \quad (12)$$

and $c_m < 675 \ln \ln m$. For $2 \leq m \leq 15$ the inequality holds with $c_m = 4$.

Corollary 10.2. $g(n) \leq (17 + 149e \ln(1 + \ln(8n^3)))n$ for all $n \geq 1$; in particular $g(n) = O(n \ln \ln n)$.

Proof. If $a_n \geq 8n^3$, Theorem 1.3 gives $2n$. Otherwise $m = a_n < 8n^3$ and, by Lemma 8.3 with a maximiser z , $\sum_{a \in A} w_z(a) \leq c_m n + \sum_{k \leq m} (w_z(k) - c_m)^+ \leq c_m n + \sum_{b \in I} (w_z(b) - 1)^+$, so $\tau^* \leq c_m n$ and Lemmas 8.4, 8.5 give $|B| < (c_m + 16)n$, with $c_m = 4$ when $m \leq 15$; in all cases $c_m \leq c_{8n^3}$ since c_m is increasing for $m \geq 16$ and $4 < c_{16}$. $\square$

Throughout this section $\lambda(t) := \ln(1 + \ln t)$, $L \geq 16$ is an integer, J is a set of L consecutive positive integers, $R := \lfloor \log_2 L \rfloor$, $H := e\lambda(L)$ and $L_0 := \ln(H + 2)$; note $H > 3$ and $L_0 > 1$ for $L \geq 16$.

Two-term Bonferroni at level k

Lemma 10.3. *For integers $r \geq 0$ and $k \geq 1$, $[r \geq k] \geq \binom{r}{k} - k \binom{r}{k+1}$.*

Proof. For $r < k$ both sides vanish; for $r \in \{k, k+1\}$ the right side is 1; for $r \geq k+2$, $\binom{r}{k} - k \binom{r}{k+1} = \binom{r}{k} (1 - \frac{k(r-k)}{k+1}) \leq 0$. $\square$

A sieve lemma with an arbitrary gain

Let D be a finite family of pairwise coprime integers $d \geq 2$ and $\nu_D(n) := \#\{d \in D : d \mid n\}$. For $1 \leq k \leq R-1$ put

$$\rho(k) := \left\lceil 16H + \frac{16kL_0}{\sqrt{\ln(2+k/H)}} \right\rceil, \quad C(k) := 2k + \rho(k). \quad (13)$$

Lemma 10.4 (gain- G sieve). *For every real $1 \leq G \leq 2H$, every $1 \leq k \leq R-1$ and every J ,*

$$G \cdot \#\{n \leq L : \nu_D(n) \geq C(k)\} \leq \#\{n \in J : \nu_D(n) \geq k\}.$$

Proof. Put $y := L^{1/(k+1)} \geq 2$, $D_0 := \{d \in D : d \leq y\}$ and $H_0 := \sum_{d \in D_0} 1/d$. Choosing a prime divisor of each $d \in D_0$ gives distinct primes $\leq y$, so $H_0 \leq \sum_{p \leq y} 1/p \leq e\lambda(y) \leq H$ by Lemma 9.5. Let e_j denote the elementary symmetric functions of the numbers $1/d$, $d \in D_0$; then $(k+1)e_{k+1} \leq H_0 e_k$ and $e_{k+r} \leq e_k e_r / \binom{k+r}{k} \leq e_k H_0^r / r!$.

Lower bound on J . Retain each element of D_0 independently with probability $\theta := 1/(8H)$, obtaining a random subfamily E . For every E , Lemma 10.3 summed over $n \in J$ gives $\#\{n \in J : \nu_E(n) \geq k\} \geq M_k(E) - kM_{k+1}(E)$, where $M_j(E) := \sum_{F \subseteq E, |F|=j} N_J(\prod F)$; since $\nu_D \geq \nu_E$ and $\mathbb{E} M_j(E) = \theta^j M_j(D_0)$, taking expectations yields $\#\{n \in J : \nu_D(n) \geq k\} \geq \theta^k (M_k(D_0) - k\theta M_{k+1}(D_0))$. A product of k elements of D_0 is at most $y^k = L/y \leq L/2$, so $N_J(a) \geq L/a - 1 \geq L/(2a)$ and $M_k(D_0) \geq \frac{L}{2} e_k$; a product of $k+1$ elements is at most $y^{k+1} = L$, so $N_J(a) \leq L/a + 1 \leq 2L/a$ and $M_{k+1}(D_0) \leq 2Le_{k+1}$. Hence $k\theta M_{k+1}(D_0) \leq 2\theta LH_0 e_k \leq \frac{L}{4} e_k$ and

$$\#\{n \in J : \nu_D(n) \geq k\} \geq \frac{Le_k}{4(8H)^k}. \quad (14)$$

Upper bound on $[1, L]$. An integer $n \leq L$ is divisible by at most k elements of $D \setminus D_0$, since $k+1$ of them would have product $> y^{k+1} = L$. Hence $\nu_D(n) \geq C(k)$ implies $\nu_{D_0}(n) \geq k + \rho(k)$, and with $r = \rho(k)$,

$$\#\{n \leq L : \nu_D(n) \geq C(k)\} \leq Le_{k+r} \leq Le_k \frac{H^r}{r!}. \quad (15)$$

The gain. Write $s = k/H$ and $\ell = \ln(2+s)$. We claim $\ln(\frac{16}{e}(1+s/\sqrt{\ell})) \geq \frac{1}{2}\sqrt{\ell}$. With $t = \sqrt{\ell}$: if $t \leq 1$ the left side is at least $\ln(16/e) > 1 \geq t/2$; if $t \geq 1$ then $s = e^{t^2} - 2$ and $1+s/t \geq e^{t^2}/(2t)$ (as $e^{t^2}/2 + t - 2 \geq 0$), so the difference of the two sides is at least $\ln(8/e) + t^2 - \ln t - t/2$, which is positive at $t = 1$ and increasing. Since $r = \rho(k) \geq 16H(1+sL_0/\sqrt{\ell}) \geq 16H(1+s/\sqrt{\ell})$, we get $\ln \frac{r}{eH} \geq \frac{1}{2}\sqrt{\ell}$ and, using $r! \geq (r/e)^r$ and $r \geq 16kL_0/\sqrt{\ell}$, $\ln \frac{r!}{H^r} \geq r \ln \frac{r}{eH} \geq 8kL_0$. On the other hand $4G \leq 8H \leq (H+2)^3$ gives $\ln(4G(8H)^k) \leq 3(k+1)L_0 \leq 6kL_0$. Therefore $4G(8H)^k H^r / r! \leq 1$, and (14), (15) give the lemma. $\square$

Dyadic encoding of fractional valuations

Let $u = (u_q)$ be weights, $s_q(n) := \min(u_q v_q(n), 1)$ and $X(n) := \sum_q s_q(n)$. For $j \geq 0$ put $N_j(n) := \#\{q : s_q(n) \geq 2^{-j}\}$ and, for $u_q > 0$, $d_{q,j} := q^{\lceil 2^{-j}/u_q \rceil}$. Then $s_q(n) \geq 2^{-j}$ if and only if $d_{q,j} \mid n$, the integers $d_{q,j}$ (q varying) are pairwise coprime, so $N_j = \nu_{D_j}$ for $D_j := \{d_{q,j}\}$; and since $s \leq \sum_{j \geq 0} 2^{-j} \lfloor s \geq 2^{-j} \rfloor \leq 2s$ for $s \in [0, 1]$,

$$X(n) \leq \sum_{j \geq 0} 2^{-j} N_j(n) \leq 2X(n). \quad (16)$$

Lemma 10.5 (weighted sieve lemma). *For every family of weights u , every $L \geq 16$ and every J ,*

$$\#\{n \leq L : X(n) \geq 149H\} \leq \#\{n \in J : X(n) \geq 1\}.$$

Proof. Put $Q := 1 + \lfloor \log_2(R-1) \rfloor$, so that $2^j \leq R-1$ for $j < Q$ and $2^Q \geq R$. With $t = \ln L$ one has $R < 2t$ and $Q < 3 + 2 \ln t < 5\lambda(L) < 2H$ for $L \geq 16$. For $j < Q$ set $a_j := C(2^j) - 1$ and for $j \geq Q$ set $a_j := R$; let $A := \sum_{j \geq 0} 2^{-j} a_j$. Since $N_j(n) \leq \omega(n) \leq R$ for $n \leq L$, if $X(n) \geq A+1$ for some $n \leq L$ then by (16) some $j < Q$ has $N_j(n) \geq C(2^j)$. Lemma 10.4 with $D = D_j$, $k = 2^j$ and $G = Q$ gives $Q \cdot \#\{n \leq L : N_j(n) \geq C(2^j)\} \leq \#\{n \in J : N_j(n) \geq 2^j\} \leq \#\{n \in J : X(n) \geq 1\}$, the last step because $N_j(n) \geq 2^j$ forces $X(n) \geq 2^{-j} N_j(n) \geq 1$. Summing over the Q indices $j < Q$, $\#\{n \leq L : X(n) \geq A+1\} \leq \#\{n \in J : X(n) \geq 1\}$.

It remains to show $A+1 \leq 149H$. From (13), $a_j \leq 2 \cdot 2^j + 16H + 16 \cdot 2^j L_0 / \sqrt{\ln(2 + 2^j/H)}$, so $\sum_{j < Q} 2^{-j} a_j \leq 2Q + 32H + 16L_0 S_Q$ with $S_Q := \sum_{j < Q} \ln(2 + 2^j/H)^{-1/2}$. The indices with $2^j \leq H$ number at most $1 + \log_2 H \leq 1 + \frac{3}{2}\sqrt{H}$ and contribute at most $\frac{5}{4}$ each, hence at most $\frac{21}{8}\sqrt{H}$; if j_0 is the first index with $2^{j_0} > H$, the j_0 -term is less than 1 and the term $j_0 + t$ ($t \geq 1$) is less than $\frac{5}{4}t^{-1/2}$, so the remaining terms contribute less than $1 + \frac{5}{2}\sqrt{Q} \leq \frac{21}{5}\sqrt{H}$. Thus $S_Q \leq 7\sqrt{H}$, and with $L_0 = \ln(H+2) \leq \sqrt{H}$ we get $16L_0 S_Q \leq 112H$. Finally $\sum_{j \geq Q} 2^{-j} R = 2R/2^Q \leq 2$, so $A \leq 4H + 32H + 112H + 2 = 148H + 2$ and $A+1 \leq 149H$ because $H > 3$. $\square$

Proof of Theorem 10.1

Write $w_z = E + S$ with $S(n) = \sum_p \min(z_p v_p(n), 1)$ as in the remark after Theorem 9.4; it suffices to prove $\sum_{k \leq m} (S(k) - c_m)^+ \leq \sum_{b \in I} (S(b) - 1)^+$. By continuity in the (finitely many) weights we may assume $z_p = a_p/N$ rational; put $r_p(n) := \min(a_p v_p(n), N)$ and $R(n) := \sum_p r_p(n) = NS(n)$. For integers n and thresholds $T \geq 0$ let $R_{<p}(n) := \sum_{q < p} r_q(n)$. Ordering the primes dividing n and letting p^* be the first with $R_{\leq p^*}(n) \geq T$, one has $\sum_{p \mid n} r_p(n) [R_{<p}(n) \geq T] = R(n) - R_{\leq p^*}(n)$ and $T \leq R_{\leq p^*}(n) < T + N$, whence

$$(R(n) - T - N)^+ \leq \sum_{p \mid n} r_p(n) [R_{<p}(n) \geq T] \leq (R(n) - T)^+. \quad (17)$$

Applying (17) with $T = (c_m - 1)N$ on $[1, m]$ and with $T = N$ on I , it suffices to show, for every prime p , $\sum_{k \leq m, p \nmid k} r_p(k) [R_{<p}(k) \geq (c_m - 1)N] \leq \sum_{b \in I, p \nmid b} r_p(b) [R_{<p}(b) \geq N]$. Writing $r_p(n) = \sum_{j \geq 1} \Delta_{p,j} [p^j \mid n]$ with $\Delta_{p,j} = \min(a_p j, N) - \min(a_p(j-1), N) \geq 0$ and using $R_{<p}(p^j i) = R_{<p}(i)$, this follows from: for every $j \geq 1$, $L := \lfloor m/p^j \rfloor$ and every run J of L consecutive integers (the quotients i of the multiples $p^j i \in I$ form a run of L or $L+1$ consecutive integers),

$$\#\{i \leq L : R_{<p}(i) \geq (c_m - 1)N\} \leq \#\{i \in J : R_{<p}(i) \geq N\}.$$

Dividing by N , $R_{<p}/N$ is the function X of Lemma 10.5 for the weights $u_q = z_q$ ($q < p$). If $L \geq 16$ the lemma applies because $c_m - 1 = 149e\lambda(m) \geq 149e\lambda(L)$; if $L < 16$ the left side is empty since $X(i) \leq \omega(i) \leq 2 < c_m - 1$ for $i \leq 15$. This proves (12) for $m \geq 16$; for $2 \leq m \leq 15$, $S(k) \leq \omega(k) \leq 2$ gives the statement with $c_m = 4$. Finally, for $m \geq 16$ and $t = \ln m$, $\lambda(m) = \ln \ln m + \ln(1 + 1/t) < \frac{3}{2} \ln \ln m$ (as $1/t < 3/8 < \frac{1}{2} \ln \ln m$), and $1 + 149e\lambda(m) < 150e\lambda(m) < 675 \ln \ln m$. $\square$

Remark 10.6. The constant $149e \approx 405$ is far from optimal: the thinning probability $1/(8H)$, the gain $G = Q$ and the dyadic budget were chosen for simplicity, and the bound of Corollary 10.2 is smaller than that of Theorem 1.2 only for astronomically large n . Section 11 optimises these choices and reaches the threshold $20 \ln \ln m$; the resulting bound $(16 + 20 \ln \ln(8n^3))n$ is below Theorem 1.2 for every n and below Theorem 1.1 for $n \gtrsim 10^{35}$. The hinge inequality with threshold 2 (Conjecture 8.1), which would give $18n$, remains open.

11 An improved constant: threshold $20 \ln \ln m$

The dyadic budget of Section 10 pays about eH at every level. Replacing the dyadic levels by a geometric sequence with ratio close to 1, and allowing an arbitrary gain in Lemma 10.4, the levels can be made to share one payment of eH , and the constant $149e \approx 405$ drops to 20.

Theorem 11.1. *For every weight z , every $m \geq 3$ and every $x \geq 0$,*

$$\sum_{k \leq m} (w_z(k) - 20 \ln \ln m)^+ \leq \sum_{b \in I} (w_z(b) - 1)^+, \quad (18)$$

and for $m \leq 2$ the inequality holds with threshold 2. Consequently $g(n) \leq (16 + 20 \ln \ln(8n^3))n$ for all $n \geq 1$.

The consequence follows exactly as in Corollary 10.2 ($20 \ln \ln m$ is increasing and exceeds 2 for $m \geq 3$). We keep the notation of Section 10: $L \geq 16$, J a set of L consecutive positive integers, $R = \lfloor \log_2 L \rfloor$, and now $h := \ln(1 + \ln L)$, $H := eh$, $h_0 := eH$. For a family D of pairwise coprime integers ≥ 2 and $1 \leq k \leq R - 1$ we use $y = L^{1/(k+1)}$, $D_0 = \{d \in D : d \leq y\}$ and the elementary symmetric functions e_r of $\{1/d : d \in D_0\}$ as before; Lemma 9.5 gives $\sum_{d \in D_0} 1/d \leq H$.

Lemma 11.2 (arbitrary gain). *Let $G \geq 1$, $A := \ln(4G) + k \ln(8H)$, $s(A) := \min\{A, 2A/\ln(2 + A/h_0)\}$ and $\rho := \lceil h_0 + s(A) \rceil$. Then*

$$G \cdot \#\{n \leq L : \nu_D(n) \geq 2k + \rho\} \leq \#\{n \in J : \nu_D(n) \geq k\}.$$

Proof. By (14) and (15) it suffices that $\rho! \geq 4G(8H)^k H^\rho$, and by $\rho! \geq (\rho/e)^\rho$ it suffices that $\rho \ln \frac{\rho}{eH} \geq A$. Put $u = A/h_0$. For the first candidate $\rho = h_0(1 + u)$ this reads $(1 + u) \ln(1 + u) \geq u$, which holds. For the second, $\rho = h_0(1 + v)$ with $v = 2u/\ln(2 + u)$, we need $(1 + v) \ln(1 + v) \geq u$. If $\ln(2 + u) \leq 2$ then $v \geq u$ and we are done. Otherwise put $y = 2 + u \geq e^2$, so $v = 2(y - 2)/\ln y$; the function $2(y - 2)/(\sqrt{y} \ln y)$ is increasing for $y \geq e^2$ (its logarithmic derivative is $\frac{1}{y-2} - \frac{1}{2y} - \frac{1}{y \ln y} > 0$) and equals $1.98 \dots$ at $y = e^2$, hence $v \geq \sqrt{y}$, $\ln(1 + v) \geq \frac{1}{2} \ln y$ and $(1 + v) \ln(1 + v) \geq v \cdot \frac{1}{2} \ln y = y - 2 = u$. Rounding up preserves the inequality, since $\rho \ln(\rho/h_0)$ is increasing for $\rho \geq h_0$. $\square$

Geometric encoding

Fix $q := 51/50$, $\delta := 1/50$, and put $k_j := \lceil q^j \rceil$; let Q be the number of $j \geq 0$ with $k_j \leq R-1$ and $\alpha_j := \delta q^{-j}$. For weights $u = (u_p)$, $X(n) = \sum_p \min(u_p v_p(n), 1)$ and $D_j := \{p^{\lceil q^{-j}/u_p \rceil} : u_p > 0\}$ (pairwise coprime), one has $\min(u_p v_p(n), 1) \geq q^{-j}$ iff $p^{\lceil q^{-j}/u_p \rceil} \mid n$, so $N_j := \nu_{D_j}$ counts the primes with $\min(u_p v_p(n), 1) \geq q^{-j}$. For $a \in (0, 1]$ and j_0 the least j with $q^{-j} \leq a$, $\sum_{j \geq j_0} \delta q^{-j} = q^{1-j_0} \geq a$; since $n \leq L$ has at most R distinct prime factors, the tail $j \geq Q$ contributes at most Rq^{1-Q} , and therefore

$$X(n) \leq \sum_{j < Q} \alpha_j N_j(n) + Rq^{1-Q} \quad (n \leq L), \quad \text{while} \quad N_j(n) \geq k_j \Rightarrow X(n) \geq k_j q^{-j} \geq 1. \quad (19)$$

Lemma 11.3. *Let $h \geq 4$, $\rho_j := \rho(H, Q, k_j)$ as in Lemma 11.2 with $G = Q$, and $B_L := \sum_{j < Q} \alpha_j (2k_j + \rho_j)$. Then $\#\{n \leq L : X(n) \geq B_L + 2\} \leq \#\{n \in J : X(n) \geq 1\}$.*

Proof. Since $k_Q = \lceil q^Q \rceil \geq R$ we have $q^Q > R-1$, and $h \geq 4$ gives $R \geq 77$, so $Rq^{1-Q} < qR/(R-1) < 1.05$. If $n \leq L$ has $N_j(n) \leq 2k_j + \rho_j - 1$ for all $j < Q$, then by (19) $X(n) < B_L - \sum_{j < Q} \alpha_j + 2 < B_L + 2$. Hence $X(n) \geq B_L + 2$ forces $N_j(n) \geq 2k_j + \rho_j$ for some $j < Q$; Lemma 11.2 with $D = D_j$, $k = k_j$, $G = Q$ and (19) give $Q \cdot \#\{n \leq L : N_j(n) \geq 2k_j + \rho_j\} \leq \#\{n \in J : N_j(n) \geq k_j\} \leq \#\{n \in J : X(n) \geq 1\}$, and summing over the Q levels proves the claim. $\square$

Lemma 11.4 (budget). *For $h \geq 4$, $B_L + 3 < 19.24h$.*

Proof. Write $\ell := \ln(8H) = \ln(8eh)$ and $A_j := \ln(4Q) + k_j \ell$, so $\rho_j \leq h_0 + s(A_j) + 1$. Since $R < eh/\ln 2$ and $Q \leq 1 + \ln R/\ln q$, one has $Q < 56h$; also $\sum_{j < Q} \alpha_j < q$ and, as $k_j \leq q^j + 1$, $2 \sum_{j < Q} \alpha_j k_j \leq 2\delta Q + 2q < 2.24h + 2.04$, while $\sum_{j < Q} \alpha_j (h_0 + 1) < q(e^2 h + 1) < 7.538h + 1.02$. The function $f(t) = \min\{t, 2t/\ln(2 + t/h_0)\}$ has $f(t)/t$ nonincreasing, hence is subadditive, so $s(A_j) \leq \ln(4Q) + k_j \ell g(k_j)$ with $g(u) := \min\{1, 2/\ln(2 + \beta u)\}$, $\beta := \ell/(e^2 h)$. The first part contributes at most $q \ln(4Q) < 1.02 \ln(224h) < 1.734h$. For the second, g is decreasing and $k_j \leq q^j + 1$, so $\sum_{j < Q} \alpha_j k_j g(k_j) \leq q + \delta + \frac{\delta}{\ln q} \int_1^R g(u) \frac{du}{u}$ with $\delta/\ln q < 1.01$; splitting the integral at $u = v_0/\beta$, $v_0 := e^2 - 2$, and using $\ln(2+v) \geq \ln v$ gives $\int_1^R g(u) \frac{du}{u} \leq J(h) := \ln \frac{v_0 e^2 h}{\ell} + 2 \ln \frac{h-3/2}{\ln v_0}$ (because $\ln(\beta R) < h - 3/2$ for $h \geq 4$). The one-variable inequality $\ell J(h)/h < 5$ holds for $h \geq 8$ analytically ($J(h) < 3 \ln h + 1.01$, $\ell < \ln h + 3.08$, and $(y + 3.08)(3y + 1.01)e^{-y}$ is decreasing in $y = \ln h$ with value < 4.68 at $y = \ln 8$) and on $[4, 8]$ by an interval computation with 80 bins, using that ℓ and J are increasing (maximum upper bound 4.966, attained on $[4.40, 4.45]$; the true maximum is 4.910 at $h = 4.513$; script in the repository). Hence $\ell \sum_{j < Q} \alpha_j k_j g(k_j) < 6.212h$. Adding the pieces, $B_L < (7.538 + 2.240 + 1.734 + 6.212)h + 3.06$, and $3.06 + 3 < 1.515h$ for $h \geq 4$, so $B_L + 3 < 19.239h$. $\square$

Proof of Theorem 11.1. Lemmas 11.3 and 11.4 give, for every L with $\ln(1 + \ln L) \geq 4$ and every J ,

$$\#\{n \leq L : X(n) \geq 19.25 h_L - 1\} \leq \#\{n \in J : X(n) \geq 1\}, \quad h_L := \ln(1 + \ln L). \quad (20)$$

Now let m satisfy $h_m := \ln(1 + \ln m) \geq 4$, put $T := 20 \ln \ln m - 1$ and argue as in the proof of Theorem 10.1: with $S = \sum_p \min(z_p v_p, 1)$, the weighted ordering identity (17) (in the normalised form with $r_p \in [0, 1]$ and $N = 1$) reduces (18) for S to the transfer inequality $\#\{i \leq L_d : R_{<p}(i) \geq T\} \leq \#\{i \in J_d : R_{<p}(i) \geq 1\}$ for every prime power $d = p^j$, $L_d = \lfloor m/d \rfloor$

and every run J_d of L_d consecutive integers. Since $\ln \ln m = h_m + \ln(1 - e^{-h_m}) \geq 0.995 h_m$ for $h_m \geq 4$, we have $T \geq 19.9h_m - 1 \geq 19.25h_{L_d} - 1$ whenever $h_{L_d} \geq 4$, and (20) applies; if $h_{L_d} < 4$ then $R_{<P}(i) \leq \omega(i) \leq \log_2 L_d < (e^4 - 1)/\ln 2 < 77.4 < T$, so the left side is empty. Peeling $E = w_z - S$ as before gives (18) for $h_m \geq 4$. If $m \geq 4$ and $h_m < 4$, then $\ln m < e^4 - 1$ and $w_z(k) \leq \Omega(k) \leq \log_2 m \leq 20 \ln \ln m$ (the function $t/\ln t$ on $[\ln 4, e^4 - 1]$ is at most $13.47 < 20 \ln 2$), so the left side of (18) vanishes; for $m = 3$, $20 \ln \ln 3 > 1 \geq w_z(k)$; for $m \leq 2$ the left side vanishes with threshold 2. $\square$

Remark 11.5. The constant is not sharp: with the exact ρ_j one finds $\sup_{h \geq 4} (B_L + 3)/h = 17.90$, so the same argument supports a threshold of about $18 \ln \ln m$; the binding steps are the bounds $\ln(224h) < 1.7h$ and $1.04 \ell(4)/4 < 1.162$ at $h = 4$. The remaining $\ln \ln m$ comes from the baseline $\rho \geq eH$ in Lemma 11.2, i.e. from comparing the tail of ν_D on $[1, L]$ above its typical size with the count at level k in J . An absolute constant would require a comparison at the typical level; the counting certificates of Section 9 suggest that this is possible, but we have not proved it.

12 The 0/1 case with an absolute threshold

The threshold $20 \ln \ln m$ of Theorem 9.4 can be replaced by the constant 4, for every prime set, by two explicit counting certificates. Throughout this section P is a finite set of primes, $\Omega_P(n) = \sum_{p \in P} \mathbf{v}_p(n)$, $\omega_P(n) = \#\{p \in P : p \mid n\}$, $e_P := \Omega_P - \omega_P = \sum_{p \in P} (\mathbf{v}_p - 1)^+$, and

$$y := \lfloor m^{1/3} \rfloor, \quad S := \{p \in P : p \leq y\}, \quad L := P \setminus S, \quad H := \sum_{p \in S} \frac{1}{p}, \quad \eta := \left(1 + \frac{1}{y}\right)H.$$

Theorem 12.1. *For every finite set of primes P , every $m \geq 1$ and every $x \geq 0$,*

$$\sum_{k \leq m} (\Omega_P(k) - 4)^+ \leq \sum_{b \in I} (\Omega_P(b) - 1)^+. \quad (21)$$

The proof uses Proposition 9.1 with the weight $z = \mathbf{1}_P$ (so $w_z = \Omega_P$), and splits according to whether $\eta \leq 2$ or $\eta > 2$. We write $A := \sum_{p \in P} \sum_{j \geq 2, p^j \leq m} \lfloor m/p^j \rfloor = \sum_{k \leq m} e_P(k)$ and, for $j = 2, 3$, $E_j := \sum_{B \subseteq S, |B|=j} \lfloor m/\prod B \rfloor = \sum_{k \leq m} \binom{\omega_S(k)}{j}$.

Lemma 12.2 (cube-root peeling). *Every $k \leq m$ has $\omega_L(k) \leq 2$, and $(\Omega_P(k) - 4)^+ \leq e_P(k) + (\omega_S(k) - 2)^+$. Hence $\sum_{k \leq m} (\Omega_P(k) - 4)^+ \leq A + \sum_{k \leq m} (\omega_S(k) - 2)^+$.*

Proof. Three distinct primes larger than y have product at least $(y+1)^3 > m$. Thus $\Omega_P(k) - 4 = e_P(k) + \omega_S(k) + \omega_L(k) - 4 \leq e_P(k) + \omega_S(k) - 2$, and $(a+b)^+ \leq a + b^+$ for $a \geq 0$. $\square$

Lemma 12.3 (sparse branch). *Suppose $\eta \leq 2$. The certificate $c_{pj} = 1$ ($p \in P, j \geq 2, p^j \leq m$), $c_{pq} = \frac{11}{21}$ ($p < q$ in S), $c_{pqr} = -\frac{1}{7}$ ($p < q < r$ in S), all other $c_d = 0$, satisfies (8) for $w = \Omega_P$ and has $V(c) \geq \sum_{k \leq m} (\Omega_P(k) - 4)^+$.*

Proof. Feasibility. Let $r = \omega_S(n)$ and $t = \omega_P(n) \geq r$. The pair and triple coefficients contribute $\phi(r) = \frac{11}{21} \binom{r}{2} - \frac{1}{7} \binom{r}{3} = r(r-1)(13-r)/42$, and $r(13-r) \leq 42$ for every integer r (with equality at $r = 6, 7$), so $\phi(r) \leq (r-1)^+ \leq (t-1)^+$. The prime-power coefficients contribute at most $e_P(n)$. If $t \geq 1$ then $e_P(n) + (t-1) = \Omega_P(n) - 1$; if $t = 0$ the sum is 0.

Value. Here $V(c) = A + \frac{11}{21}E_2 - \frac{1}{7}(E_3 + \binom{|S|}{3})$. Every triple in S has product at most $y^3 \leq m$, so $\binom{|S|}{3} \leq E_3$. Next, $3E_3 = \sum_{\{p,q\} \subseteq S} \sum_{r \in S \setminus \{p,q\}} \lfloor m/pqr \rfloor \leq H \sum_{\{p,q\} \subseteq S} m/pq$, and for $p, q \in S$

one has $pq \leq y^2$, hence $\lfloor m/pq \rfloor \geq y$ and $m/pq < \lfloor m/pq \rfloor + 1 \leq (1 + 1/y)\lfloor m/pq \rfloor$; therefore $3E_3 \leq \eta E_2 \leq 2E_2$. Consequently

$$V(c) \geq A + \frac{11}{21}E_2 - \frac{2}{7}E_3 \geq A + \left(\frac{11}{21} - \frac{2}{7} \cdot \frac{2}{3}\right)E_2 = A + \frac{1}{3}E_2.$$

Finally $(r-2)^+ \leq \frac{1}{3}\binom{r}{2}$ for all integers $r \geq 0$ (the difference is $(r-3)(r-4)/6$ for $r \geq 3$), so $\sum_{k \leq m} (\omega_S(k) - 2)^+ \leq \frac{1}{3}E_2$, and Lemma 12.2 finishes the proof. $\square$

Lemma 12.4 (dense branch). *Suppose $\eta > 2$. The affine certificate $c_1 = -1$, $c_{p^j} = 1$ ($p \in P$, $j \geq 1$, $p^j \leq m$) satisfies (8) for $w = \Omega_P$ and has $V(c) \geq \sum_{k \leq m} (\Omega_P(k) - 4)^+$.*

Proof. Feasibility is Corollary 9.2. The value is $V(c) = \sum_{k \leq m} \Omega_P(k) - (m+1)$, and since $\Omega - (\Omega - 4)^+ = \min(\Omega, 4)$ it suffices to show $\sum_{k \leq m} \min(\Omega_P(k), 4) \geq m+1$. For $y \leq 4$ one has $\eta \leq 0, \frac{3}{4}, \frac{10}{9}, \frac{25}{24}$ respectively, so $y \geq 5$ and $H > 2y/(y+1) \geq 5/3$. Adding the primes of S one at a time until the reciprocal sum first reaches $3/2$ gives $Q \subseteq S$ with $\frac{3}{2} \leq H_Q := \sum_{p \in Q} 1/p < 2$ (the last term is at most $1/2$) and $|Q| \leq \pi(y) \leq y$. For integers $r \geq 0$, $\min(r, 4) \geq r - \frac{1}{7}\binom{r}{2}$ (for $r \geq 5$ the difference is $(r-7)(r-8)/14 \geq 0$). Hence, with $E_1(Q) = \sum_{p \in Q} \lfloor m/p \rfloor$ and $E_2(Q) = \sum_{\{p,q\} \subseteq Q} \lfloor m/pq \rfloor$,

$$\sum_{k \leq m} \min(\omega_Q(k), 4) \geq E_1(Q) - \frac{1}{7}E_2(Q) \geq mH_Q - |Q| - \frac{m}{14}H_Q^2 \geq m\left(\frac{3}{2} - \frac{9}{56}\right) - y = \frac{75}{56}m - y,$$

because $h \mapsto h - h^2/14$ is increasing on $[\frac{3}{2}, 2]$. As $m \geq y^3$ and $y \geq 5$, $\frac{19}{56}m \geq y+1$, so $\sum_{k \leq m} \min(\Omega_P(k), 4) \geq \sum_{k \leq m} \min(\omega_Q(k), 4) \geq m+1$. $\square$

Proof of Theorem 12.1. If $\eta \leq 2$ apply Proposition 9.1 with the certificate of Lemma 12.3; if $\eta > 2$ with that of Lemma 12.4. (For $m < 8$ one has $y = 1$, $S = \emptyset$, $\eta = 0$, and the sparse certificate consists of the prime-power terms only.) $\square$

Remark 12.5. (a) The threshold is $2+2$: two from $\omega_L \leq 2$, which the choice $y = m^{1/3}$ forces, and two from $(r-2)^+ \leq \binom{r}{2}/3$. Taking $y = m^{1/2}$ would give $\omega_L \leq 1$ but destroy the bound $\binom{|S|}{3} \leq E_3$. The coefficients $\frac{11}{21}, -\frac{1}{7}$ are extremal for the pair-triple ansatz: the pointwise constraint at $r = 6$ gives $3a - 1 \leq 4b$, and the value bound $\frac{11}{21} - \frac{2}{7} \cdot \frac{2}{3} = \frac{1}{3}$ is an identity, so this ansatz reaches exactly $\eta = 2$. Numerically the inequality (21) holds with threshold 2 in every instance tested (Section 8), and fails with threshold 1.

(b) Theorem 12.1 concerns 0/1 weights only. It does not by itself improve the bound on $g(n)$, because the duality of Section 8 requires the hinge inequality for all fractional weights. The obvious averaging does not transfer it: for $z_p \in [0, 1]$ and $P_t := \{p : z_p \geq t\}$ one has $\int_0^1 (\Omega_{P_t}(n) - 1)^+ dt = w_z(n) - \max_{p|n} z_p$, which exceeds $(w_z(n) - 1)^+$ (e.g. $n = pq$ with $z_p = z_q = \frac{1}{2}$), so the averaged certificate violates (8).

(c) The anchor certificates of Lemma 9.3 cannot give an absolute threshold: for $Y = 2^{41}$, $m = (8Y)^4$ and P the primes in $(Y, 8Y]$, no choice of anchors $R \subseteq P$ (even with all positive prime-power coefficients added) reaches threshold 3, while the pair-triple certificate above handles this P at threshold 2; the computation is in the engine transcript.

13 The fractional case: reduction to a sparse core

We now return to arbitrary weights $z_p \in [0, 1]$ and to the function $S(n) = \sum_p \min(z_p v_p(n), 1)$ of Section 10; recall that the hinge inequality for S with threshold c implies it for w_z (the peel $w_z = E + S$). Write

$$\alpha_{p,j} := \min(jz_p, 1) - \min((j-1)z_p, 1) \quad (j \geq 1), \quad \text{so that} \quad \alpha_{p,j} \geq 0, \quad \sum_j \alpha_{p,j} \leq 1, \quad S(n) = \sum_p \sum_{j \geq 1} \alpha_{p,j} [p^j \mid n]$$

We call the $\alpha_{p,j}$ the *atoms* of z ; the results of this section use only $\alpha_{p,j} \geq 0$ and $\sum_j \alpha_{p,j} \leq 1$. Fix m and split $S = S_0 + S_1$, where S_0 collects the atoms with $p^j \leq m/64$ and S_1 the rest, and put

$$H_{64}(m, z) := \sum_{p^j \leq m/64} \frac{\alpha_{p,j}}{p^j} \quad (\text{the mean of } S_0 \text{ over the integers}).$$

Lemma 13.1 (large atoms). *If $m \geq 4096$ then $S_1(k) \leq 1$ for every $k \leq m$. Consequently $(S(k) - 65)^+ \leq (S_0(k) - 64)^+$ for $k \leq m$, while $(S_0(b) - 1)^+ \leq (S(b) - 1)^+$ for every b .*

Proof. If $p^j, q^\ell > m/64$ with $p \neq q$ both divide $k \leq m$, then $p^j q^\ell \mid k$ and $m \geq p^j q^\ell > m^2/4096 \geq m$, a contradiction. So the atoms of S_1 active at k belong to one prime, and their sum is at most 1. $\square$

Theorem 13.2 (dense branch). *If $H_{64}(m, z) \geq \frac{17}{16}$ then, for every $x \geq 0$,*

$$\sum_{k \leq m} (S(k) - 65)^+ \leq \sum_{b \in I} (S(b) - 1)^+, \quad (22)$$

and hence the same holds for w_z in place of S .

Proof. For $m \leq 4096$ the left side vanishes ($S(k) \leq \omega(k) \leq 12 < 65$), so let $m > 4096$. By Lemma 13.1 and the affine certificate $c_1 = -1$, $c_{p^j} = \alpha_{p,j}$ ($p^j \leq m/64$), which satisfies (8) for S_0 and a fortiori for S ,

$$\sum_{b \in I} (S(b) - 1)^+ \geq \sum_{b \in I} (S_0(b) - 1)^+ \geq \sum_{k \leq m} S_0(k) - m,$$

so it suffices to show $\sum_{k \leq m} \min(S_0(k), 64) \geq m$, because $S_0 - (S_0 - 64)^+ = \min(S_0, 64)$ and $(S - 65)^+ \leq (S_0 - 64)^+$. Put $h_p := \sum_{p^j \leq m/64} \alpha_{p,j}/p^j \leq 1/p \leq \frac{1}{2}$. Adding primes one at a time until the sum first reaches $\frac{17}{16}$ gives a set Q with $\frac{17}{16} \leq H_Q := \sum_{p \in Q} h_p \leq \frac{25}{16}$. Let $T_Q(k) \leq S_0(k)$ be the part of $S_0(k)$ coming from the primes of Q , and $U_p(k) \in [0, 1]$ the contribution of a single $p \in Q$. Since $p^j \leq m/64$ gives $\lfloor m/p^j \rfloor \geq \frac{63}{64}m/p^j$,

$$\sum_{k \leq m} T_Q(k) = \sum_{p \in Q} \sum_{p^j \leq m/64} \alpha_{p,j} \left\lfloor \frac{m}{p^j} \right\rfloor \geq \frac{63}{64} H_Q m \geq \frac{1071}{1024} m.$$

For the second moment, $\sum_{k \leq m} U_p(k)^2 \leq \sum_{k \leq m} U_p(k) \leq m h_p$ and, for $p \neq q$, $\sum_{k \leq m} U_p(k) U_q(k) = \sum_{j, \ell} \alpha_{p,j} \alpha_{q,\ell} \lfloor m/p^j q^\ell \rfloor \leq m h_p h_q$; hence $\sum_{k \leq m} T_Q(k)^2 \leq m(H_Q + H_Q^2) \leq \frac{1025}{256} m$. Finally $\min(t, 64) \geq t - t^2/256$ for all $t \geq 0$ (the right side is at most t , and at most 64 with maximum at $t = 128$), so

$$\sum_{k \leq m} \min(S_0(k), 64) \geq \sum_{k \leq m} \min(T_Q(k), 64) \geq \left(\frac{1071}{1024} - \frac{1025}{65536} \right) m = \frac{67519}{65536} m \geq m. \quad \square$$

Corollary 13.3 (reduction to a sparse core). *Suppose that for every $m > 4096$ and every system of atoms $\alpha_{p,j} \geq 0$ with $\sum_j \alpha_{p,j} \leq 1$, supported on $p^j \leq m/64$ and with $\sum \alpha_{p,j}/p^j < \frac{17}{16}$, the function $S_0 = \sum \alpha_{p,j} [p^j \mid \cdot]$ satisfies*

$$\sum_{k \leq m} (S_0(k) - 64)^+ \leq \sum_{b \in I} (S_0(b) - 1)^+ \quad \text{for every } x \geq 0. \quad (23)$$

Then (22) holds for all weights, all m and all x , and consequently $g(n) \leq 81n$ for all n .

Proof. For $m \leq 4096$ the left side of (22) is 0; for $m > 4096$ and $H_{64} \geq \frac{17}{16}$ apply Theorem 13.2; otherwise (23) and Lemma 13.1 give (22). The bound on $g(n)$ follows as in Corollary 10.2: the proof of Theorem 8.2 with threshold 65 in place of 2 gives $g(n) \leq (65 + 16)n$. $\square$

Remark 13.4. (a) The constants are one point on a curve: peeling at m/R costs one unit of threshold and needs $m \geq R^2$, and the same computation closes the dense branch for $H_R \geq H_{10}$ whenever $\frac{R-1}{R}H_{10} - (H_{10} + \frac{1}{2} + (H_{10} + \frac{1}{2})^2)/(4R) > 1$; for $R = 64$ this allows $H_{10} = \frac{133}{128}$.

(b) The sparse core (23) is the hard half of the problem, not the easy one: it is exactly the regime in which the affine certificate fails (Corollary 9.2). Its left side vanishes unless some $k \leq m$ has $S_0(k) > 64$, which requires $\omega(k) > 64$, so (23) is vacuous for m below the product of the first 65 primes ($6.108 \cdot 10^{127}$); it cannot be tested numerically at this threshold, and any finite check must be run at a scaled-down threshold.

(c) Three natural ways of transferring Theorem 12.1 to fractional weights fail. Positive combinations of the 0/1 inequalities for the level sets $\{p : z_p \geq t\}$ do not control the right side of (22), because the level-set hinges $\sum_t (\Omega_{P_t} - 1)^+$ can be positive where $(S - 1)^+ = 0$. The abstract principle “an inequality between positive combinations of hinges that holds at all 0/1 vertices holds on $[0, 1]^P$ ” is false: with nine coordinates, seven copies of the full set on the left and the 36 pairs on the right, $7(t - 4)^+ \leq \binom{t}{2}$ holds for every support size t , but $z \equiv \frac{1}{2}$ gives $\frac{7}{2} > 0$. And no certificate supported on sets of a bounded number of atoms can dominate $(S - C)^+$ pointwise for all weights: in the equal-weight regime such a certificate is a polynomial of bounded degree in the number of active atoms, which cannot vanish on an initial segment and grow linearly afterwards. These are obstructions to particular arguments, not evidence against (23), which holds in every instance we could compute at scaled thresholds.

(d) One nonnegative certificate is available in general: order the atoms, lay their weights end to end on $[0, Z]$, and let c_A be the measure of the set of t for which the unit window $[t, t + 1]$ meets exactly the atoms of A ; then $c \geq 0$, c satisfies (8) for S (the divisor sum at n equals $\sum_i (\ell_i - 1)^+$ over the maximal runs ℓ_i of active atoms, which is at most $(S(n) - 1)^+$), and its value is $\sum_{k \leq m} \sum_i (\ell_i(k) - 1)^+$ with no penalty terms. Whether some ordering makes this value exceed $\sum_{k \leq m} (S_0(k) - 64)^+$ on the sparse core is the question left open here.

What the two-sided counts alone cannot do

The certificate principle of Proposition 9.1 uses the window I only through $|I| = m$ and the two-sided counts $\lfloor m/d \rfloor \leq N_I(d) \leq \lfloor m/d \rfloor + 1$. The following example shows that these constraints, taken by themselves and without the arithmetic values of the base counts $\lfloor m/d \rfloor$, cannot yield an absolute threshold, even for 0/1 weights.

Proposition 13.5 (abstract two-sided counting is insufficient). *Let q be a prime power, let Π be a projective plane of order q with point set X , $|X| = v = q^2 + q + 1$, and put $m = qv + 1$. Let μ be the multiset of m subsets of X consisting of the v lines and $(q - 1)v + 1$ empty sets, and*

ν the multiset of m subsets consisting of each singleton q times and the full set X once. For every nonempty $A \subseteq X$ write $N_\mu(A)$, $N_\nu(A)$ for the number of members containing A . Then $0 \leq N_\nu(A) - N_\mu(A) \leq 1$ for every nonempty A , while

$$\sum_{B \in \mu} (|B| - C)^+ - \sum_{B \in \nu} (|B| - 1)^+ = v(q - C) + 1 > 0 \quad \text{for every } C \leq q,$$

and the mean $\sum_{B \in \mu} |B|/m = v(q + 1)/(qv + 1)$ is below $1 + 1/q$.

Proof. A singleton lies on $q + 1$ lines and is counted $q + 1$ times in ν ; two points lie on exactly one line and are contained only in X ; a set of at least three points lies on at most one line and is contained only in X . Every line has $q + 1$ points, so $\sum_\mu (|B| - C)^+ = v(q + 1 - C)$, and $\sum_\nu (|B| - 1)^+ = v - 1$. $\square$

Here μ plays the role of $\{1, \dots, m\}$ (with $N_\mu(A)$ in place of $\lfloor m/d_A \rfloor$, d_A the least common multiple of the moduli in A) and ν that of the window. Consequently no proof of the hinge inequality with an absolute threshold can rely solely on $|I| = m$, the two-sided window counts for all divisors, and the feasibility of a certificate: the arithmetic values of the base table $N_K(d) = \lfloor m/d \rfloor$ must enter. They do enter already at first order: at most $\lfloor m/(q+1) \rfloor - \lfloor m/(q+2) \rfloor = q - 1$ moduli d have $\lfloor m/d \rfloor = q + 1$, whereas the configuration above needs $q^2 + q + 1$ atoms with that count, so it is not realised by any prime powers. (For $q = 5$ the abstract configuration beats the threshold 4 of Theorem 12.1, whose proof is a counting certificate; the theorem holds because its certificate is evaluated on the true table.) Also, the hinge sum is not monotone under $N_I(d) \geq N_K(d)$: for $m = 11021$, $P = \{101, 103, 107\}$ and I centred at $101 \cdot 103 \cdot 107$ one has $\sum_K (\omega_P - 1)^+ = 3$ but $\sum_I (\omega_P - 1)^+ = 2$, although every $N_I(d) \geq \lfloor m/d \rfloor$. So the sparse core cannot be handled by a monotonicity argument at small harmonic mass either.

Two quantitative facts about the sparse core are worth recording. First, since $S_0 \leq \omega$, the left side of (23) vanishes unless some $k \leq m$ has more than 64 distinct prime factors, so (23) is vacuous for m below the product of the first 65 primes, $6.108 \cdot 10^{127}$; in particular the numerical evidence of Section 8 says nothing about it. Second, the left side is always tiny: with $a_p(k) = \sum_{p^j \leq m/64} \alpha_{p,j} [p^j \mid k] \in [0, 1]$ and $h_p = \sum_{p^j \leq m/64} \alpha_{p,j} / p^j$ one has $\sum_{k \leq m} e_\ell((a_p(k))_p) \leq m e_\ell((h_p)_p) \leq m H_{64}^\ell / \ell!$ for the elementary symmetric functions e_ℓ (distinct primes have coprime moduli), and $(t - C)^+ \leq e_{C+1}$ of the summands of t when they lie in $[0, 1]$, so

$$\sum_{k \leq m} (S_0(k) - 64)^+ \leq \frac{m H_{64}^{65}}{65!} < 6.24 \cdot 10^{-90} m \quad \text{when } H_{64} < \frac{17}{16},$$

while a single $k \leq m$ with $S_0(k) > 64$ forces $\sum_I (S_0 - 1)^+ > 63$, because the window contains a multiple b of k and $S_0(b) \geq S_0(k)$. A counterexample to (23) would therefore be a window of length $m \geq 6.1 \cdot 10^{127}$ that is deficient, simultaneously for astronomically many moduli, within the ± 1 freedom of the counts; we have not been able to rule this out, and the question is left open here.

The sparse core up to 10^{2887}

The arithmetic input that the two-sided counts lack is supplied, in a range of m that is finite but enormous, by the multiples of a single integer with many small prime factors.

Lemma 13.6 (packing). *Let $x_1, \dots, x_N \in (0, 1]$ with $\sum_i x_i > 64$. Then there are pairwise disjoint index sets $G_1, \dots, G_{32}$ with $\frac{4}{3} < \sum_{i \in G_r} x_i \leq 2$ for every r .*

Proof. Call x_i large if $x_i > \frac{2}{3}$. Pair the large items two by two; each pair has mass in $(\frac{4}{3}, 2]$. Then form further groups greedily from the remaining items (at most one large one, and items of mass at most $\frac{2}{3}$), placing the odd large item, if any, first: add items until the mass exceeds $\frac{4}{3}$; each such group has mass at most $\frac{4}{3} + \frac{2}{3} = 2$ (the group containing the large item has mass at most $1 + \frac{2}{3}$, since the large item is added while the mass is 0). Every completed group has mass in $(\frac{4}{3}, 2]$, the leftover has mass at most $\frac{4}{3}$, so the completed groups have total mass more than $64 - \frac{4}{3} > 31 \cdot 2$, and there are at least 32 of them. $\square$

Theorem 13.7 (arithmetic shadows). *Let $m > 4096$ and let S_0 be as above. If $S_0(k) > 64$ for some $k \leq m$, then for every window I of m consecutive integers*

$$\sum_{b \in I} (S_0(b) - 1)^+ \geq \frac{32}{3} (1 - 6^{-31}) \frac{m}{k^{1/32}} \geq \frac{32}{3} (1 - 6^{-31}) m^{31/32}.$$

Proof. Write $a_p(k) = \sum_{p^j \leq m/64, p^j | k} \alpha_{p,j} \in [0, 1]$, so $S_0(k) = \sum_p a_p(k)$. Apply Lemma 13.6 to the numbers $a_p(k) > 0$: this gives disjoint sets of primes $G_1, \dots, G_{32}$ with $\sum_{p \in G_r} a_p(k) > \frac{4}{3}$. For each prime p let q_p be the largest power $p^j \leq m/64$ with $\alpha_{p,j} > 0$ and $p^j | k$, and put $D_r := \prod_{p \in G_r} q_p$. The D_r are pairwise coprime, $\prod_r D_r | k$, and each $D_r \geq 6$ (a group contains at least two primes, since one prime carries mass at most 1). If $D_r | n$ then every atom of a prime of G_r that is active at k is active at n , so $a_p(n) \geq a_p(k)$ for $p \in G_r$; hence if exactly $t \geq 1$ of the D_r divide n then $S_0(n) > \frac{4t}{3}$ and $(S_0(n) - 1)^+ > \frac{4t}{3} - 1 \geq \frac{t}{3}$. Thus the nonnegative certificate $c_{D_r} = \frac{1}{3}$ satisfies (8) for S_0 , and by Proposition 9.1

$$\sum_{b \in I} (S_0(b) - 1)^+ \geq \frac{1}{3} \sum_{r=1}^{32} \left\lfloor \frac{m}{D_r} \right\rfloor \geq \frac{1}{3} (1 - 6^{-31}) \sum_{r=1}^{32} \frac{m}{D_r},$$

because $m/D_r \geq \prod_{s \neq r} D_s \geq 6^{31}$. Finally $\sum_r 1/D_r \geq 32(\prod_r D_r)^{-1/32} \geq 32 k^{-1/32}$ by the arithmetic–geometric mean inequality. $\square$

Corollary 13.8. *The sparse-core inequality (23) holds for every atom system and every window whenever $4096 < m \leq 10^{2887}$. Consequently the hinge inequality with threshold 65 holds for all weights and all $m \leq 10^{2887}$, and*

$$g(n) \leq 81 n \quad \text{for all } n \leq 10^{962}.$$

Proof. If the left side of (23) is positive, some $k \leq m$ has $S_0(k) > 64$, and Theorem 13.7 together with the moment bound above gives $R \geq \frac{32}{3} (1 - 6^{-31}) m^{31/32}$ and $L < 6.24 \cdot 10^{-90} m$; so $L \leq R$ as soon as $m^{1/32} \leq \frac{32}{3} (1 - 6^{-31}) / (6.24 \cdot 10^{-90})$, which holds for $m \leq 10^{2887}$. The hinge inequality follows from Theorem 13.2 and Lemma 13.1, and the bound on $g(n)$ from the proof of Theorem 8.2 with threshold 65, since $a_n < 8n^3 \leq 10^{2887}$ for $n \leq 10^{962}$ (and $2n$ integers suffice when $a_n \geq 8n^3$). $\square$

Remark 13.9. Corollary 13.8 proves the sparse core by a nonnegative certificate in a finite but enormous range; Section 14 removes the restriction on m with a certificate whose coefficients have both signs.

14 The sparse core for all m : a signed clipped-prefix certificate

Throughout this section fix an integer $m > 4096$, a window I of m consecutive positive integers, and a finitely supported atom system $\alpha_{p,j} \geq 0$ with $\sum_j \alpha_{p,j} \leq 1$ for every prime p ; write

$$S_0(n) = \sum_p a_p(n), \quad a_p(n) = \sum_{j: p^j | n} \alpha_{p,j} = A_{p, \nu_p(n)}, \quad A_{p,j} := \sum_{i \leq j} \alpha_{p,i} \quad (A_{p,0} = 0),$$

$$H := \sum_{p,j} \frac{\alpha_{p,j}}{p^j} < \frac{17}{16}, \quad L := \sum_{k \leq m} (S_0(k) - 64)^+, \quad R := \sum_{b \in I} (S_0(b) - 1)^+.$$

The support condition $p^j \leq m/64$ of Section 13 is not needed here.

Theorem 14.1 (signed certificate). $R \geq \frac{141}{128}L$. In particular the sparse-core inequality (23) holds for every $m > 4096$, every atom system with $H_{64}(m, z) < \frac{17}{16}$ and every window I .

The certificates of Sections 9–13 have nonnegative coefficients, and the obstructions recorded there (Proposition 13.5, and the reflected-window barrier in the repository) concern such certificates. The certificate below has coefficients of both signs: a positive coefficient on each “carrier” modulus P , and negative coefficients on products Pq which clip the contribution of the primes outside P . It is feasible at every positive integer, and its value on the window is bounded below by the two-sided counts of multiples together with one arithmetic input, a factorial-moment bound (Lemma 14.4).

Dyadic rounding and removal by modulus cost

For each prime p and each j with $A_{p,j} > 0$ let $t_{p,j}$ be the largest number 2^{-h} , $h \geq 0$, with $2^{-h} \leq A_{p,j}$, so that $t_{p,j} \leq A_{p,j} < 2t_{p,j}$. Keep, for each distinct rounded value, only the first exponent at which it occurs; this yields finitely many pairs (q, t) with $q = p^j$, in which q and t both increase strictly along j . Call (q, t) *retained* if

$$q^{16/t} \leq m \quad (\text{note that } 16/t \in \mathbb{N}). \quad (24)$$

Let $b_p(n)$ be the largest retained value t at p whose modulus divides n ($b_p(n) = 0$ if there is none), and $B(n) := \sum_p b_p(n)$. The differences of consecutive retained values at p form a new atom system with dyadic cumulative values and per-prime total at most 1, and $b_p(n) \leq a_p(n)$ (a retained $q = p^j \mid n$ has $t \leq A_{p,j} \leq a_p(n)$). Put

$$H_B := \sum_{\text{retained } q} \frac{\text{inc}_q}{q}, \quad R_B := \sum_{b \in I} (B(b) - 1)^+, \quad L_B := \sum_{k \leq m} (B(k) - 16)^+,$$

where $\text{inc}_q \geq 0$ is the increment of the retained value at q over the previous retained value at the same prime. Every retained (q, t) satisfies $\log q \leq \frac{t}{16} \log m$ by (24), and this remains true after other levels have been removed, because the retained increments telescope to t .

Lemma 14.2 (rounding). $H_B \leq H$, $R_B \leq R$ and $L \leq 2L_B$.

Proof. $R_B \leq R$ since $B \leq S_0$. For the means take a common multiple Q of all original moduli: S_0 and B are periodic modulo Q , the average of $[q \mid \cdot]$ over $1, \dots, Q$ is $1/q$, so the averages are H and H_B , and $B \leq S_0$ gives $H_B \leq H$. Fix $k \leq m$. For each prime with $a_p(k) > 0$ let t_p be the

rounded value of $A_{p,v_p(k)}$, so $a_p(k) < 2t_p$, and let q_p be the first modulus at which the rounded value t_p occurs; then $q_p \mid k$. If (q_p, t_p) is retained then $b_p(k) \geq t_p$ and $a_p(k) < 2b_p(k)$. Otherwise $q_p^{16/t_p} > m$, i.e. $t_p < 16 \log q_p / \log m$ and $a_p(k) < 32 \log q_p / \log m$; the q_p of these primes are powers of distinct primes whose product divides $k \leq m$, so $\sum \log q_p \leq \log m$ and these primes contribute less than 32 in total. Hence $S_0(k) < 2B(k) + 32$ and $(S_0(k) - 64)^+ \leq 2(B(k) - 16)^+$; sum over $k \leq m$. $\square$

Prefixes at threshold 16

Order all retained levels by decreasing value t , breaking ties by (p, j) . Call $k \leq m$ *hot* if $B(k) > 16$. For hot k list its effective levels—at each prime with $b_p(k) > 0$, the largest retained level dividing k —in this order, with values $a_1 \geq \dots \geq a_s$ and partial sums s_i ; let $P_i(k)$ be the product of the first i moduli and

$$w_{k,i} := |[s_{i-1}, s_i) \cap (2, 3]|, \quad \text{so that} \quad \sum_i w_{k,i} = 1.$$

A *carrier* is a $P = P_i(k)$ with $w_{k,i} > 0$ for some hot k . Its levels are the p -parts of P , listed in the fixed order, so its *mass* $\mu(P) := B(P) = s_i$, its *last value* $\theta_P := a_i$ (a dyadic number) and s_{i-1} depend only on P ; positivity of $w_{k,i}$ gives $s_i > 2$ and $s_{i-1} < 3$, hence $\mu(P) \in (2, 4)$ and $w_{k,i} \leq \theta_P$. With $d_k := B(k) - 16$ put

$$M(P) := \sum_{k \text{ hot}, i: P_i(k)=P} d_k w_{k,i}, \quad \text{so} \quad \sum_P M(P) = L_B, \quad K_P := \lfloor m/P \rfloor, \quad c_P := \frac{M(P)}{K_P}.$$

Lemma 14.3 (hinge). *For $x_1, \dots, x_N \in [0, 1]$ and an integer $C \geq 0$, $(\sum_i x_i - C)^+ \leq e_{C+1}(x_1, \dots, x_N)$.*

Proof. With the other coordinates fixed, e_{C+1} has the form $A + (u+v)D + uvE$ in two coordinates u, v , where $E \geq 0$ is e_{C-1} of the remaining coordinates ($E = 0$ if $C = 0$). Moving u, v at fixed sum until one of them reaches 0 or 1 decreases uv , so it does not increase e_{C+1} , and it leaves the left side unchanged. Each move reduces the number of coordinates in $(0, 1)$, so finitely many moves lead to a vector with $q = \lfloor \sum_i x_i \rfloor$ ones, one coordinate $\tau \in [0, 1)$ and zeros, at which $e_{C+1} = \binom{q}{C+1} + \tau \binom{q}{C}$. This is ≥ 0 if $q < C$, equals τ if $q = C$, and is $\geq q - C + \tau$ if $q > C$ (fix C of the q ones and adjoin each of the other $q - C$), which is the hinge in each case. $\square$

Lemma 14.4 (moment bound). *For all integers $N, r \geq 1$, $\sum_{j \leq N} e_r((b_p(j))_p) \leq N H_B^r / r!$.*

Proof. $b_p(j) = \sum_{\text{retained } q|j} \text{inc}_q$, the increments telescoping. Expanding e_r chooses r distinct primes and one retained level at each; the moduli are coprime, so $\sum_{j \leq N} [\prod q \mid j] = \lfloor N / \prod q \rfloor \leq N / \prod q$. Hence the sum is at most $N e_r(h)$ for $h = (\text{inc}_q/q)_q$, and $r! e_r(h) \leq (\sum h)^r = H_B^r$. $\square$

Lemma 14.5 (carried mass). *Every carrier satisfies $P < m^{1/4}$, so $K_P \geq 1$. If $\theta = \theta_P$ and $r := 12/\theta + 1$, then*

$$M(P) \leq \varepsilon(\theta) \frac{m}{P} \quad \text{and} \quad c_P \leq 2\varepsilon(\theta), \quad \text{where} \quad \varepsilon(\theta) := \theta^{2-r} \frac{H_B^r}{r!}. \quad (25)$$

Proof. Summing $\log q \leq \frac{t}{16} \log m$ over the levels of P gives $\log P \leq \frac{\mu(P)}{16} \log m < \frac{1}{4} \log m$. Let $k = Pj$ be hot with $P_i(k) = P$. At the primes of P the effective level of k is the level in P ; at a prime $p \nmid P$ we have $b_p(j) = b_p(k) \leq \theta$, because these levels come after position i in the decreasing order. Hence $B(k) = \mu(P) + \sum_{p \nmid P} b_p(j)$ and, as $\mu(P) < 4$,

$$d_k \leq \left(\sum_{p \nmid P} b_p(j) - 12 \right)^+ \leq \theta^{1-r} e_r((b_p(j))_{p \nmid P}) \leq \theta^{1-r} e_r((b_p(j))_p),$$

by Lemma 14.3 applied to $x_p = b_p(j)/\theta \in [0, 1]$ and $C = 12/\theta$ (an integer, θ being dyadic). Multiply by $w_{k,i} \leq \theta$ and sum over the hot k with $P_i(k) = P$: the map $k \mapsto j$ is injective and $j \leq K_P$, so Lemma 14.4 gives

$$M(P) \leq \theta^{2-r} \sum_{j \leq K_P} e_r((b_p(j))_p) \leq \theta^{2-r} K_P \frac{H_B^r}{r!} \leq \varepsilon(\theta) \frac{m}{P},$$

and $c_P = M(P)/K_P \leq 2\varepsilon(\theta)$ because $\lfloor u \rfloor \geq u/2$ for $u \geq 1$. □

Carriers through a point

For a positive integer n and a dyadic θ put

$$T_\theta(n) := \sum_p \min(b_p(n), \theta), \quad N_\theta(n) := \#\{p : b_p(n) \geq \theta\}, \quad \text{so that} \quad \theta N_\theta(n) \leq T_\theta(n).$$

Lemma 14.6 (carriers with a given last level). *Let q be a retained level of value θ with $q \mid n$. The number of carriers $P \mid n$ with last level q is at most $2^{(3+T_\theta(n))/\theta}$.*

Proof. Such a P is q times a set of retained levels at distinct other primes, each dividing n , each of value at least θ (they precede q in the order), of total value less than 3; by unique factorisation P determines this set. Values $\geq \theta$ are of the form $\theta 2^i$, $i \geq 0$, and at one prime they are distinct, so the sum of $2^{-\text{value}/\theta}$ over the levels available at one prime is at most $\sum_{i \geq 0} 2^{-2^i} \leq \sum_{i \geq 0} 2^{-i-1} = 1$. The generating sum of $2^{-\text{total}/\theta}$ over all choices of at most one available level at each prime is therefore at most $2^{N_\theta(n)}$, and every admissible choice contributes at least $2^{-3/\theta}$; so there are at most $2^{3/\theta + N_\theta(n)} \leq 2^{(3+T_\theta(n))/\theta}$ of them (dropping the order restriction only enlarges the count). □

Lemma 14.7 (values at a point). *For every n , $\sum_{\text{retained } q \mid n} t_q \leq 2B(n)$.*

Proof. At one prime the retained levels dividing n have distinct dyadic values at most $b_p(n)$, so their sum is at most $b_p(n)(1 + \frac{1}{2} + \frac{1}{4} + \dots) = 2b_p(n)$; sum over p . □

Lemma 14.8 (a uniform numerical bound). *For $\theta = 2^{-h}$, $h \geq 0$, and $H_B \leq \frac{17}{16}$,*

$$G(\theta) := \frac{\varepsilon(\theta)}{\theta} 2^{23/\theta} \leq \frac{17}{416}. \tag{26}$$

Proof. With $t = 1/\theta \in \mathbb{N}$ and $r = 12t + 1$ we have $G(\theta) = t^{12t} H_B^{12t+1} 2^{23t} / (12t + 1)!$, and $(12t)! \geq (12t/e)^{12t}$ (from $e^n \geq n^n/n!$) gives

$$G(\theta) \leq \frac{H_B}{12t + 1} \left(2^{23} \left(\frac{e H_B}{12} \right)^{12} \right)^t.$$

Since $e < \frac{11}{4}$ (sum the exponential series through $1/4!$ and bound the tail by $\frac{1}{120} \cdot \frac{6}{5}$, giving $e < \frac{65}{24} + \frac{1}{100} < \frac{11}{4}$) and $H_B \leq \frac{17}{16}$, we get $eH_B/12 < \frac{187}{768} < \frac{1}{4}$ and so $2^{23}(eH_B/12)^{12} < 2^{23} \cdot 2^{-24} = \frac{1}{2}$ (exactly, $2^{23}(187/768)^{12} = 1828518162230556187140793681/5019318332045454244023631872 < \frac{1}{2}$). Hence $G(\theta) \leq 17/(16(12t+1)2^t) \leq \frac{17}{416}$, the right side decreasing in t . $\square$

The certificate

For a carrier P with last value θ_P and a positive integer n put

$$U_P(n) := \sum_{p \nmid P} \min(b_p(n), \theta_P), \quad F(n) := 3 \sum_P c_P [P \mid n] \left(1 - \frac{U_P(n)}{16}\right). \quad (27)$$

Lemma 14.9 (pointwise feasibility). $F(n) \leq (B(n) - 1)^+$ for every positive integer n .

Proof. If $P \mid n$ then $b_p(n) \geq b_p(P) \geq \theta_P$ for every $p \mid P$, so $B(n) \geq \mu(P) > 2$; thus $F(n) = 0$ when $B(n) \leq 2$. Let $B(n) > 2$. For $P \mid n$ each prime of P contributes exactly θ_P to $T_{\theta_P}(n)$, so $U_P(n) = T_{\theta_P}(n) - \theta_P \omega(P)$ with $\theta_P \omega(P) \leq \mu(P) < 4$. If the bracket in (27) is positive then $U_P(n) < 16$, hence $T_{\theta_P}(n) < 20$; and the bracket is at most 1. Discard the nonpositive terms and group the remaining carriers by their last level q , a retained level dividing n of value θ_q . By (25), Lemma 14.6 with $T_{\theta_q}(n) < 20$, (26) and Lemma 14.7,

$$F(n) \leq 3 \sum_{q \mid n} 2\varepsilon(\theta_q) 2^{23/\theta_q} = 6 \sum_{q \mid n} \theta_q G(\theta_q) \leq \frac{102}{416} \sum_{q \mid n} t_q \leq \frac{51}{104} B(n) \leq \frac{B(n)}{2} \leq B(n) - 1. \quad \square$$

A prime of P may have $b_p(n) > b_p(P)$ at a multiple n of P ; the cap at θ_P in U_P makes this irrelevant, and this is why the primes of P are excluded from U_P .

Lemma 14.10 (value on the window). F is a finite signed divisor sum $\sum_D c'_D [D \mid \cdot]$ whose positive moduli are carriers $P < m^{1/4}$ and whose negative moduli have the form $Pq < m^{5/16}$; its coefficients depend only on the atom system and m ; and

$$\sum_{n \in I} F(n) \geq \frac{141}{64} L_B.$$

Proof. For a dyadic θ the function $\min(b_p(n), \theta)$ telescopes over the retained levels at p : $\min(b_p(n), \theta) = \sum_{q \mid n} \beta_q^\theta$ with $\beta_q^\theta := \min(t_q, \theta) - \min(t_{q^-}, \theta) \geq 0$, where q^- is the previous retained level at p (value 0 if there is none); moreover $\beta_q^\theta \leq \text{inc}_q$, so $H_\theta := \sum_q \beta_q^\theta / q \leq H_B$. Since the primes in U_P lie outside P , $[P \mid n] U_P(n) = \sum_{p \nmid P} \sum_q \beta_q^{\theta_P} [Pq \mid n]$ with $\gcd(P, q) = 1$, and every retained q has $q \leq m^{t_q/16} \leq m^{1/16}$, so $Pq < m^{5/16} \leq m$. For $d \leq m$ every window of m consecutive integers satisfies $\lfloor m/d \rfloor \leq N_I(d) \leq m/d + 1 \leq 2m/d$. The positive part of the term of P sums to $3c_P N_I(P) \geq 3c_P K_P = 3M(P)$; its negative part is at most

$$\frac{3}{16} c_P \sum_q \beta_q^{\theta_P} N_I(Pq) \leq \frac{3}{16} \cdot \frac{M(P)}{K_P} \cdot \frac{2m}{P} \sum_q \frac{\beta_q^{\theta_P}}{q} \leq \frac{3}{16} \cdot 4M(P) H_{\theta_P} \leq \frac{3}{4} H_B M(P),$$

using $K_P \geq m/(2P)$. Summing over the carriers, $\sum_I F \geq 3(1 - H_B/4) \sum_P M(P) \geq 3(1 - \frac{17}{64}) L_B = \frac{141}{64} L_B$. Bounding the terms before combining equal moduli is legitimate, $\sum_I F$ being linear in the terms. $\square$

Proof of Theorem 14.1. By Lemmas 14.2, 14.9 and 14.10, $R \geq R_B \geq \sum_{n \in I} F(n) \geq \frac{141}{64} L_B \geq \frac{141}{128} L$. $\square$

Theorem 14.11 (linear bound). *For every weight z , every $m \geq 1$ and every $x \geq 0$ the hinge inequality (22) with threshold 65 holds, and*

$$g(n) \leq 81n \quad \text{for every } n \geq 1.$$

Proof. Theorem 14.1 supplies the hypothesis of Corollary 13.3. $\square$

Remark 14.12. (a) Theorem 14.1 does not use the support condition $p^j \leq m/64$, so it holds for every finitely supported atom system with $H < \frac{17}{16}$. Its binding numerical step is $eH_B/12 < \frac{1}{4}$, i.e. $H_B < 3/e \approx 1.1036$, so the argument tolerates a mean slightly above $\frac{17}{16}$. (b) The certificate uses the window only through $|I| = m$ and the two-sided counts of multiples, but its value bound rests on Lemma 14.4, which uses $\lfloor N/d \rfloor \leq N/d$ for coprime moduli; this is the arithmetic input that Proposition 13.5 shows to be necessary, and it fails in the projective-plane configuration there. (c) The nonnegativity of the coefficients is essential in the barriers for window-adapted certificates (repository, rounds 13–14): those arguments force coefficients to vanish through inequalities of the form $\sum_{E|D} c_E \leq 0$, which do not constrain signed coefficients. (d) Theorem 13.7 and Corollary 13.8 remain of interest as a proof of the sparse core by a nonnegative certificate in the range $m \leq 10^{2887}$. (e) The threshold 65 and the constant 81 are not optimised; Conjecture 8.1 (threshold 2, which would give $18n$) and Erdős’s $2n$ remain open.

Formalisation

The proof of Theorem 14.11 has been formalised in Lean 4 (toolchain `v4.34.0-rc1`, Mathlib commit `de5ce8a9`) and is distributed with the repository (directory `lean/`). The development is statement-first: the definitions of this section (dyadic rounding, retained levels, carriers, the certificate F) and the objects of Sections 5, 8 and 13 are Lean definitions; each lemma of this section and each of Lemmas 13.1, 8.3, 8.4, 8.5, Theorems 1.3, 13.2 and the threshold-65 variant of Theorem 8.2 is a separate theorem with its own proof file; and the final statement

`Erdos708Final.g.le_81n` : for every $n \geq 1$, every set A of n integers ≥ 2 and every x , some $B \subseteq \{x+1, \dots, x+\max A\}$ with $|B| \leq 81n$ has $\prod A \mid \prod B$

is derived from them. The kernel’s axiom audit (`#print axioms`) reports only `propext`, `Classical.choice` and `Quot.sound`; no `sorry`, `native.decide` or custom axiom occurs. The Lean text was written by GPT-6 Astra in three sessions (about 160 minutes in total) from the statements of this paper and recompiled independently by the orchestrating agent; the sparse-core theorem and three of its lemmas were also verified by the Prove2Me platform under a different Mathlib version. The formalisation does not use the support condition $p^j \leq m/64$ in Theorem 14.1, in accordance with Remark (a) above.

15 A smaller threshold: $g(n) \leq 33n$

The constants of Section 14 were chosen for simplicity. Tightening them along the same route gives the hinge inequality with the absolute threshold 17, and hence, through Theorem 8.2 with threshold 17 in place of 2 (the rounding cost of Lemma 8.4 adds $16n$), the bound $g(n) \leq 33n$. Throughout, an *atom system* is a finitely supported family $\alpha_{p,j} \geq 0$ (p prime, $j \geq 1$) with

$\sum_j \alpha_{p,j} \leq 1$ for every p , and $S(n) = \sum_{p,j} \alpha_{p,j} [p^j \mid n]$; this includes $S(n) = \sum_p \min(z_p v_p(n), 1)$ for every weight z .

Theorem 15.1 (threshold 17). *For every atom system, every $m \geq 1$ and every window I of m consecutive positive integers,*

$$\sum_{k \leq m} (S(k) - 17)^+ \leq \sum_{b \in I} (S(b) - 1)^+.$$

Consequently $g(n) \leq 33n$ for every $n \geq 1$.

Fix the constants

$$Q = 2^{16}, \quad H^* = \frac{1025}{1024}, \quad \gamma = \frac{65536}{65535}, \quad K = \frac{35}{16},$$

and for a system $f = \sum_p f_p$ with nonnegative prime-power increments write $H(f) = \sum_{p,j} (f_p(p^j) - f_p(p^{j-1}))/p^j$.

Lemma 15.2 (hinges and moments). *For $0 \leq u_i \leq 1$ and an integer $r \geq 1$, $(\sum_i u_i - (r-1))^+ \leq e_r(u)$ and $(\sum_i u_i - \frac{7}{2})^+ \leq \frac{1}{2}e_4(u)$. If f_p has nonnegative prime-power increments and $H = H(f)$, then $\sum_{n \leq N} e_r((f_p(n))_p) \leq NH^r/r!$. If $f_p(p^v) \leq g_p(p^v)$ for all p, v then $H(f) \leq H(g)$.*

Proof. The polynomials $e_r(u) - \sum u_i + r - 1$ and $e_4(u) - 2 \sum u_i + 7$ are multi-affine, hence on $[0, 1]^N$ convex combinations of their vertex values; at a vertex with j ones these are $\binom{j}{r} - j + r - 1 \geq 0$ and $\binom{j}{4} - 2j + 7 \geq 0$ (check $j \leq 4$; increasing afterwards). Together with $e_r \geq 0$ this gives both hinges. The moment bound is Lemma 14.4 (expand e_r into one level per prime; distinct primes give coprime moduli). For the last claim, $\sum_j (f_p(p^j) - f_p(p^{j-1}))/p^j = (1 - \frac{1}{p}) \sum_{v \geq 1} f_p(p^v)/p^v$; compare termwise. $\square$

Lemma 15.3 (large atoms). *Let S_0 keep only the atoms with $p^j \leq m/Q$. For every $c \geq 7$ and $k \leq m$, $(S(k) - c)^+ = (S_0(k) - c)^+$. Moreover the left side of Theorem 15.1 vanishes when $m < 2^{48}$.*

Proof. If a removed atom $p^j > m/Q$ divides $k \leq m$, write $k = p^{v_p(k)}u$; then $u < Q$, and since $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 > Q$, u has at most six prime factors, so $S(k) \leq \omega(k) \leq 7$ and both sides vanish. Otherwise $S(k) = S_0(k)$. The product of the first thirteen primes exceeds 2^{48} , so $S(k) \leq \omega(k) \leq 12$ when $m < 2^{48}$. $\square$

Henceforth $m \geq 2^{48}$ and $H := H(S_0)$.

Lemma 15.4 (dense branch). *If $H \geq H^*$ then Theorem 15.1 holds.*

Proof. Let $V = (H^*/H)S_0$, so $H(V) = H^*$ and $V \leq S_0$. For $d \leq m/Q$, $\lfloor m/d \rfloor \geq (1 - \frac{1}{Q})m/d$, whence $\sum_{k \leq m} V(k) \geq (1 - \frac{1}{Q})mH^*$; Lemma 15.2 gives $\sum_{k \leq m} (V(k) - 7)^+ \leq mH^{*8}/8!$, and $(1 - \frac{1}{Q})H^* - H^{*8}/8! > 1$ (indeed $(1 - \frac{1}{Q})H^* - 1 = 64511/2^{26} > 2^{-11}$ and $H^{*8}/8! < 1/20160$). Hence $\sum_{k \leq m} \min(V(k), 7) \geq m$ and, as $\min(S_0, 17) \geq \min(V, 7)$, $\sum_{k \leq m} \min(S_0(k), 17) \geq m$. The affine certificate $F(n) = S_0(n) - 1$ is pointwise at most $(S_0(n) - 1)^+$ and sums over I to at least $\sum_{k \leq m} S_0(k) - m \geq \sum_{k \leq m} (S_0(k) - 17)^+$; conclude with Lemma 15.3 and $S_0 \leq S$. $\square$

Lemma 15.5 (rounding with loss 6). *Assume $H < H^*$. For dyadic $t = 2^{-h}$ let $q_p(t)$ be the least prime power p^j at which the p -component of S_0 is at least t , and retain $(q_p(t), t)$ exactly when $q_p(t)^{3/t} \leq m$. Let $b_p(n)$ be the largest retained t with $q_p(t) \mid n$ (0 if none) and $B = \sum_p b_p$. Then $B \leq S_0$, $H(B) \leq H$, $S_0(k) \leq 2B(k) + 6$ for $k \leq m$, and with $L_B := \sum_{k \leq m} (B(k) - \frac{11}{2})^+$,*

$$\sum_{k \leq m} (S_0(k) - 17)^+ \leq 2L_B.$$

All atoms of b_p , and of $\min(b_p, \theta)$, have moduli at most $m^{1/3}$.

Proof. As in Lemma 14.2: a p -component of S_0 at k with rounded value t is less than $2t$; if the level is retained it is at most $2b_p(k)$; otherwise $q_p(t) \mid k$ and $t < 3 \log q_p(t) / \log m \leq 3 v_p(k) \log p / \log m$, and these contributions sum to at most $6 \log k / \log m \leq 6$. Retained moduli satisfy $q \leq m^{t/3} \leq m^{1/3}$. $\square$

Lemma 15.6 (carriers of mass $1 + \theta$). *For every hot $k \leq m$ (i.e. $B(k) > \frac{11}{2}$) order its nonzero effective levels by decreasing value, ties by increasing prime, and let the carrier C of k be the shortest prefix of mass exceeding 1; write P_C for the product of its moduli and θ_C for its last value. Then $\mu(C) = 1 + \theta_C$ and $P_C \leq m^{(1+\theta_C)/3} \leq m^{2/3}$. Let M_C be the sum of $B(k) - \frac{11}{2}$ over the k with carrier C , $\nu_C = \lfloor m/P_C \rfloor$ and $c_C = M_C/\nu_C$; then $\sum_C M_C = L_B$. With $\varepsilon_1 = \gamma H^{*4}/(2 \cdot 4!)$ and, for dyadic $\ell \geq 2$, $r_\ell = 9\ell/2$ and $\varepsilon_\ell = \gamma \ell^{r_\ell-1} H^{*r_\ell}/r_\ell!$, one has $c_C \leq \varepsilon_\ell$ whenever $\theta_C = 1/\ell$.*

Proof. Every selected value is an integer multiple of θ_C , the mass before the last level is at most 1 and the total exceeds 1, so the total is exactly $1 + \theta_C$; the bound on P_C follows from the retention rule. For $k = P_C u$ with carrier C , every unselected prime has $b_p(k) = b_p(u) \leq \theta_C$. If $\theta_C = 1$ the residual excess is $(\sum_{p \nmid P_C} b_p(u) - \frac{7}{2})^+ \leq \frac{1}{2} e_4$; if $\theta_C = 1/\ell \leq \frac{1}{2}$ it is $\theta_C (\sum u_p - (r_\ell - 1))^+$ with $u_p = b_p(u)/\theta_C \in [0, 1]$, since $(\frac{11}{2} - 1 - \frac{1}{\ell})\ell = r_\ell - 1$; apply Lemma 15.2, sum over $u \leq \nu_C$ (the map $k \mapsto u$ is injective) and use the moment bound with $H(B) \leq H < H^*$. $\square$

Lemma 15.7 (the certificate and the counting reduction). *Let $\delta_{p,j}^{(\theta)} = \min(b_p(p^j), \theta) - \min(b_p(p^{j-1}), \theta)$, $U_C(n) = \sum_{p \nmid P_C} \min(b_p(n), \theta_C)$ and*

$$F(n) = 24 \sum_C c_C [P_C \mid n] \left(1 - \frac{U_C(n)}{K}\right) = \sum_{d \mid n} \lambda_d, \quad \lambda_d = 24 \sum_{C: P_C=d} c_C - \frac{24}{K} \sum_{\substack{C, p \nmid P_C, j \\ P_C p^j = d}} c_C \delta_{p,j}^{(\theta_C)}.$$

Every negative modulus is at most m . For dyadic ℓ put $C_\ell(N) = [X^{\ell+1}](1 + X + X^2 + X^4 + \dots + X^\ell)^N$ and

$$R_\ell = \max_{0 \leq N \leq \lfloor (K+1)\ell+1 \rfloor} \frac{C_\ell(N) (1 - \max(0, \frac{N-\ell-1}{\ell})/K)^+}{\max(\frac{1}{\ell}, \frac{N-\ell}{\ell})}.$$

Then $F(n) \leq (24 \sum_{\ell=1,2,4,\dots} \varepsilon_\ell R_\ell) (B(n) - 1)^+$ for every positive integer n .

Proof. Negative moduli are $P_C p^j \leq m^{2/3} m^{1/3}$. If $B(n) \leq 1$ no carrier divides n . Fix $\theta = 1/\ell$ and let N be the number of primes with $b_p(n) \geq \theta$: a carrier of last value θ dividing n selects at each such prime no level or one level of scaled value in $\{1, 2, 4, \dots, \ell\}$ with total $\ell + 1$, and a set of choices determines at most one carrier, so there are at most $C_\ell(N)$ of them. For $P_C \mid n$, $U_C(n) = T_\theta(n) - \theta \omega(P_C) \geq \max(0, N\theta - 1 - \theta)$ with $T_\theta(n) = \sum_p \min(b_p(n), \theta)$, and $B(n) - 1 \geq \max(\theta, N\theta - 1)$. Drop the negative summands, use $c_C \leq \varepsilon_\ell$ and sum over the scales. $\square$

Lemma 15.8 (the numerical constant). $24 \sum_{\ell=1,2,4,\dots} \varepsilon_\ell R_\ell < \frac{4}{5}$. Consequently $F(n) \leq \frac{4}{5}(B(n) - 1)^+$ for every n .

Proof. For $\ell \leq 64$ the quantities $24\varepsilon_\ell R_\ell$ are bounded exactly (all 417 relevant values of N , the coefficients $C_\ell(N)$ from the recurrence $sg_s = \sum_{j \in \{1,2,4,\dots,\ell\}, j \leq s} ((N+1)j - s)g_{s-j}$ for $g = f^N$) by $\frac{51}{100}, \frac{6}{25}, \frac{3}{100}, \frac{1}{250}, \frac{1}{2000}, \frac{1}{16000}, \frac{1}{100000}$ at $\ell = 1, 2, 4, 8, 16, 32, 64$. For the tail put $f = \frac{2633}{2000}$, $E = \frac{87}{32} > e$ and $\zeta = (2EH^*/9)^{9/2}$: then $C_\ell(N) \leq 4^{\ell+1}f^N$, $\varepsilon_\ell \leq (\gamma/\ell)\zeta^\ell$ for $\ell \geq 2$, and the exact rational inequalities $4^{16}f^{51}(2EH^*/9)^{72} < 1$, $16f^4(2EH^*/9)^9 < \frac{9}{16}$ give $24\varepsilon_\ell R_\ell(N) \leq 96\gamma(\frac{3}{4})^\ell$ for $N < 2\ell$ and $\leq 192\gamma f/(K\ell^2)$ for $N \geq 2\ell$ (using $qf^{-q} \leq 2$ for the integer $q = (K+1)\ell + 1 - N$ when $\ell \geq 128$). Summing over $\ell = 128 \cdot 2^j$ gives less than $\frac{1}{100}$, and $\frac{317829}{400000} < \frac{4}{5}$. $\square$

Lemma 15.9 (value on the window). For every window I of m consecutive positive integers, $\sum_{b \in I} F(b) \geq \frac{62248}{30583} L_B$.

Proof. $N_I(P_C) \geq \nu_C$; every negative modulus is at most m , so $N_I(P_C q) \leq 2m/(P_C q)$; $m/P_C \geq m^{1/3} \geq 2^{16}$ gives $m/(P_C \nu_C) \leq \gamma$; and the mean of the increments of U_C is at most H^* . Hence $\sum_I F \geq 24(1 - 2\gamma H^*/K) \sum_C M_C = \frac{62248}{30583} L_B$. $\square$

Proof of Theorem 15.1. For $m < 2^{48}$ the left side vanishes (Lemma 15.3); for $H \geq H^*$ apply Lemma 15.4. Otherwise, by Lemmas 15.3, 15.5, 15.8 and 15.9,

$$\sum_{k \leq m} (S(k) - 17)^+ \leq 2L_B \leq \frac{30583}{31124} \sum_I F \leq \frac{30583}{38905} \sum_{b \in I} (B(b) - 1)^+ \leq \sum_{b \in I} (S(b) - 1)^+.$$

The bound $g(n) \leq 33n$ follows from Theorem 8.2 with threshold 17: the fractional cover value is at most $17n$ and rounding adds fewer than $16n$ (Lemmas 8.4 and 8.5; $2n$ suffice when $a_n \geq 8n^3$ by Theorem 1.3). $\square$

Formalisation

Theorem 15.1, its nine lemmas and the threshold-17 variant of the reduction chain are formalised in the repository (directory `lean/Erdoes708/H17`); the final theorem `Erdoes708H17.g.le_33n` depends only on `propext`, `Classical.choice` and `Quot.sound`. The proof of this section was found by GPT-6 (web interface, effort “Pro”) in a 34-minute run from the statements of Section 14, its constants were re-verified in exact arithmetic, it was refereed by a Claude Opus 5 agent, and the Lean text was written by GPT-6 Astra in a 116-minute session and recompiled independently by the orchestrator. The tightest numerical steps are $4^{16}f^{51}(2EH^*/9)^{72} < 1$ (0.6% margin) and the total $\frac{317829}{400000} < \frac{4}{5}$.

16 The rounding cost revisited: $g(n) \leq 19n$

The reductions of Sections 8, 14 and 15 pay $16n$ for the passage from the fractional cover to an integral one (Lemmas 8.4 and 8.5); with the threshold 17 this rounding cost exceeds the threshold term itself. The bound $|\mathcal{P}| < 16n$ of Lemma 8.5 is crude: an input $a \leq m$ has at most one prime factor exceeding $\sqrt{m}$, so all but $\pi(\sqrt{m})$ of the primes of $\mathcal{P}$ are “large” and there are at most n of those. Combined with the k -split theorem for $k = 3$, which disposes of $m \geq 9n^2$ outright, this replaces the cost $16n$ by $2n$. We state the reduction for an arbitrary real threshold.

Theorem 16.1 (the $(c+2)n$ reduction). *Let $c \geq 1$ be real and suppose that for every atom system, every $m \geq 1$ and every window I of m consecutive positive integers*

$$\sum_{k \leq m} (S(k) - c)^+ \leq \sum_{b \in I} (S(b) - 1)^+. \quad (28)$$

Then $g(A, x) \leq \lceil cn \rceil + 2n$ for every A with $|A| = n$ and every $x \geq 0$; in particular $g(n) \leq (c+2)n + 1$, and $g(n) \leq (c+2)n$ whenever cn is an integer.

Corollary 16.2. $g(n) \leq 19n$ for every $n \geq 1$.

Proof. Theorem 15.1 is (28) with $c = 17$. □

The proof of Theorem 16.1 uses three elementary lemmas. As before, $m = a_n$, $I = \{x + 1, \dots, x + m\}$, $\mathcal{P}$ is the set of primes dividing $\prod A$, and $\tau^* = \tau^*(A, x)$ is the value of the linear programme (4).

Lemma 16.3 (counting the primes). $|\mathcal{P}| \leq n + \pi(\sqrt{m})$.

Proof. If $p \in \mathcal{P}$ and $p > \sqrt{m}$, choose $a \in A$ with $p \mid a$. Two distinct primes exceeding $\sqrt{m}$ cannot both divide $a \leq m$, since their product exceeds m ; so a determines p , and the primes of $\mathcal{P}$ exceeding $\sqrt{m}$ number at most $|A| = n$. The remaining primes of $\mathcal{P}$ are at most $\sqrt{m}$. □

Lemma 16.4 (strict rounding). *There is a set $B \subseteq I$ with $\prod A \mid \prod B$ and $|B| < \tau^* + |\mathcal{P}|$.*

Proof. Let y^* be an optimal vertex of (4) and $B = \{b : y_b^* > 0\}$ as in Lemma 8.4; then $\prod A \mid \prod B$. Let $F = \{b : 0 < y_b^* < 1\}$, so that $|F| \leq |\mathcal{P}|$ by the vertex argument of Lemma 8.4. Then

$$|B| - \tau^* = \#\{b : y_b^* = 1\} + |F| - \left(\#\{b : y_b^* = 1\} + \sum_{b \in F} y_b^* \right) = \sum_{b \in F} (1 - y_b^*),$$

which is $< |F| \leq |\mathcal{P}|$ if $F \neq \emptyset$ and equals $0 < |\mathcal{P}|$ if $F = \emptyset$ (note $|\mathcal{P}| \geq 1$ because every $a \in A$ exceeds 1). □

Lemma 16.5 (primes up to $3n$). $\pi(3n) \leq n + 1$ for every $n \geq 1$.

Proof. Exactly n of the integers in $[1, 3n]$ are coprime to 6: each block of six consecutive integers contains two of them, and for odd n the final half-block $\{3n - 2, 3n - 1, 3n\}$ contains exactly one, namely $3n - 2$. Every prime other than 2 and 3 is among these n integers, and so is 1, which is not prime. Hence $\pi(3n) \leq (n - 1) + 2$. □

Proof of Theorem 16.1. If $m \geq 9n^2$, Theorem 1.4 with $k = 3$ gives a cover of size at most $3n \leq \lceil cn \rceil + 2n$. Otherwise $\sqrt{m} < 3n$, so $|\mathcal{P}| \leq n + \pi(3n) \leq 2n + 1$ by Lemmas 16.3 and 16.5. As in the proof of Theorem 8.2 (with c in place of 2), (28) applied to the maximiser z of Lemma 8.3 gives $\tau^* \leq cn$: indeed $\sum_{a \in A} w_z(a) \leq cn + \sum_{k \leq m} (w_z(k) - c)^+ \leq cn + \sum_{b \in I} (w_z(b) - 1)^+$, and the left side minus the last sum is the dual objective, which equals τ^* . Lemma 16.4 now gives an integer $|B| < cn + 2n + 1$, i.e. $|B| \leq \lceil cn \rceil + 2n$. □

Theorem 16.6 (asymptotic refinement). $g(n) \leq 18n + \pi((18n)^{19/34}) - 1 = 18n + O(n^{19/34} / \ln n)$ for every $n \geq 1$.

Proof. Theorem 1.4 with $k = 18$: if $m^{17} \geq (18n)^{19}$ then $18n$ elements suffice, and $18n$ is at most the displayed bound since $\pi(t) \geq 1$ for $t \geq 2$. Otherwise $\sqrt{m} < (18n)^{19/34}$, and Lemmas 16.3 and 16.4 with $\tau^* \leq 17n$ give $|B| < 17n + n + \pi((18n)^{19/34})$. The error term is $O(t/\ln t)$ with $t = (18n)^{19/34}$ by Chebyshev's bound. $\square$

The coincidence with the conditional bound $18n$ of Theorem 8.2 is not an error: that theorem spends $2n$ on the threshold and $16n$ on rounding, whereas Theorem 16.6 spends $17n$ and $n + o(n)$.

The rounding step cannot be improved to a sublinear loss for the linear programme (4) as it stands:

Proposition 16.7 (the integrality gap is linear). *For every $n \geq 1$ and every integer $K \geq 2$ there is an instance (A, x) with $|A| = n$, $\tau^*(A, x) = n/K$ and $g(A, x) = n$.*

Proof. Choose n primes $p_1 < \dots < p_n$ in an interval $[Q, \frac{4}{3}Q]$ with $Q \geq 6n$ (such an interval exists because $\sum 1/p$ diverges while $\sum_j (n-1)(3/4)^j$ converges), and let $A = \{p_1, \dots, p_n\}$, $m = p_n$, $t_i = \lfloor m/2 \rfloor + i$ for $i = 1, \dots, n$. Then $m - p_1 < t_i \leq p_1$ for all i , because $p_1 > \frac{3}{4}m$ and $t_i \leq m/2 + n \leq \frac{5}{6}Q < p_1$. Let $x \equiv p_i^K - t_i \pmod{p_i^{K+1}}$ for all i (Chinese remainder theorem). Then $v_{p_i}(x + t_i) = K$ exactly, and $x + t_i$ is the only multiple of p_i in I since $t_i - p_i \leq 0$ and $t_i + p_i > m$. Every cover must contain the n elements $x + t_i$, so $g(A, x) = n$. The fractional cover $y_{x+t_i} = 1/K$ is feasible with value n/K , and the dual weights $z_{p_i} = 1/K$ have $w_z(x + t_i) = 1$ and $w_z = 0$ elsewhere on I , so Lemma 8.3 gives $\tau^* \geq n/K$. $\square$

Thus a uniform bound $g(A, x) \leq \tau^* + o(n)$, or $g(A, x) \leq \alpha\tau^* + C$ with fixed α, C , is false; whether $g(A, x) \leq \tau^* + n$ holds for every instance is open, and with Theorem 15.1 it would give $g(n) \leq 18n$. The instances of Proposition 16.7 have $g(A, x) = n \leq 2n$, so they do not bear on Erdős's conjecture.

Provenance and formalisation

Lemmas 16.3–16.5, Theorems 16.1 and 16.6 and Proposition 16.7 were found by GPT-6 Astra (OpenAI, through the codex CLI) in a fifteen-minute run on 2026-09-07, refereed by a Claude Opus 5 subagent (verdict: no mathematical error; presentational repairs) and checked by the authors' exact-arithmetic scripts (the instance of Proposition 16.7 with $n = 2$, $K = 3$, $A = \{13, 17\}$, $x = 2274114691$ has $g(A, x) = 2$ and $\tau^* = 2/3$ by the dynamic programme and the exact dual). The reduction is formalised in Lean 4 (v4.34.0-rc1, Mathlib de5ce8a9) in the modules `Erdos708/H17/Chain19/` (`Primes`, `Rounding`, `Packing`, `Ksplit`, `Bound`, `Results`; written by GPT-6 Astra in fourteen minutes, recompiled from scratch by the authors): the theorems `primes_card_le`, `pi_le`, `strict_rounding`, `ksplit3` ($k = 3$ in Theorem 1.4), `bound_of_hinge` (Theorem 16.1 for real $c \geq 1$) and `g_le_19n`, whose statement is literally that of `Erdos708H17.g_le_33n` of Section 15 with $19n$ in place of $33n$, depend only on the axioms `propext`, `Classical.choice`, `Quot.sound`: for every $n \geq 1$, every $A \subseteq \mathbb{N}$ with $|A| = n$ and $\min A \geq 2$, and every x , there is $B \subseteq [x + 1, x + \max A]$ with $|B| \leq 19n$ and $\prod A \mid \prod B$. So does the conditional `g_le_12n_of_hinge`, which derives $g(n) \leq 12n$ from the hinge inequality with threshold $\frac{97}{10}$ (a candidate proof of that inequality is under review at the time of writing). Theorem 16.6 and Proposition 16.7 are not formalised.

17 Threshold $\frac{97}{10}$: $g(n) \leq 12n$

The proof of Section 15 loses a factor 2 and an additive 6 in the dyadic rounding, and its residual hinge has the fixed order $\frac{7}{2}$. Rounding to four mantissas per dyadic band (ratio $\frac{5}{4}$), choosing the moment order separately for each carrier mass, and replacing the large- m proviso by an arithmetic property of the hot points (a carrier's cofactor has at least four prime factors, hence is at least 210) bring the threshold down to $\frac{97}{10}$. Through Theorem 16.1 this gives $g(n) \leq 12n$.

Theorem 17.1 (threshold $\frac{97}{10}$). *For every atom system, every $m \geq 1$ and every window I of m consecutive positive integers,*

$$\sum_{k \leq m} (S(k) - \frac{97}{10})^+ \leq \sum_{b \in I} (S(b) - 1)^+.$$

Corollary 17.2. $g(n) \leq 12n$ for every $n \geq 1$.

Proof. Theorem 16.1 with $c = \frac{97}{10}$ gives $g(n) \leq \lceil \frac{97}{10}n \rceil + 2n \leq 10n + 2n$. $\square$

Fix the constants

$$Q = 2^{16}, \quad H^* = \frac{1025}{1024}, \quad \Gamma = \frac{211}{210}, \quad K = \frac{8}{3}, \quad \lambda = \frac{51}{10}, \quad \rho = \frac{5}{4}, \quad D_0 = \frac{47}{16}, \quad T = \frac{541}{100},$$

so that $D_0 + \rho T = \frac{97}{10} =: c$. As in Section 15, $H(f) = \sum_{p,j} \beta_{p,j}/p^j$ for a system $f = \sum_p f_p$ with nonnegative prime-power increments $\beta_{p,j}$, and e_r is the r -th elementary symmetric polynomial.

Lemma 17.3 (moments and optimised hinges). *(i) If each $f_p \leq 1$ then $\sum_{n \leq N} e_r((f_p(n))_p) \leq NH(f)^r/r!$, and pointwise domination of finite atom systems implies domination of H . (ii) Let $r \geq 2$ and $a \geq r - 1$ be real, $v = \lfloor ra/(r - 1) \rfloor$ and $A_r(a) = (v - a)/\binom{v}{r}$. Then $(\sum_i u_i - a)^+ \leq A_r(a) e_r(u)$ for all finite vectors $u_i \in [0, 1]$. In particular $(\sum_i u_i - 7)^+ \leq e_8(u)$.*

Proof. (i) is Lemma 15.2. (ii) The function $A_r(a)e_r(u) - (\sum u_i - a)^+$ is affine minus convex, hence concave, in each coordinate separately, so its minimum over the cube is attained at a vertex. At a vertex with h ones the claim reads $(h - a)^+ \leq A_r(a)\binom{h}{r}$; for $h < r$ the left side vanishes because $h \leq r - 1 \leq a$. For $h \geq r$ put $g(h) = (h - a)^+/\binom{h}{r}$; where both numerators are positive, $g(h + 1) \leq g(h)$ if and only if $h \geq ra/(r - 1) - 1$, so g is maximised at $h = v$ (a tie between adjacent integers does not affect the maximum), and $g(v) = A_r(a)$. For $r = 8$, $a = 7$ one gets $v = 8$ and $A_8(7) = 1$. $\square$

Lemma 17.4 (large atoms and the dense branch). *Let S_0 consist of the atoms $p^j \leq m/Q$ of S and $H = H(S_0)$. Then $(S(k) - c')^+ = (S_0(k) - c')^+$ for all $k \leq m$ and all $c' \geq 7$. If $H \geq H^*$ then Theorem 17.1 holds.*

Proof. If an omitted atom $q = p^j > m/Q$ divides $k \leq m$, the cofactor $k/q < Q = 65536 < 510510 = 2 \cdot 3 \cdots 17$ has at most six distinct prime factors, so $\omega(k) \leq 7$ and $S(k) \leq 7 \leq c'$, while $S_0(k) \leq S(k)$; both sides are 0. For the dense branch put $V = (H^*/H)S_0 \leq S_0$. Every retained modulus is at most m/Q , so $\sum_{k \leq m} V(k) \geq (1 - Q^{-1})mH^*$, and by Lemma 17.3 $\sum_{k \leq m} \min(V(k), 7) \geq m((1 - Q^{-1})H^* - H^{*8}/8!) > m$ (the constant is 1.00093...). Since $c > 7$, $\sum_{k \leq m} \min(S_0(k), c) \geq m$, hence $\sum_{k \leq m} (S_0(k) - c)^+ \leq \sum_{k \leq m} S_0(k) - m \leq \sum_{b \in I} S_0(b) - m \leq \sum_{b \in I} (S(b) - 1)^+$, using that every modulus has at least $\lfloor m/d \rfloor$ multiples in I . $\square$

From now on $H < H^*$ and $m \geq 2$ (for $m = 1$ the left side of the theorem is 0).

Lemma 17.5 (four-mantissa rounding, loss $\frac{47}{16}$). *Let $\mathcal{D} = \{1\} \cup \bigcup_{h \geq 3} \{\frac{4}{2^h}, \frac{5}{2^h}, \frac{6}{2^h}, \frac{7}{2^h}\}$ (adjacent levels have ratio at most $\frac{5}{4}$). For $t \in \mathcal{D}$ let $q_p(t)$ be the least power of p at which the cumulative p -part of S_0 reaches t , and retain (p, t) exactly when $q_p(t) \leq m^{\eta(t)}$, where $\eta(t) = \frac{1}{3}$ for $t > \frac{1}{2}$ and $\eta(t) = \frac{4t}{9}$ for $t \leq \frac{1}{2}$. Put $b_p(n) = \max(\{t : (p, t) \text{ retained}, q_p(t) \mid n\} \cup \{0\})$ and $B = \sum_p b_p$. Then $B \leq S_0$, $H(B) \leq H < H^*$, every atom of B has modulus at most $m^{1/3}$, and for $k \leq m$*

$$S_0(k) \leq \rho B(k) + D_0, \quad \text{hence} \quad \sum_{k \leq m} (S_0(k) - c)^+ \leq \rho L_B, \quad L_B := \sum_{k \leq m} (B(k) - T)^+.$$

Proof. Only finitely many levels are retained ($t \leq \frac{1}{2}$ and $q_p(t) \geq 2$ force $t \geq 9 \ln 2 / (4 \ln m)$), b_p is nondecreasing in the valuation, and $b_p \leq S_{0,p}$ by construction. Fix $k \leq m$ and $f_p = S_{0,p}(k) > 0$; round f_p down to the largest level $t_p \leq f_p$, so $f_p \leq \frac{5}{4} t_p$. Since the cumulative p -part reaches t_p at $p^{v_p(k)}$, $q_p(t_p) \leq p^{v_p(k)}$, so $q_p(t_p) \mid k$. If (p, t_p) is retained then $b_p(k) \geq t_p$ and $f_p - \rho b_p(k) \leq 0$. Otherwise $p^{v_p(k)} \geq q_p(t_p) > m^{\eta(t_p)}$, so $a_p := v_p(k) \ln p / \ln m > \eta(t_p)$, where $\sum_p a_p \leq 1$. If $t_p > \frac{1}{2}$ then $a_p > \frac{1}{3}$ and $f_p \leq 1$; there are at most two such primes, say $h \in \{0, 1, 2\}$. For every other unretained level $t_p < \frac{9}{4} a_p$ and $f_p < \frac{45}{16} a_p$. Hence $S_0(k) - \rho B(k) \leq h + \frac{45}{16}(1 - \frac{h}{3}) \leq \frac{47}{16}$. Finally $c - D_0 = \rho T$, so $(S_0(k) - c)^+ \leq \rho(B(k) - T)^+$. $\square$

Lemma 17.6 (carriers; no small- m exception). *Call $k \leq m$ hot if $B(k) > T$. Order its positive levels $b_p(k)$ nonincreasingly (ties by the prime) and let the carrier C be the shortest prefix of mass > 1 , recording primes, levels t_p and moduli $q_p(t_p)$; put $s_C = \sum_{p \in C} t_p$, $\theta_C = \min_{p \in C} t_p$, $P_C = \prod_{p \in C} q_p(t_p)$. Then $1 < s_C \leq 1 + \theta_C \leq 2$, every level of k outside C is at most θ_C , $P_C \leq m^{2/3}$, and $P_C q \leq m$ for every atom q of B . With $M_C = \sum_{k \leq m, \text{carrier}(k)=C} (B(k) - T)$, $\nu_C = \lfloor m/P_C \rfloor$ and $c_C = M_C/\nu_C$ (carriers with $M_C > 0$ only): $\sum_C M_C = L_B$, $\nu_C \geq 210$ and $m/(P_C \nu_C) < 1 + 1/\nu_C \leq \Gamma$.*

Proof. The mass bounds are those of Lemma 15.6. Exponent budget: if $\theta_C > \frac{1}{2}$ the carrier has exactly two primes with budget $\frac{1}{3}$ each. If no level exceeds $\frac{1}{2}$, the budget is $\frac{4}{9} s_C \leq \frac{4}{9}(1 + \theta_C) \leq \frac{2}{3}$. Otherwise there is exactly one level $a \geq \frac{5}{8}$; if $\theta_C \leq \frac{3}{8}$ the remaining mass is at most $1 + \frac{3}{8} - \frac{5}{8} = \frac{3}{4}$ and the budget at most $\frac{1}{3} + \frac{4}{9} \cdot \frac{3}{4} = \frac{2}{3}$; if $\frac{3}{8} < \theta_C \leq \frac{1}{2}$ every other level is at least $\frac{7}{16}$, and $\frac{9}{8} + \frac{7}{16} > 1$ leaves room for exactly one, with budget at most $\frac{1}{3} + \frac{2}{9} < \frac{2}{3}$. Thus $P_C \leq m^{2/3}$, and $q \leq m^{1/3}$ gives $P_C q \leq m$. At a source point k of C the mass outside C exceeds $T - s_C \geq T - 2 = \frac{341}{100} > 3$, and each prime contributes $b_p \leq 1$, so at least four distinct primes outside C divide k/P_C ; hence $k/P_C \geq 2 \cdot 3 \cdot 5 \cdot 7 = 210$, and $\nu_C \geq k/P_C \geq 210$ because $P_C \mid k \leq m$. $\square$

Lemma 17.7 (mass-dependent coefficients). *With $\theta = \theta_C$, $s = s_C$ and $a = (T - s)/\theta$,*

$$c_C \leq \epsilon_{\theta,s} := \min_{2 \leq r \leq \lfloor a \rfloor + 1} \theta A_r(a) \frac{(H^*/\theta)^r}{r!}.$$

Proof. Write a source point as $k = P_C u$ with $u \leq \nu_C$. Outside C the valuations of k and u agree and the levels are at most θ , so $M_C \leq \sum_{u \leq \nu_C} (\sum_{p \notin C} \min(b_p(u), \theta) - (T - s))^+$. Divide the capped functions by θ (each p -part is then at most 1, and the mean is at most H^*/θ) and apply Lemma 17.3(ii) then (i); divide by ν_C . No floor loss occurs here: the sum runs over the actual integers $u \leq \nu_C$. $\square$

Lemma 17.8 (the certificate and the counting reduction). *Put $U_C(n) = \sum_{p \notin C} \min(b_p(n), \theta_C)$ and*

$$F(n) = \lambda \sum_C c_C [P_C \mid n] \left(1 - \frac{U_C(n)}{K}\right),$$

a finite signed atom family whose negative moduli $P_C q$ are all at most m . Write a level $\theta \neq 1$ uniquely as j/L with $j \in \{4, 5, 6, 7\}$, $L \in \{8, 16, \dots\}$, and 1 as $(j, L) = (4, 4)$; let $G_{j,L}(X) = 1 + \sum_{t \in \mathcal{D}, t \geq j/L} X^{Lt}$ and $D_{j,L,d}(N) = [X^{L+d}](G_{j,L}(X)^N - (G_{j,L}(X) - X^j)^N)$, and

$$A_{j,L} = \max_{0 \leq N \leq \lfloor (K+1)L/j+1 \rfloor} \sum_{d=1}^j D_{j,L,d}(N) \epsilon_{j/L, 1+d/L} \frac{(1 - \max(0, N\theta - 1 - d/L)/K)^+}{\max(d/L, N\theta - 1)}.$$

Then $F(n) \leq \lambda(A_{4,4} + \sum_{L \geq 8} \sum_{j=4}^7 A_{j,L})(B(n) - 1)^+$ for every positive integer n .

Proof. Every level $\geq \theta = j/L$ is a multiple of $1/L$ (a level $j'/2^{h'}$ with $2^{h'} > L$ is at most $7/(2L) < 4/L \leq \theta$), so $s = 1 + d/L$ with $1 \leq d \leq j$. Fix n and let $N = \#\{p : b_p(n) \geq \theta\}$. A carrier dividing n with last level θ and mass $1 + d/L$ is determined by its level assignment (the modulus $q_p(t)$ depends only on (p, t)), which gives each of the N primes either nothing or a level $\geq \theta$, uses θ at least once, and has total degree $L + d$; such assignments are counted by $D_{j,L,d}(N)$, and ignoring unavailable levels only enlarges the count. For such a carrier $U_C(n) \geq \theta(N - |C|) \geq N\theta - s$ and $B(n) - 1 \geq \max(s - 1, N\theta - 1)$. For $N > \lfloor (K+1)L/j+1 \rfloor$ one has $N\theta - 1 - d/L > K$ and the numerator vanishes. Discard the negative terms of F , bound each c_C by Lemma 17.7 and sum; if $B(n) \leq 1$ no carrier divides n . $\square$

Lemma 17.9 (the finite scales). $A_{4,4} < \frac{24680}{10^6}$ and $\sum_{j=4}^7 A_{j,L} < \frac{128711}{10^6}, \frac{25773}{10^6}, \frac{5665}{10^6}, \frac{1025}{10^6}, \frac{150}{10^6}, \frac{21}{10^6}, \frac{3}{10^6}$ for $L = 8, 16, \dots, 512$; the exact sum of these 29 scales lies in $[186024001631, 186024001632] \cdot 10^{-12} < \frac{187}{1000}$.

Proof. Exact integer coefficient extraction in $G_{j,L}^N - (G_{j,L} - X^j)^N$ through degree $L + j$, over all admissible N and all moment orders r in Lemma 17.7, in rational arithmetic (script `exact_verifier.py`; the maximising N and the chosen r for every mass are recorded in `constants.json`). $\square$

Lemma 17.10 (the tail $L \geq 1024$). For $z > 1$ and $u_i \in [0, 1]$, $z^{\sum u_i} \leq \prod_i (1 + (z - 1)u_i)$, whence, with $y^+ \leq z^y/(e \ln z)$ and the proof of Lemma 17.7, $c_C \leq \frac{\theta z}{e \ln z} (e^{(z-1)H^*} z^{1-T})^{1/\theta}$. With $z = \frac{9}{2}$, $a_0 = \frac{4377}{100000}$ and the exact comparisons $e < \frac{87}{32}$, $(\frac{87}{32})^3 < (\frac{9}{2})^2$, $e > \frac{8}{3}$ (so $e \ln \frac{9}{2} > 4$), $e^{7H^*/2} < \frac{133}{4}$ and $(\frac{133}{4})^{100} < a_0^{100} (\frac{9}{2})^{441}$, this gives the uniform bound $c_C \leq \frac{9\theta}{8} a_0^{1/\theta}$, which may replace $\epsilon_{\theta,s}$ in $A_{j,L}$. Let $\mathcal{G}_j(x) = 1 + \sum_{a=j}^7 x^a + \sum_{r \geq 1} \sum_{a=4}^7 x^{a2^r}$ and take $(x_4, F_4) = (\frac{583}{1000}, \frac{632591}{500000})$, $(x_5, F_5) = (\frac{321}{500}, \frac{159089}{125000})$, $(x_6, F_6) = (\frac{69}{100}, \frac{1278827}{1000000})$, $(x_7, F_7) = (\frac{73}{100}, \frac{639137}{500000})$; then $\mathcal{G}_j(x_j) \leq F_j$, $F_j \geq \frac{5}{4}$ and $a_0^3 x_j^{-3j} F_j^{11} < 1$. With $D_j = \sum_{d=1}^j x_j^{-d}$, $\theta = j/L$ and $h = \frac{4}{3}$: every coefficient in $D_{j,L,d}(N)$ is at most $x_j^{-(L+d)} F_j^N$ (the exponents Lt , $t \in \mathcal{D}$, $t \geq j/L$, are among those of $\mathcal{G}_j$); for $N\theta - 1 \leq h$ the contribution to $A_{j,L}$ is at most $\frac{9j}{8} D_j (\frac{3}{4})^{L/j}$ (using $(\frac{4}{5})^4 < (\frac{3}{4})^3$), and for $N\theta - 1 > h$ it is at most $\frac{2187}{4096} \theta^2 F_j D_j$ (maximise $w F_j^{-w/\theta}$, use $e > \frac{8}{3}$ and $\ln F_j \geq \ln \frac{5}{4} > \frac{2}{9}$). Summing over $L = 1024 \cdot 2^r$, $r \geq 0$, with $t = (\frac{3}{4})^{146}$,

$$\sum_{L \geq 1024} \sum_{j=4}^7 A_{j,L} \leq \frac{9}{8} \left(\sum_{j=4}^7 j D_j \right) \frac{t}{1-t} + \frac{2187}{4096} \cdot \frac{4}{3 \cdot 1024^2} \sum_{j=4}^7 j^2 F_j D_j < \frac{3}{1000}.$$

Consequently $F(n) \leq \frac{51}{10} \cdot \frac{19}{100} (B(n) - 1)^+ = \frac{969}{1000} (B(n) - 1)^+ \leq (S(n) - 1)^+$.

Proof. All displayed comparisons are exact rational inequalities checked by the verifier; the exact tail bound is in $[2846921822, 2846921823] \cdot 10^{-12}$. The last inequality uses $B \leq S_0 \leq S$. $\square$

Lemma 17.11 (value on the window). $\sum_{b \in I} F(b) \geq \lambda(1 - 2\Gamma H^*/K)L_B = \frac{718539}{573440}L_B > \frac{5}{4}L_B$.

Proof. $N_I(P_C) \geq \nu_C$; each negative modulus $P_C q \leq m$ has $N_I(P_C q) \leq m/(P_C q) + 1 \leq 2m/(P_C q)$; the increments of U_C have mean at most $H(B) < H^*$. Hence $\sum_I F \geq \lambda \sum_C c_C(\nu_C - 2mH^*/(KP_C)) \geq \lambda(1 - 2\Gamma H^*/K) \sum_C M_C$ by Lemma 17.6; the excess over $\frac{5}{4}$ is $\frac{1739}{573440}$. $\square$

Proof of Theorem 17.1. By Lemmas 17.4, 17.5, 17.11 and 17.10,

$$\sum_{k \leq m} (S(k) - c)^+ = \sum_{k \leq m} (S_0(k) - c)^+ \leq \frac{5}{4}L_B \leq \sum_{b \in I} F(b) \leq \frac{969}{1000} \sum_{b \in I} (B(b) - 1)^+ \leq \sum_{b \in I} (S(b) - 1)^+.$$

If there are no hot points, $L_B = 0$ and Lemma 17.5 suffices; the dense case is Lemma 17.4. $\square$

Provenance, verification and formalisation

The proof of this section was found by GPT-6 (web interface, effort “Pro”) in a 45-minute run from the statements of Section 15 and the four structural levers listed in the brief; it was refereed blind by a Claude Opus 5 agent, which recomputed all 29 finite scales and the tail from the definitions in exact rational arithmetic (agreeing with the author’s `constants.json` exactly), validated $D_{j,L,d}$ against brute-force enumeration, enumerated the 14176 admissible carrier level multisets (maximal exponent budget exactly $\frac{2}{3}$), and stress-tested Lemmas 17.5–17.6 on synthetic atom systems at $m \approx 10^{18}$ – 10^{20} without finding a violation. The overall slack is 3.9%; the tightest steps are $(\frac{133}{4})/(\frac{9}{2})^{4.41} < a_0$ (0.012%), $a_0^3 x_6^{-18} F_6^{11} < 1$ (0.19%) and the window excess (0.24%); none of them may be re-derived in floating point. The reduction from Theorem 17.1 to $g(n) \leq 12n$ is kernel-checked (`Erdos708H17Chain.Chain19.g_le_12n_of_hinge`, Section 16). Theorem 17.1 itself is formalised in Lean 4 (v4.34.0-rc1, Mathlib de5ce8a9) in the 108 modules of `lean/Erdos708/H97`, written by GPT-6 Astra in a single-agent session of 3 h 42 min: the nine lemmas as statement cards with separate proof modules, the 29 finite scales of Lemma 17.9 as exact coefficient certificates (the large scales encoded as natural-number digits with a proved carry bound and checked by `decide` only; no native evaluation), the tail of Lemma 17.10 with all its rational and logarithmic constants, and the six referee clarifications. The theorems `Erdos708H97.hinge97` (Theorem 17.1) and `Erdos708H97.g_le_12n` (Corollary 17.2, obtained by feeding `hinge97` into the kernel-checked chain of Section 16) depend only on `propext`, `Classical.choice` and `Quot.sound`; the whole development was recompiled from scratch by the orchestrating agent. Thus $g(n) \leq 12n$ is verified end to end.

18 Threshold $\frac{56501}{6400}$ and the barrier at 2

Eight mantissas per dyadic band instead of four, a retention rule tuned level by level, prime-density estimates in the window value, and a count of carriers that keeps track of their cardinality bring the threshold of Section 17 down to $\frac{56501}{6400} = 8.828\dots$, hence $g(n) \leq 11n$. The same run also produced an explicit window on which the hinge inequality with threshold $\frac{19}{10}$ fails, and a family showing that *no* absolute threshold below 2 can hold; so the route of Sections 8–17 cannot, as it stands, give a bound below $4n$. Unlike Sections 16 and 17, the results of this section have been refereed and verified in exact arithmetic but *not* formalised (see the end of the section).

Theorem 18.1 (threshold $\frac{56501}{6400}$). *For every atom system, every $m \geq 1$ and every window I of m consecutive positive integers, $\sum_{k \leq m} (S(k) - \frac{56501}{6400})^+ \leq \sum_{b \in I} (S(b) - 1)^+$.*

Corollary 18.2. $g(n) \leq 11n$ for every $n \geq 1$.

Proof. Theorem 16.1 with $c = \frac{56501}{6400} < 9$ gives $g(n) \leq \lceil cn \rceil + 2n \leq 9n + 2n$. $\square$

Throughout, $Q = 2^{16}$, $H^* = \frac{1025}{1024}$, $\rho = \frac{9}{8}$, $T = \frac{401}{75}$, $D_0 = \frac{3601}{1280}$, so that $c := D_0 + \rho T = \frac{56501}{6400}$. Lemma 17.3 is used as before, together with the exponential moment bound: for $z > 1$ and $f_p \leq 1$, $\frac{1}{N} \sum_{n \leq N} z^{\sum_p f_p(n)} \leq \exp((z-1)H(f))$ (from $z^u \leq 1 + (z-1)u$ on $[0, 1]$ and the multiple count). Lemma 17.4 (large atoms, $c \geq 7$; dense branch) is unchanged, so we assume $H(S_0) < H^*$ and $m \geq 2$.

Lemma 18.3 (eight-mantissa retention). *Let $\mathcal{D} = \{1\} \cup \{j/2^h : 8 \leq j \leq 15, h \geq 4\}$ (adjacent ratio at most $\frac{9}{8}$). Put $\eta(t) = \frac{16t}{27}$ for $t \leq \frac{1}{8}$ and, for $t = v/64 > \frac{1}{8}$, $\eta(t) = E_v/720$ with*

$$E_{9,10} = 60, E_{11} = 71, E_{12} = 72, E_{13} = 76, E_{14} = 80, E_{15} = 88, E_{16} = 96, E_{18,20,\dots,28} = 120, E_{30,32} = 160, E_{36,4}$$

Retain (p, t) iff $q_p(t) \leq m^{\eta(t)}$, and define b_p, B as in Lemma 17.5. Then $B \leq S_0$, $H(B) \leq H(S_0)$, every atom of B has modulus at most $m^{1/3}$, and every shortest nonincreasing prefix of levels of mass > 1 has $P_C \leq m^{2/3}$; hence every negative modulus $P_C q$ below is at most m .

Proof. $\eta \leq \frac{1}{3}$ and $\eta(t) \leq \frac{16t}{27}$ at every level. If the last (smallest) level θ of the prefix is at most $\frac{1}{8}$, the budget is at most $\frac{16}{27}s \leq \frac{16}{27} \cdot \frac{9}{8} = \frac{2}{3}$. Otherwise all levels are multiples of $\frac{1}{64}$; removing the last level $v/64$ leaves a mass $w/64$ with $65 - v \leq w \leq 64$, and the exact knapsack $D_v(w) = \max_{u \geq v, u \leq w} (D_v(w-u) + E_u)$ over multisets of levels $\geq v$ gives $\max_{65-v \leq w \leq 64} (D_v(w) + E_v) = 480$ for each of the 24 possible v ; $480/720 = \frac{2}{3}$. $\square$

Lemma 18.4 (rounding loss $\frac{3601}{1280}$). $S_0(k) \leq \rho B(k) + D_0$ for $k \leq m$; hence $\sum_{k \leq m} (S_0(k) - c)^+ \leq \rho L_B$ with $L_B = \sum_{k \leq m} (B(k) - T)^+$.

Proof. As in Lemma 17.5, a prime with positive error has its rounded level t_p unretained, so $a_p = v_p(k) \ln p / \ln m > \eta(t_p)$ with $\sum_p a_p \leq 1$, and the error is at most the successor level of t_p (at most 1 at the top). For $t_p \leq \frac{1}{8}$ the error is at most $\rho t_p = \frac{243}{128} \eta(t_p)$. For the larger levels the cost $720\eta(t_p) = E_v$ is an integer and the reward (successor level in units of $\frac{1}{64}$) is r_v ; the strict inequality forces total large-level cost at most 719. The exact knapsack $R(w) = \max_{E_v \leq w} (R(w - E_v) + r_v)$ gives $\max_{w \leq 719} (1440R(w) + 243(720 - w)) = 259272$, attained at $w = 696$, $R = 176$ (levels $1, 1, \frac{7}{16}, \frac{1}{4}$), and $259272/92160 = \frac{3601}{1280}$. $\square$

Lemma 18.5 (carriers and coefficients). *Define carriers, $\theta_C, s_C, P_C, \nu_C, M_C, c_C$ and U_C as in Lemmas 17.6–17.8 (carriers with $M_C > 0$ only; $\sum_C M_C = L_B$; at a source point every level outside C is at most θ_C). With $a = (T - s_C)/\theta_C$, $c_C \leq \epsilon_{\theta_C, s_C} := \min_{2 \leq r \leq \lfloor a \rfloor + 1} \theta A_r(a) (H^*/\theta)^r / r!$.*

Proof. Lemma 17.7. $\square$

Lemma 18.6 (prime density and the value on the window). $\pi(\nu)/\nu \leq \frac{9}{40}, \frac{3}{20}, \frac{1}{8}$ for $\nu \geq 210, 2310, 30030$ respectively. Let $\delta(\theta) = \frac{9}{40}$ for $\theta \geq \frac{7}{8}$, $\frac{3}{20}$ for $\frac{3}{4} \leq \theta < \frac{7}{8}$, $\frac{1}{8}$ for $\theta < \frac{3}{4}$; then $\pi(\nu_C)/\nu_C \leq \delta(\theta_C)$ for every carrier. For $K_\theta > H^* + \delta(\theta)\theta$ put $\lambda_\theta = \rho K_\theta / (K_\theta - H^* - \delta(\theta)\theta)$ and $F(n) = \sum_C \lambda_{\theta_C} c_C [P_C \mid n] (1 - U_C(n)/K_{\theta_C})$. Then $\sum_{b \in I} F(b) \geq \rho L_B$.

Proof. $\vartheta(x) \leq x \ln 4$ (central binomial coefficients) and partial summation from $\pi(64) = 18$ give $\pi(x) \ln x / x < \frac{17}{10}$ for $x \geq 32768$; below that, sieve checks on $[210, 32768)$, $[2310, 2^{17})$, $[30030, 2^{21})$ give the three bounds (the ratio decreases between primes). At a source point k of C the mass outside C exceeds $T - s_C \geq T - 1 - \theta_C$, and every outside level lies in $\mathcal{D}$ and is at most θ_C

(Lemma 18.5). If $\theta_C \geq \frac{7}{8}$ the outside mass exceeds $\frac{401}{75} - 2 > 3$ with each prime contributing at most 1: at least four outside primes. If $\frac{3}{4} \leq \theta_C < \frac{7}{8}$, the outside levels are at most $\frac{13}{16}$ (the largest level of $\mathcal{D}$ below $\frac{7}{8}$) and the mass exceeds $\frac{401}{75} - 1 - \frac{13}{16} > 3.53 > 4 \cdot \frac{13}{16}$: at least five. If $\theta_C < \frac{3}{4}$, the levels are at most $\frac{11}{16}$ and the mass exceeds $\frac{401}{75} - 1 - \frac{11}{16} > 3.66 > 5 \cdot \frac{11}{16}$: at least six. Hence $k/P_C \geq 210, 2310, 30030$ and $\nu_C \geq k/P_C$. For the value: $N_I(P_C) \geq \nu_C$; every atom q of U_C has $P_C q \leq m$, so $N_I(P_C q) \leq \lfloor \nu_C/q \rfloor + 1$; the primes of these atoms are at most ν_C and each carries total weight at most θ_C , so $\sum_{b \in I} [P_C | b] U_C(b) \leq \nu_C H^* + \theta_C \pi(\nu_C)$. Therefore $\sum_I \lambda_{\theta_C} c_C [P_C | b] (1 - U_C(b)/K_{\theta_C}) \geq \lambda_{\theta_C} c_C \nu_C (1 - (H^* + \delta \theta_C)/K_{\theta_C}) = \rho M_C$; sum over C . $\square$

Lemma 18.7 (pointwise feasibility). *Write a level as $\theta = j/L$ ($8 \leq j \leq 15$, $L \geq 16$ a power of two; level 1 as $(8, 8)$), and for n let $N = \#\{p : b_p(n) \geq \theta\}$. A carrier of cardinality k , mass $1 + d/L$ and minimum level θ dividing n satisfies $U_C(n) \geq (N - k)\theta$ and $B(n) - 1 \geq d/L + (N - k)\theta$. With $P_{j,L}(X) = \sum_{t \in \mathcal{D}, t \geq j/L} X^{Lt}$ and $E_{k,d} = [X^{L+d}](P_{j,L}^k - (P_{j,L} - X^j)^k)$, the number of such carriers is at most $\binom{N}{k} E_{k,d}$, and*

$$A_\theta = \max_{N \leq \lfloor K_\theta/\theta \rfloor + \lfloor 1/\theta \rfloor + 2} \sum_{k,d} \binom{N}{k} E_{k,d} \epsilon_{\theta, 1+d/L} \frac{(1 - (N - k)\theta/K_\theta)^+}{d/L + (N - k)\theta}$$

satisfies $F(n) \leq (\sum_\theta \lambda_\theta A_\theta)(B(n) - 1)^+$. For $\theta \geq \frac{1}{2}$ (carriers of two primes, plus the triples of level $\frac{1}{2}$ when $\theta = \frac{1}{2}$) the smaller bound $A_\theta = \max_{N,v} [N(N - 1)(e_\theta/2 + E_\theta(v))p_2 + \binom{N}{3} h_\theta p_3] / (Nv - 1)$ may be used, where $e_\theta = \epsilon_{\theta, 2\theta}$ ($\theta > \frac{1}{2}$), $h_{1/2} = \epsilon_{1/2, 3/2}$, $E_\theta(v) = \sum_{t \in \mathcal{D}, \theta < t \leq v} \epsilon_{\theta, \theta+t}$, $p_k = (1 - (N - k)\theta/K_\theta)^+$.

Proof. As in Lemma 17.8, a carrier is determined by its k primes (among the N eligible ones) and their levels, with the minimum level θ used at least once and total degree $L + d$. For $\theta \geq \frac{1}{2}$ the numerator is, for fixed N , an affine sum of functions of the individual caps v_i of the eligible primes while the denominator is $\sum_i (v_i - 1/N)$, so the ratio is bounded by its value when all caps are equal. $\square$

Lemma 18.8 (the 57 finite scales and the tail). *With $40K_\theta$ given by the table ($j = 8, \dots, 15$; only $j = 8$ at $L = 8$)*

L	$40K_\theta$	$10^6 \sum_j \lambda_\theta A_\theta <$
8	200	48879
16	125, 113, 125, 138, 150, 162, 175, 187	549612
32	61, 72, 78, 73, 79, 87, 97, 91	271163
64	50, 54, 57, 58, 60, 58, 57, 66	99084
128	45, 46, 47, 48, 47, 50, 48, 49	20073
256	43, 43, 43, 43, 44, 44, 44, 45	1577
512	41, 41, 42, 42, 42, 42, 42, 42	26
1024	41, 41, 41, 41, 41, 41, 41, 41	1

the finite scales contribute less than $\frac{990415}{10^6}$. For $L \geq 2048$ take $K_\theta = \frac{41}{40}$ (so $\lambda_\theta < 50$); with $a_0 = \frac{120347}{2500000}$ the exponential moment at $z = \frac{9}{2}$ gives $c_C \leq \rho \theta a_0^{1/\theta}$ (from $e \ln \frac{9}{2} > 4$, $e^{7H^*/2} < \frac{133}{4}$, $(\frac{133}{4})^{75} < a_0^{75} (\frac{9}{2})^{326}$), and with $\mathcal{P}_j(X) = \sum_{a=j}^{15} X^a + \sum_{h \geq 1} \sum_{a=8}^{15} X^{a2^h}$, the pairs $(x_j, \bar{P}_j, \sigma_j)$: $(\frac{473}{625}, \frac{421847}{10^6}, \frac{157}{200})$, $(\frac{3883}{5000}, \frac{42539}{10^5}, \frac{827}{1000})$, $(\frac{7939}{10^4}, \frac{42847}{10^5}, \frac{859}{1000})$, $(\frac{4047}{5000}, \frac{107879}{250000}, \frac{22}{25})$, $(\frac{8233}{10^4}, \frac{216251}{500000}, \frac{111}{125})$, $(\frac{4181}{5000}, \frac{86661}{200000}, \frac{441}{500})$, $(\frac{8481}{10^4}, \frac{215799}{500000}, \frac{863}{1000})$, $(\frac{8593}{10^4}, \frac{213963}{500000}, \frac{83}{100})$ satisfy $\mathcal{P}_j(x_j) \leq \bar{P}_j < 1$ and $(a_0 x_j^{-j})^{40} \leq \sigma_j^{40} (1 - \bar{P}_j)^{41}$; the

contribution of the scales $L \geq 2048$ is then less than $\frac{1}{1000}$. Consequently $F(n) \leq \frac{198283}{200000}(B(n) - 1)^+ \leq (S(n) - 1)^+$.

Proof. The finite scales are exact integer coefficient extractions and rational maximisations (moment bounds rounded upward to a common power-of-two denominator). For the tail, write $n = N - k$; positive terms need $n\theta < K_\theta$, i.e. $n \leq M := \lceil K_\theta/\theta \rceil - 1 \leq \frac{41}{40} \cdot \frac{L}{j}$; the identity $\sum_{k \geq 0} \binom{n+k}{k} P^k = (1-P)^{-n-1}$ and $\sum_{n \leq M} (1-\bar{P})^{-n-1} \leq \bar{P}^{-1}(1-\bar{P})^{-M-1}$ bound the contribution of (L, j) by $50\rho j (\sum_{d \leq j} x_j^{-d}) \sigma_j^{L/j} / (\bar{P}_j(1-\bar{P}_j))$; summing over $L = 2048 \cdot 2^h$ with $2^h \geq h+1$ gives the stated bound. $\square$

Proof of Theorem 18.1. By Lemmas 17.4, 18.4, 18.6 and 18.8, $\sum_{k \leq m} (S(k) - c)^+ = \sum_{k \leq m} (S_0(k) - c)^+ \leq \rho L_B \leq \sum_I F \leq \frac{198283}{200000} \sum_I (B - 1)^+ \leq \sum_I (S - 1)^+$; the dense branch and $m = 1$ are covered by Lemma 17.4. $\square$

Proposition 18.9 (no threshold below 2). *Let $r \geq 2$, let $p_1 < \dots < p_r$ be primes in an interval $[M, \sqrt{2}M]$ with $M > 2$, let $m = p_{r-1}p_r$ and let I be the window of length m centred at $P = p_1 \dots p_r$ (all weights 1). Then $\sum_{k \leq m} (S(k) - c)^+ = \binom{r}{2}(2 - c)$ and $\sum_{b \in I} (S(b) - 1)^+ = r - 1$ for $1 \leq c \leq 2$, so the hinge inequality with threshold c fails whenever $c < 2 - \frac{2}{r}$. Such clusters exist for every r ; hence no absolute threshold $c < 2$ holds. Explicitly, $p_1, \dots, p_{21} = 503, \dots, 641$, $m = 404471$, $x = P - 202236$ give $21 > 20$ for $c = \frac{19}{10}$.*

Proof. $m < 2p_1p_2$ and $m < p_1p_2p_3$ (because $p_1p_2 \geq M^2 > m/2$ and $p_3 > M > 2$), so in $[1, m]$ each of the $\binom{r}{2}$ pair products occurs exactly once and no integer has three of the primes; the left side is $\binom{r}{2}(2 - c)$. Every pair product divides P and exceeds the window radius, so P is the only element of I with two of the primes, where $S(P) = r$; the right side is $r - 1$. If only boundedly many primes lay in every interval $[2^{h/2}, 2^{(h+1)/2})$, then $\sum_p 1/p$ would converge, contradicting Euler; so intervals $[M, \sqrt{2}M]$ with at least r primes exist for every r , and $M > 2$ can be assumed. The instance was verified by direct enumeration. $\square$

Remark 18.10. Through Theorem 16.1, threshold c gives $\lceil cn \rceil + 2n$; Proposition 18.9 shows that the hinge inequality itself cannot go below $c = 2$, so the present route cannot give a bound below $4n$ (and $(2 + 2)n = 4n$ would need the still-open threshold 2, cf. Conjecture 8.1). Erdős's $2n$ requires either a rounding step with loss $o(n)$ for a different relaxation (Proposition 16.7 rules it out for (4) itself) or a different reduction.

Provenance and verification

The proof of Theorem 18.1 and Proposition 18.9 was found by GPT-6 (web interface, “Pro”) in a 50-minute run from the statements of Section 17. It was refereed blind by a Claude Opus 5 agent, which recomputed the 57 scales and the tail from the definitions in exact arithmetic ($990409.9 \cdot 10^{-6}$ and $801.7 \cdot 10^{-6}$ against the claimed $990415 \cdot 10^{-6}$ and 10^{-3} ; overall headroom 0.88%), both knapsacks, the prime-density ranges, both sides of the explicit instance by direct enumeration, and $1.7 \cdot 10^6$ carriers over 6081 pointwise profiles; it found no error but two derivations that were incomplete as written (the outside-prime counts of Lemma 18.6, which need $b_p \leq \theta_C$ and the discreteness of $\mathcal{D}$ and are load-bearing; and the summation range of the tail), repaired above. The tightest steps are $(\frac{133}{4})^{75} < a_0^{75}(\frac{9}{2})^{326}$ (ratio 0.99988) and the $\frac{3}{20}$ density bound. **The results of this section are not formalised:** the Lean development of the repository ends with Section 17, and $g(n) \leq 12n$ is the strongest bound verified by the kernel; $g(n) \leq 11n$ rests on the refereed proof above.

19 What remains: the threshold interval $[2, \frac{56501}{6400}]$

By Theorem 18.1 and Proposition 18.9, the set of admissible thresholds is a closed half-line $[c^*, \infty)$ with

$$2 \leq c^* \leq \frac{56501}{6400} = 8.828\dots$$

(the set is upward closed, and closed because for each instance both sides depend continuously on c). This section records what is known about the interior of that interval: an exact truncation available at *every* threshold, the inequality with threshold 2 for systems with few primes and for systems of large mean, and the precise statement of what is left. Nothing here narrows the interval.

Remark 19.1 (the barrier family is not tight at 2). On the family of Proposition 18.9 the two sides are $\binom{r}{2}(2-c)$ and $r-1$, so equality holds at $c = 2 - \frac{2}{r}$ and at $c = 2$ the left side vanishes while the right side is $r-1$. The family therefore forces $c^* \geq 2$ only in the limit $r \rightarrow \infty$, and provides no evidence whatever that $c^* = 2$; the true threshold may lie anywhere in $[2, \frac{56501}{6400}]$.

Lemma 19.2 (primorial truncation). *Let $d \geq 0$ be an integer and let $Q_d = p_1 \cdots p_d$ be the d -th primorial ($Q_0 = 1$, $Q_1 = 2$, $Q_2 = 6$, $Q_3 = 30$, $Q_4 = 210$, $Q_5 = 2310$, $Q_6 = 30030$, $Q_7 = 510510$). Given an atom system S and $m \geq 1$, let S_0 consist of the atoms p^j with $p^j Q_d \leq m$. Then for every $k \leq m$ and every real $c \geq d$,*

$$(S(k) - c)^+ = (S_0(k) - c)^+,$$

so $\sum_{k \leq m} (S(k) - c)^+ = \sum_{k \leq m} (S_0(k) - c)^+$; and $S_0 \leq S$ gives $\sum_{b \in I} (S_0(b) - 1)^+ \leq \sum_{b \in I} (S(b) - 1)^+$ for every window. In particular a counterexample to (H_c) yields one supported on $p^j \leq m/Q_d$. The primorial is optimal: for every $Q' > Q_d$ there are S and m for which the identity fails at $c = d$.

Proof. Suppose an omitted atom $q = p^j$ (that is, $qQ_d > m$) divides $k \leq m$, and write $k = qt$. Then $t \leq m/q < Q_d$, so $\omega(t) \leq d-1$: an integer with d distinct prime factors is at least the product of the d smallest primes, namely Q_d . Since q is a prime power, $\omega(k) \leq \omega(t) + 1 \leq d$. The per-prime capacity now gives

$$S(k) = \sum_{p|k} \sum_j \alpha_{p,j} [p^j | k] \leq \sum_{p|k} 1 = \omega(k) \leq d \leq c,$$

and $S_0 \leq S$, so both hinges vanish at k . At every other $k \leq m$ no omitted atom of positive weight divides k , so $S(k) = S_0(k)$. For optimality take q the least prime exceeding Q_d , $m = Q_d q$, and unit weights on $p_1, \dots, p_d$ and q : then $S(m) = d+1$ and the hinge at $k = m$ equals 1, while any cutoff m/Q' with $Q' > Q_d$ omits q and the hinge collapses to 0. $\square$

The truncation of Lemma 17.4 is the case $d = 7$ with 2^{16} in place of $Q_7 = 510510$, and is weaker ($2^{16} < Q_7$); we keep it in Section 17 because that is the form which has been formalised. The point of Lemma 19.2 is that it is available at *every* threshold: at $c = 2$ one may truncate at $p^j \leq m/6$, whereas no argument of the 2^{16} type truncates at all below $c = 7$. It is a prerequisite for the two propositions below and for any attack on $c < 7$. Its price is that the floor $\lfloor m/q \rfloor \geq \frac{Q_d}{Q_d+1} \frac{m}{q}$ degrades from $1 - 2^{-16}$ to $\frac{6}{7}$ at $d = 2$, and that the truncated system may still carry every prime power up to $m/6$.

Proposition 19.3 (threshold 2 for at most four primes). *Let S be an atom system in which at most four primes carry positive weight. Then $\sum_{k \leq m} (S(k) - 2)^+ \leq \sum_{b \in I} (S(b) - 1)^+$ for every $m \geq 1$ and every window I of m consecutive positive integers.*

Proof. Padding with zero functions, write $S = f_1 + f_2 + f_3 + f_4$ with $f_i(n) = \sum_j \alpha_{p_i, j} [p_i^j \mid n] \in [0, 1]$, put $g_{ij} = (f_i + f_j - 1)^+$ and $G = \frac{1}{3} \sum_{i < j} g_{ij}$. Each f_i occurs in exactly three of the six pairs, so $\sum_{i < j} g_{ij} \geq \sum_{i < j} (f_i + f_j - 1) = 3S - 6$; the sum is also nonnegative, whence $G \geq (S - 2)^+$. For the other half, the six pairs split into the three perfect matchings $\{12, 34\}$, $\{13, 24\}$, $\{14, 23\}$, and $(a - 1)^+ + (b - 1)^+ \leq (a + b - 1)^+$ for $a, b \geq 0$; applying this to each matching and averaging gives $G \leq (S - 1)^+$.

Fix a pair p, q and let $A_a = \sum_{j \leq a} \alpha_{p, j}$, $B_b = \sum_{j \leq b} \alpha_{q, j}$ be the cumulative weights, so that $f_p(n) = A_{\min(v_p(n), J_p)}$ where J_p is the largest exponent occurring at p . With $\varphi(u) = (u - 1)^+$ put

$$\beta_{a, b} = \varphi(A_a + B_b) - \varphi(A_{a-1} + B_b) - \varphi(A_a + B_{b-1}) + \varphi(A_{a-1} + B_{b-1}).$$

Then $\beta_{a, b} \geq 0$: with $\delta = A_a - A_{a-1} \geq 0$ the function $u \mapsto \varphi(u + \delta) - \varphi(u)$ is nondecreasing because φ is convex, and $B_b \geq B_{b-1}$. Since $A_a \leq 1$ and $B_b \leq 1$, the terms with index 0 vanish, and telescoping in both variables gives the finite expansion $g_{pq}(n) = \sum_{a, b \geq 1} \beta_{a, b} [p^a q^b \mid n]$. Every modulus d satisfies $\#\{b \in I : d \mid b\} \geq \lfloor m/d \rfloor$, so $\sum_{k \leq m} g_{pq}(k) \leq \sum_{b \in I} g_{pq}(b)$. Summing the six pairs,

$$\sum_{k \leq m} (S(k) - 2)^+ \leq \sum_{k \leq m} G(k) \leq \sum_{b \in I} G(b) \leq \sum_{b \in I} (S(b) - 1)^+. \quad \square$$

The certificate G depends only on the atom system, not on m or on the window. It does not survive a fifth prime: for five weights equal to $\frac{1}{2}$ one has $(S - 2)^+ = \frac{1}{2}$ while every pair term vanishes.

Proposition 19.4 (threshold 2 for systems of large mean). *Let S be an atom system, $m \geq 1$, and let S_0 be its truncation at $p^j Q_d \leq m$ as in Lemma 19.2. Write $H(f) = \sum_{p, j} \beta_{p, j} / p^j$ for the mean of a system with nonnegative prime-power increments $\beta_{p, j}$. If, for an integer $c \geq 2$ and a real $h > 0$,*

$$H(S_0) \geq h \quad \text{and} \quad \frac{Q_c}{Q_c + 1} h - \frac{h^c}{(c + 1)!} \geq 1,$$

then (H_c) holds for S , m and every window. In particular, taking $c = 2$, $Q_2 = 6$ and $h = h_0 := \frac{18 - \sqrt{30}}{7} = 1.78896 \dots$, every system whose $m/6$ -truncation has mean at least h_0 satisfies the hinge inequality with threshold 2; the rational value $\frac{179}{100}$ suffices, since $\frac{6}{7} \cdot \frac{179}{100} - \frac{1}{6} \left(\frac{179}{100}\right)^2 - 1 = \frac{113}{420000} > 0$.

Proof. Put $V = (h/H(S_0)) S_0$, so that $V \leq S_0 \leq S$, the per-prime capacities are still at most 1, and $H(V) = h$ exactly; note that the hypothesis is used only through this rescaling, so no upper bound on $H(S_0)$ is needed. For $q \leq m/Q_c$ we have $\ell = \lfloor m/q \rfloor \geq Q_c$ and $m/q < \ell + 1$, hence $\lfloor m/q \rfloor \geq \frac{Q_c}{Q_c + 1} \frac{m}{q}$ and $\sum_{k \leq m} V(k) \geq \frac{Q_c}{Q_c + 1} m h$. For $u_i \in [0, 1]$ and an integer $c \geq 2$ one has $(\sum_i u_i - c)^+ \leq e_c(u)/(c + 1)$ (the difference is minimised at a vertex of the cube after the usual compression, where it reduces to $\binom{k}{c} \geq (c + 1)(k - c)^+$), so Lemma 17.3 gives $\sum_{k \leq m} (V(k) - c)^+ \leq m h^c / (c + 1)!$. Subtracting, $\sum_{k \leq m} \min(V(k), c) \geq m$, and $V \leq S$ gives $\sum_{k \leq m} \min(S(k), c) \geq m$. Now the affine certificate $F = S - 1$ is pointwise at most $(S - 1)^+$, and

$$\sum_{b \in I} F(b) = \sum_{p, j} \alpha_{p, j} \#\{b \in I : p^j \mid b\} - m \geq \sum_{k \leq m} S(k) - m \geq \sum_{k \leq m} S(k) - \sum_{k \leq m} \min(S(k), c) = \sum_{k \leq m} (S(k) - c)^+,$$

while $\sum_{b \in I} F(b) \leq \sum_{b \in I} (S(b) - 1)^+$. $\square$

The same argument with $V = \tau S$ ($0 \leq \tau \leq 1$, no truncation) shows that $\mu - \frac{1}{6}H(S)^2 \geq 1$ implies (H_2) , where $\mu = \frac{1}{m} \sum_{k \leq m} S(k)$; and under the hypotheses below the choice $\tau = 1$ is optimal.

Proposition 19.5 (the remaining case). *The hinge inequality with threshold 2 holds for all atom systems, all m and all windows if and only if it holds for every S , m , x subject to: all atoms satisfy $p^j \leq m/6$; at least five primes carry positive weight; $m \geq 66$; $\sum_{k \leq m} (S(k) - 2)^+ > 0$; $H(S) < h_0 = \frac{18-\sqrt{30}}{7}$; and $\mu - \frac{1}{6}H(S)^2 < 1$ with $\mu = \frac{1}{m} \sum_{k \leq m} S(k)$.*

Proof. One direction is trivial. Conversely, let (S, m, x) violate (H_2) and replace S by its $m/6$ truncation, which by Lemma 19.2 leaves the left side unchanged and does not increase the right side. At most four active primes is excluded by Proposition 19.3. A positive left side forces some $k \leq m$ with $S(k) > 2$, hence $\omega(k) \geq 3$; with five active primes all of whose atoms are at most $m/6$, the fifth smallest active prime is at least 11, so $m \geq 66$. Mean at least h_0 is excluded by Proposition 19.4 applied with $h = h_0$, and $\mu - \frac{1}{6}H^2 \geq 1$ by the remark following it. $\square$

The region described by Proposition 19.5 is not empty — for instance $m = 210$ with the atoms 2, 4, 3, 13, 19, 23, 29 of weights $\frac{2}{3}, \frac{1}{3}, 1, 1, 1, \frac{2}{3}, 1$ satisfies every hypothesis — so the proposition is a genuine reduction and not a disguised proof. Since truncation gives $\mu \geq \frac{6}{7}H$, the last two hypotheses together already force $H < h_0$ or $H > h_1 = \frac{18+\sqrt{30}}{7}$, the second being excluded only by the rescaling in the proof of Proposition 19.4. Analogous reductions hold at every integer threshold $c \geq 3$, with Q_c in place of 6 and $m \geq Q_{c+1}$, but without the five-prime clause: no analogue of Proposition 19.3 is known for $c \geq 3$.

Beyond four primes

Proposition 19.3 is the base case of an induction: an atom system is finitely supported, so (H_2) holds in general as soon as it holds for every bounded number of active primes. The pair certificate itself does not survive a fifth prime, but two different mechanisms do, and a third is provably unavailable.

Lemma 19.6 (Boolean antichain bound). *Let $2 \leq a < b < c < d$ be pairwise coprime integers and $Q \geq 2$ an integer. Let $\mathcal{M}_Q$ be the family of inclusion-minimal nonempty $A \subseteq \{a, b, c, d\}$ with $\prod A / \max A \geq Q$. Then $Q \sum_{A \in \mathcal{M}_Q} (\prod A)^{-1} \leq 1$, with maximum $\frac{101}{105}$ attained at $(\{2, 3, 5, 7\}, Q = 2)$.*

Proof. A pair is feasible exactly when its smaller member is $\geq Q$; a triple exactly when the product of its two smaller members is $\geq Q$, and is minimal exactly when it contains no feasible pair; the quadruple is feasible exactly when $abc \geq Q$ and minimal exactly when $bc < Q$. This gives nine cases according to the position of Q relative to a, b, c, ab, ac, bc, abc , and in each the displayed bound is ≤ 1 : all rows but the first are bounded by $\frac{1}{c} + \frac{3}{d} \leq \frac{4}{c} \leq \frac{4}{5}$ (pairwise coprimality forces $c \geq 5$), and the first row is $a e_2(\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d})$, which is at most $\frac{101}{105}, \frac{131}{140}, \frac{179}{210}, \frac{6}{b} \leq 1$ according as $b = 3, b = 4, b = 5, b \geq 6$. $\square$

Theorem 19.7 (five primes, one multilevel). *Let $S = T + f$ where $T(n) = \sum_{i=1}^4 w_i [q_i \mid n]$ with $w_i \in [0, 1]$ and $q_1, \dots, q_4$ powers of four distinct primes, and $f(n) = \sum_j \alpha_j [p^j \mid n]$ with $\alpha_j \geq 0$, $\sum_j \alpha_j \leq 1$, for a fifth prime p . Then (H_2) holds for S , every m and every window. In particular it holds for five arbitrary primes carrying one arbitrary prime power each.*

Proof. For $a \geq 0$ and $f \in [0, 1]$ the layer identity $(T + f - a)^+ - (T - a)^+ = \int_0^1 [f > t] [T > a - t] dt$ holds (both sides equal $(\min(f, 1) - (a - T)^+)^+$; the hypothesis $f \leq 1$ is the per-prime capacity and is the only one needed). Because $f(n) = A_{v_p(n)}$ with $A_j = \sum_{h \leq j} \alpha_h$ nondecreasing and $A_0 = 0$, for each $t > 0$ the indicator $[f(n) > t]$ is either identically zero or equals $[p^{j(t)} \mid n]$ with $j(t) = \min\{j : A_j > t\} \geq 1$; write $d = p^{j(t)} \geq 2$. Put $c = 1 - t$ and let $Q(t) = \min\{\prod_{i \in B} q_i : \sum_{i \in B} w_i > c\}$ when such a B exists. If $T(n) > c + 1$ then deleting any single w_i leaves a value $> c$, so the set of indices dividing n is feasible for Lemma 19.6 and contains a minimal member; hence $[T > c + 1] \leq \sum_{A \in \mathcal{M}_{Q(t)}} [q_A \mid n]$ and, by Lemma 19.6, $\#\{k \leq M : T(k) > c + 1\} \leq M \sum_A q_A^{-1} \leq M/Q(t)$, an integer bound, so $\leq \lfloor M/Q(t) \rfloor$. Since d is coprime to every q_i we have $T(dk) = T(k)$, so taking $M = \lfloor m/d \rfloor$,

$$\sum_{k \leq m} [d \mid k] [T(k) > 2 - t] \leq \left\lfloor \frac{m}{dQ(t)} \right\rfloor,$$

while $[dQ(t) \mid n] \leq [d \mid n] [T(n) > 1 - t]$ shows the window contains at least that many points contributing to the right side. Integrating over t (the integrand is a step function with finitely many cells) gives $\sum_{k \leq m} \Delta_2(T, f)(k) \leq \sum_{b \in I} \Delta_1(T, f)(b)$, and adding Proposition 19.3 for T gives (H_2) for S . $\square$

Remark 19.8 (domination after summation, not pointwise). The certificate assembled in the proof — a finite nonnegative combination of divisor indicators, namely $\frac{1}{3} \sum_{i < j} (w_i + w_j - 1)^+ [q_i q_j \mid n]$ plus one term $(v_\ell - u_\ell) [d_\ell Q_\ell \mid n]$ per cell — dominates $(S - 1)^+$ *pointwise* but dominates $(S - 2)^+$ only *after summation over* $[1, m]$. The pointwise statement is false for it: with atoms $\frac{257}{720}, \frac{421}{720}, \frac{563}{720}, \frac{611}{720}$ at 2, 3, 5, 7 and $\frac{293}{720}, \frac{427}{720}$ at 11, 11^2 , the certificate takes the value $\frac{1948}{2160}$ at $n = 12705$ while $(S(n) - 2)^+ = \frac{2625}{2160}$. This is not a defect but the mechanism, and it is what reconciles Theorem 19.7 with Proposition 19.10 below.

Theorem 19.9 (any number of primes under a weight cap). *Let $N \geq 2$ and $\lambda = \frac{2}{N-1}$. If at most N primes are active and each has total weight at most λ , then (H_2) holds.*

Proof. Pad to $f_0, \dots, f_{N-1} \in [0, \lambda]$ and set $G(f) = \sum_{r=0}^{N-1} \max(0, \min(\frac{\lambda}{N}, \min_i (f_i - \lambda a_{ri})))$ with $a_{ri} = ((r + i) \bmod N)/N$. Then $G(f) = \lambda \mathbb{P}(Z_i \leq f_i \ \forall i)$ where $Z_i = \lambda \{U + i/N\}$ and U is uniform on $[0, 1]$; the union bound gives $G \geq \lambda(1 - \sum_i (1 - f_i/\lambda)) = \sum_i f_i - 2$, and $G \geq 0$, so $G \geq (\sum_i f_i - 2)^+$. On branch r , writing $U = r/N + t$, all the threshold inequalities force $\sum_i f_i \geq 1 + N\lambda t$ because $\sum_i a_{ri} = \frac{N-1}{2}$, so the admissible t -length is at most $(\sum_i f_i - 1)^+/(N\lambda)$; multiplying by λ and summing the N branches gives $G \leq (\sum_i f_i - 1)^+$. Partitioning each branch at the finitely many cumulative levels turns G into a nonnegative combination of divisor indicators, and the interval counts finish as before. $\square$

At five equal half-weights this certificate has value $\frac{1}{2}$, exactly the lower hinge, where the pair certificate of Proposition 19.3 collapses to 0.

Proposition 19.10 (no pointwise nonnegative certificate at six primes). *There is no $G(f) = \sum_t c_t \prod_{i: t_i > 0} [f_i \geq t_i]$ with all $c_t \geq 0$ on $\{0, \frac{1}{2}, 1\}^6$ satisfying $(\sum_i f_i - 2)^+ \leq G \leq (\sum_i f_i - 1)^+$ everywhere. Equivalently, for the six-prime system with atoms $\frac{1}{2}$ at p_i and $\frac{1}{2}$ at p_i^2 , no nonnegative divisor certificate $C(n) = \sum_d \gamma_d [d \mid n]$ satisfies $(S - 2)^+ \leq C \leq (S - 1)^+$ pointwise.*

Proof. The upper bound at the states of total weight at most 1 forces the constant, singleton and $(\frac{1}{2}, \frac{1}{2})$ -pair coefficients to vanish. Let P, T, H be the total coefficient mass of support 2, 3 and ≥ 4 .

The all-ones state gives $P + T + H \leq 5$. Each of the twenty states with exactly three coordinates equal to 1 has lower hinge 1, and a threshold vector of support s is dominated by exactly $\binom{6-s}{3-s}$ of them, so summing gives $4P + T \geq 20$. But $4P + T = 4(P + T + H) - 3T - 4H \leq 20$, forcing $T = H = 0$. Every surviving support-2 coefficient has a threshold equal to 1 in some coordinate, so G vanishes at the all-half state, where the lower hinge is 1. $\square$

Proposition 19.10 is an obstruction to a *method*, not evidence against (H_2) : by Remark 19.8 a certificate need only dominate the left side after summation, and the exhaustive searches reported below found no violation of (H_2) itself. What it does say is that past five primes any certificate must be signed, or must give up pointwise left domination, as Theorem 19.7 does.

Corollary 19.11 (the surviving five-prime case). *If (H_2) fails for a system with at most five active primes, then at least two of those primes carry more than one nonzero atom, and at least one carries total weight greater than $\frac{1}{2}$.*

Proof. Theorem 19.7 excludes systems with at most one multilevel prime, and Theorem 19.9 with $N = 5$, where $\lambda = \frac{1}{2}$, excludes systems all of whose primes have total weight at most $\frac{1}{2}$. $\square$

Two remarks delimit these results. The weight cap of Theorem 19.9 is load-bearing and not an artefact: at $N = 6$ with six weights equal to $\frac{1}{2} > \frac{2}{5} = \lambda$ the cyclic certificate falls below the lower hinge. And the obstruction of Proposition 19.10 genuinely begins at six primes: a pointwise nonnegative certificate does exist on $\{0, \frac{1}{2}, 1\}^5$. Pairwise coprimality in Lemma 19.6 is likewise essential, since $(a, b, c, d) = (2, 3, 4, 5)$ with $Q = 2$ gives $\frac{71}{60} > 1$.

Remark 19.12 (the bridge to unrestricted five primes). The elimination proof of Theorem 19.7 needs the antichain bound only for the four moduli actually present. For an arbitrary multilevel system on four primes P the corresponding statement is: for every integer $Q \geq 2$, the coordinatewise-minimal exponent vectors e with $D(e)/\max_i p_i^{e_i} \geq Q$, where $D(e) = \prod p_i^{e_i}$, satisfy $K_P(Q) := Q \sum_e D(e)^{-1} \leq 1$. It is now proved for $P = \{2, 3, 5, 7\}$ at every $Q \geq 2$, by an exact sweep of the 1 763 328 exponent states of the ladder below $Q_0 = 4\,900\,014\,488\,437\,221\,682$ (which fall into 35 109 minimality intervals) together with the uniform tail $K_P(Q) \leq Q^{-1/5} \prod_{p \in P} (1 - p^{-1/10})^{-1} \leq 1$ for $Q \geq Q_0$, valid for every quadruple; the maximum over all $Q \geq 2$ is exactly $\frac{101}{105}$, attained only at $Q = 2$ (the runner-up is $Q = 4$ with $\frac{299}{315}$, so the maximiser is not simply the least Q). Because Q_0 does not depend on P and no ladder is larger than that of $\{2, 3, 5, 7\}$, the same finite computation decides any named quadruple in seconds — it does so for all 70 four-element subsets of the first eight primes — and no computation is needed when every prime of P exceeds 10^{15} , since then $A_P^5 \leq (31/30)^{20} < 2$. Consequently Theorem 19.7 upgrades: (H_2) holds for arbitrary multilevel systems supported on $\{2, 3, 5, 7, p\}$, and on the analogous supports for those 70 quadruples, for every fifth prime p . What remains open is the *uniform* statement over the infinitely many quadruples containing a small prime, and, as before, the passage from five primes to six. Even granted in full, it would not iterate: the five-prime analogue is false, since $2e_2(\frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{11}) = \frac{194}{165} > 1$.

Corollary 19.13. *If (H_2) fails for a system with at most five active primes, then $\{2, 3, 5, 7\}$ is not among its primes; in particular the smallest admissible support $\{2, 3, 5, 7, 11\}$ cannot carry a failure.*

The incidence graph

The reduction of Sections 8–18 passes through the linear programme, and Proposition 18.9 caps everything it can give at $4n$. The following bound uses no relaxation at all, only the count $\#\{b \in I : d \mid b\} \geq \lfloor a_n/d \rfloor$, and is therefore not subject to that cap. Let $\mathcal{P}(A)$ be the set of primes dividing $\prod A$, let H be the bipartite graph on the n inputs and $\mathcal{P}(A)$ with an edge (a, p) when $p \mid a$, let c be its number of components and

$$\beta = \sum_{a \in A} \omega(a) - |\mathcal{P}(A)| + c - n$$

its cycle rank. (Primes dividing a single input are pendant vertices, so β and c are unchanged if they are deleted; β is the cycle rank of the shared-prime subgraph.)

Theorem 19.14 (collision–cycle bound). *$g(A, x) \leq n + \beta$ for every A and every $x \geq 0$, with no hypothesis on a_n .*

Proof. Every $a \in A$ satisfies $a \leq a_n$, so $\#\{b \in I : a \mid b\} \geq \lfloor a_n/a \rfloor \geq 1$; choose any $b(a) \in I$ divisible by a and let $S = \{b(a) : a \in A\}$, a set. The fibres $C_b = \{a : b(a) = b\}$ partition A , and distinct fibres have distinct anchors. Write $r_{C,p}(j) = \#\{a \in C : p^j \mid a\}$. If $r_{C,p}(j) > 0$ then $p^j \mid b_C$, so at least $\#\{C : r_{C,p}(j) > 0\}$ elements of S are divisible by p^j , whence $c_p(j) - s_p(j) \leq \sum_C (r_{C,p}(j) - 1)^+ \leq \sum_C (r_{C,p}(1) - 1)^+$ by monotonicity in j ; taking the maximum over j gives $\delta_p(S) \leq \sum_C (r_{C,p}(1) - 1)^+$. Summing over $p \in \mathcal{P}(A)$,

$$\sum_p \delta_p(S) \leq \sum_C \left(\sum_{a \in C} \omega(a) - |\mathcal{P}(C)| \right) = \sum_C (|C| - c_C + \beta_C),$$

by Euler’s formula in each fibre’s own incidence graph H_C . Identifying the prime vertices of the H_C to rebuild H removes one vertex at a time and either merges two components or creates a cycle, so $\sum_C \beta_C \leq \beta$. Hence $|S| + \sum_p \delta_p(S) \leq \sum_C (1 + |C| - c_C + \beta_C) \leq n + \sum_C \beta_C \leq n + \beta$, and Lemma 17.3’s companion anchor bound $g \leq |S| + \sum_p \delta_p(S)$ finishes the proof. $\square$

Corollary 19.15. *If $\beta \leq n$, equivalently if $\sum_{a \in A} (\omega(a) - 2) \leq |\mathcal{P}(A)| - c$, then $g(A, x) \leq 2n$.*

Corollary 19.15 contains the earlier criterion that $g \leq 2n$ whenever every component of the incidence graph carries at most one cycle: a pseudoforest has $\beta \leq c \leq n$. The coefficient of β in Theorem 19.14 cannot be lowered on the instances below.

Proposition 19.16 (sharpness). *Let $t \geq 2$ and let $w_1 < \dots < w_t < q_u < q_v$ be primes all within a factor $\sqrt{2}$ with $q_u^2 > w_t q_v =: m$. Put $A = \{w_i q_u, w_i q_v : i \leq t\}$, the edge set of $K_{2,t}$, so that $n = 2t$, $c = 1$ and $\beta = t - 1$. If the window of length m can be placed so that $M = w_1 \dots w_t q_u q_v$ is its only element divisible by two of the demanded primes and neither q_u^2 nor q_v^2 has a multiple in it, then $g(A, x) = n + \beta = 3t - 1$.*

Proof. A cover containing M needs $t - 1$ further multiples of q_u , $t - 1$ of q_v and one of each w_i , giving $3t - 1$; a cover avoiding M needs at least t multiples of each of q_u, q_v and t more, giving $3t$. $\square$

Such a placement is not automatic and has to be exhibited. It exists for $t = 2, \dots, 6$, giving the exact values $g = 5, 8, 11, 14, 17$ at $n = 4, 6, 8, 10, 12$; for instance $t = 2$ with $(w_1, w_2, q_u, q_v) = (59, 61, 71, 73)$ and $x = 18\,652\,287$ gives $g = 5$, and $t = 3$ with $(67, 73, 79; 97, 101)$ and $x =$

3 785 449 127 gives $g = 8$. In particular $g(10) \geq 14$ and $g(12) \geq 17$; the ratio $g/n = \frac{3}{2} - \frac{1}{n}$ along the family stays below the $(2 - o(1))n$ of the Erdős–Surányi lower bound, as it must, but these are the largest exact values we have certified at these sizes. Since $\beta = t - 1$ grows with t , the coefficient of β in Theorem 19.14 cannot be replaced by any $\theta < 1$ at these sizes, and cannot be so replaced at all if the placement exists for every t — which we have not proved.

Theorem 19.14 does not settle the conjecture, and the reason is quantitative rather than structural: on the family of all $\binom{\ell}{3}$ triple products of ℓ primes one has $|\mathcal{P}(A)| = \ell \asymp n^{1/3}$ and $\beta/n \rightarrow 2$, so the bound gives only $3n$ there. What is missing is an arithmetic saving on primitive connected instances whose cycle rank exceeds n .

One further negative datum bears on Section 18: the signed carrier architecture of that section cannot simply be re-run at threshold 2. The atom system with the single point $2^{340}3^{214}5^{77}7^{49}$ defeats both the published retention table and a retuned one, the supremum of the relevant quantity being $\frac{5}{256}$ and $\frac{117}{256}$ respectively, in each case far below what a threshold-2 closure would need.

Provenance

Lemma 19.2, Propositions 19.3–19.5 and Remark 19.1 were found by GPT-6 (web interface, “Pro”) in a 41-minute run on 2026-09-08, whose declared targets — a proof of the hinge inequality with threshold 2, or a refutation at some $c \geq 2$ — were *not* reached. Lemma 19.6, Theorems 19.7 and 19.9, Remarks 19.8 and 19.12 and Proposition 19.10 come from a second 50-minute run of the same system on the same day, which likewise did not reach its target: the largest number of primes for which (H_2) is proved without further hypotheses remains four, and no counterexample was found. That run was refereed by a Claude Opus 5 agent which re-derived the layer identity, swept the antichain bound over 6.1 million (quadruple, Q) instances, re-verified the cyclic certificate on 56 877 grid points, rebuilt the explicit five-prime certificate over its full period 25 410 (reproducing its pointwise deficit at $n = 12705$), and gave an independent exact rational Farkas certificate for Proposition 19.10; it also supplied the two structural obstructions in Remark 19.12. The system’s own exhaustive audits cover 190 246 980 window instances on five to eight primes without a violation. They were refereed blind by a Claude Opus 5 agent, which recomputed every constant in exact arithmetic, verified the four-prime certificate exhaustively over all weight vectors in $\{0, \frac{1}{6}, \dots, 1\}^4$ on the primes 2, 3, 5, 7 for all $m \leq 210$ and all windows (105 884 100 instances), established the optimality of the primorial cutoff and the sharper bound $m \geq 66$, and confirmed that the region of Proposition 19.5 is non-empty. None of the results of this section is formalised. The bound of Theorem 19.14 was found by GPT-6 (web interface, “Pro”) in a 46-minute run on 2026-09-08 aimed at the conjecture itself, which it did not reach; the run also restated three results of the introduction and of Section 6 as if new, and those are not repeated here. Proposition 19.16, the reduction of the run’s core-graph corollary to Corollary 19.15, and the exact values at $n \leq 12$ are due to the refereeing Claude Opus 5 agent, which re-derived the proof step by step and recomputed every optimum by an independent dynamic programme. The orchestrating agent then re-verified the sharpness family at $t = 2, 3$ and found that the primes first proposed for it admit no admissible window at $t = 2$, which is why the placement is stated as a hypothesis above and exhibited case by case. The proof of $K_{\{2,3,5,7\}}$ and the resulting upgrade of Theorem 19.7 were found by GPT-6 Astra (OpenAI, through the codex CLI) in a 41-minute single-agent run on 2026-09-08 whose declared targets, a proof or refutation of (H_2) , were again not reached. A Claude Opus 5 agent refereed them, recomputing the frontier sweep independently in three implementations, extending it to $Q \leq 10^{60}$ without change to the maximum, verifying that

the finite range and the analytic tail join with more than five orders of slack, and proving the statement for all 70 four-subsets of the first eight primes and for all quadruples of primes above 10^{15} ; it also verified the certificate of the upgraded theorem over full periods on 46 multilevel systems. None of this is formalised.

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